Unit-3

IOT Data link layer and Network layer protocols

- **PHY/MAC Layer**
  - 3GPP MTC
  - IEEE 802.11
  - IEEE 802.15
  - Wireless HART
  - Z Wave
  - Bluetooth Low Energy
  - Zigbee Smart Energy
  - DASH7

- **Network Layer**
  - IPv4
  - IPv6
  - 6LoWPAN
  - 6TiSCH
  - ND
  - DHCP
  - ICMP
  - RPL
  - CORPL
  - CARP

IOT Data link layer and Network layer protocols

PHY Layer:

- physical transmission of data
- hardware
- electrical specifications
- modulation,
- coding,
- frequency bands,
- transmission power, and
- signal propagation

IOT Data link layer and Network layer protocols

MAC Layer:

- timing
- synchronization
- channel access methods
- avoiding collisions
- efficient utilization of the available bandwidth

IOT Data link layer and Network layer protocols

MAC Layer:

- Channel accessing Methods
  - contention based access (e.g., CSMA/CA - Carrier Sense Multiple Access with Collision Avoidance),
  - time-based access (e.g., TDMA - Time Division Multiple Access), or
  - scheduled access (e.g., FDMA - Frequency Division Multiple Access)

IOT Data link layer and Network layer protocols

MAC Layer:

the MAC layer may handle functions such as

- packet fragmentation,
- error detection,
- acknowledgments, and
- retransmissions

IOT Data link layer and Network layer protocols

1. 3GPP MTC

What is 3GPP ?

ARIB
www.arib.or.jp
The Association of Radio Industries and Businesses, Japan

ATIS
www.atis.org
The Alliance for Telecommunications Industry Solutions, USA

CCSA
www.ccsa.org.cn
China Communications Standards Association

ETSI
www.etsi.org
The European Telecommunications Standards Institute

TSDSI
http://tsdsi.org/
Telecommunications Standards Development Society, India



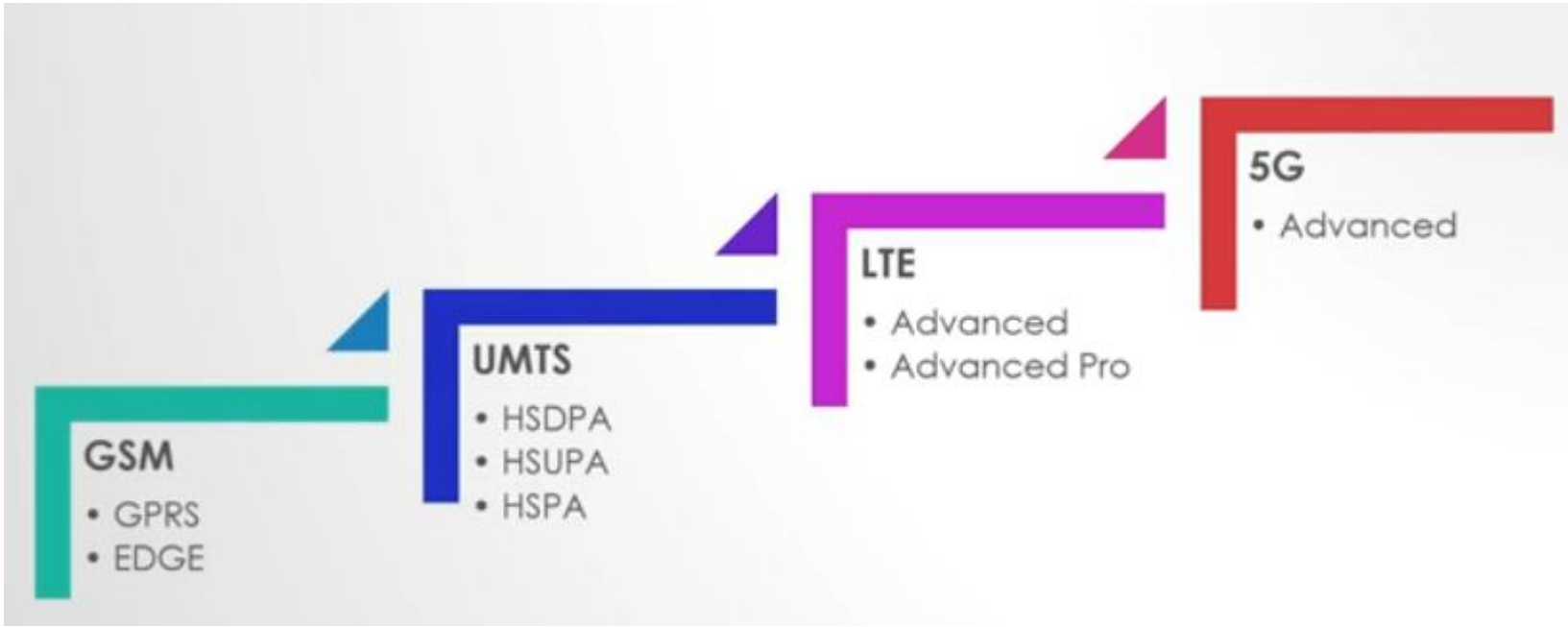
TTA
www.tta.or.kr
Telecommunications Technology Association, Korea

TTC
www.ttc.or.jp
Telecommunication Technology Committee, Japan

IOT Data link layer and Network layer protocols

1. 3GPP MTC

3GPP Supports



IOT Data link layer and Network layer protocols

1. 3GPP MTC



- 3GPP (3rd Generation Partnership Project) MTC (Machine-Type Communications) is designed to support communication between machines, like IoT devices, over cellular networks.

IOT Data link layer and Network layer protocols

1. 3GPP MTC

Key aspects of 3GPP MTC in IoT

i. **Low-Power Consumption**

ii. **Enhanced Coverage**

iii. **Support for Massive Device Connectivity**

iv. **Low-Cost Device Design**

v. **Reduced Data Transmission Requirements**

vi. **Enhanced Security for IoT**

vii. **Efficient Network Resource Usage**

viii. **Improved Latency Control**

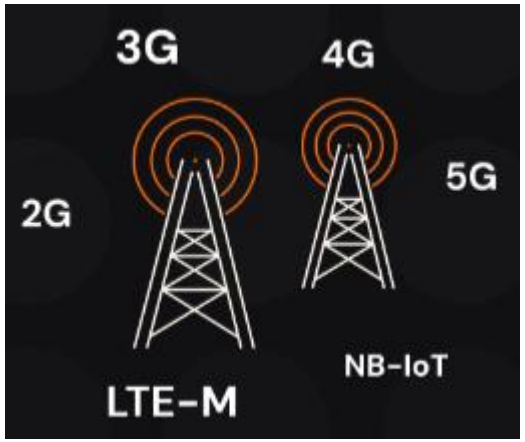
ix. **Support for Mobility and Stationary Devices**

IOT Data link layer and Network layer protocols

1. 3GPP MTC

Key 3GPP Technologies Under MTC:

- **LTE-M**: (Long Term Evoluation for Machines) :Designed to provide low-power, wide-area connectivity within LTE networks.
- **NB-IoT**: A narrowband technology that supports extended coverage and low data rate, making it ideal for low-cost and low-power IoT devices.



IOT Data link layer and Network layer protocols

1. 3GPP MTC



What are the key features of LTE-M?

FEATURE	LTE-M PROTOCOL
Deployment	In-band LTE
Minimum carrier bandwidth	1400 kHz
Data rate	Up to 1000 kbps
Transmission range	5 km
Security	AES 256
Modulation	QPSK, 16-QAM and 64-QAM
Duplexing	FD & HD (type B), FDD & TDD
Power saving	PSM, ext. I-DRX, C-DRX
Power class	23 dBm, 20 dBm
Battery life time	10 years
Link budget	146 db

IOT Data link layer and Network layer protocols

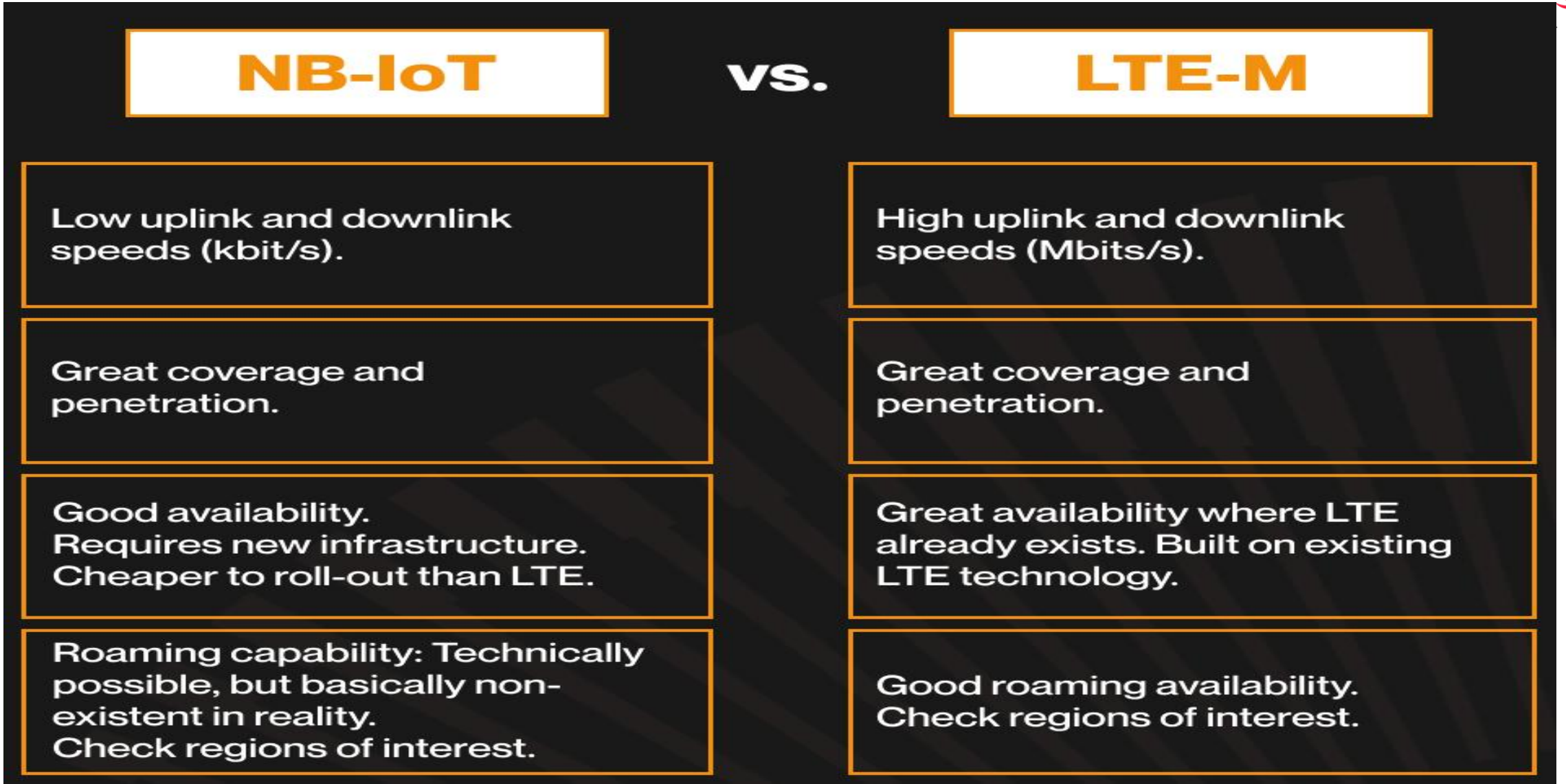
1. 3GPP MTC

NB-IoT evolution: LTE Cat-NB1 vs Cat-NB2

Feature	LTE Cat-NB1 (NB-IoT release 13)	LTE Cat-NB2 (NB-IoT release 14)
Uplink data rate	Up to 66 kbps	Up to 159 kbps
Downlink data rate	Up to 26 kbps	Up to 127 kbps
Bandwidth	180 KHz carrier	180 kHz carrier with improved modulation
Power consumption	Very low, supports PSM and eDRX	Low, supports PSM and eDRX with enhancements
Coverage	Good, deep indoor and rural areas	Good, with improved performance in challenging environments
Latency	Higher	Lower
Efficiency	Standard	Enhanced
Positioning	Basic (Cell-ID)	Enhanced (OTDOA, E-CID)

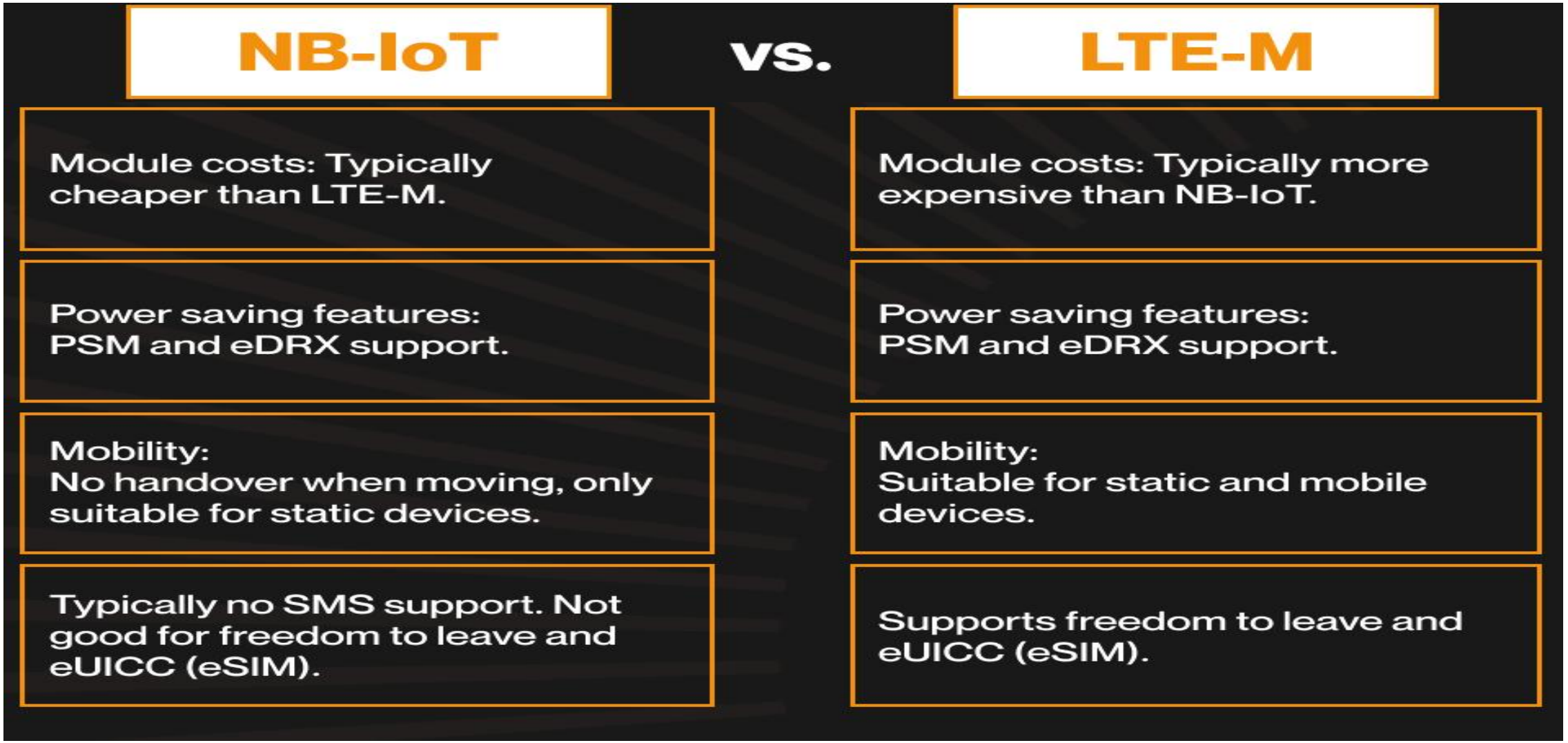
IOT Data link layer and Network layer protocols

1. 3GPP MTC



IOT Data link layer and Network layer protocols

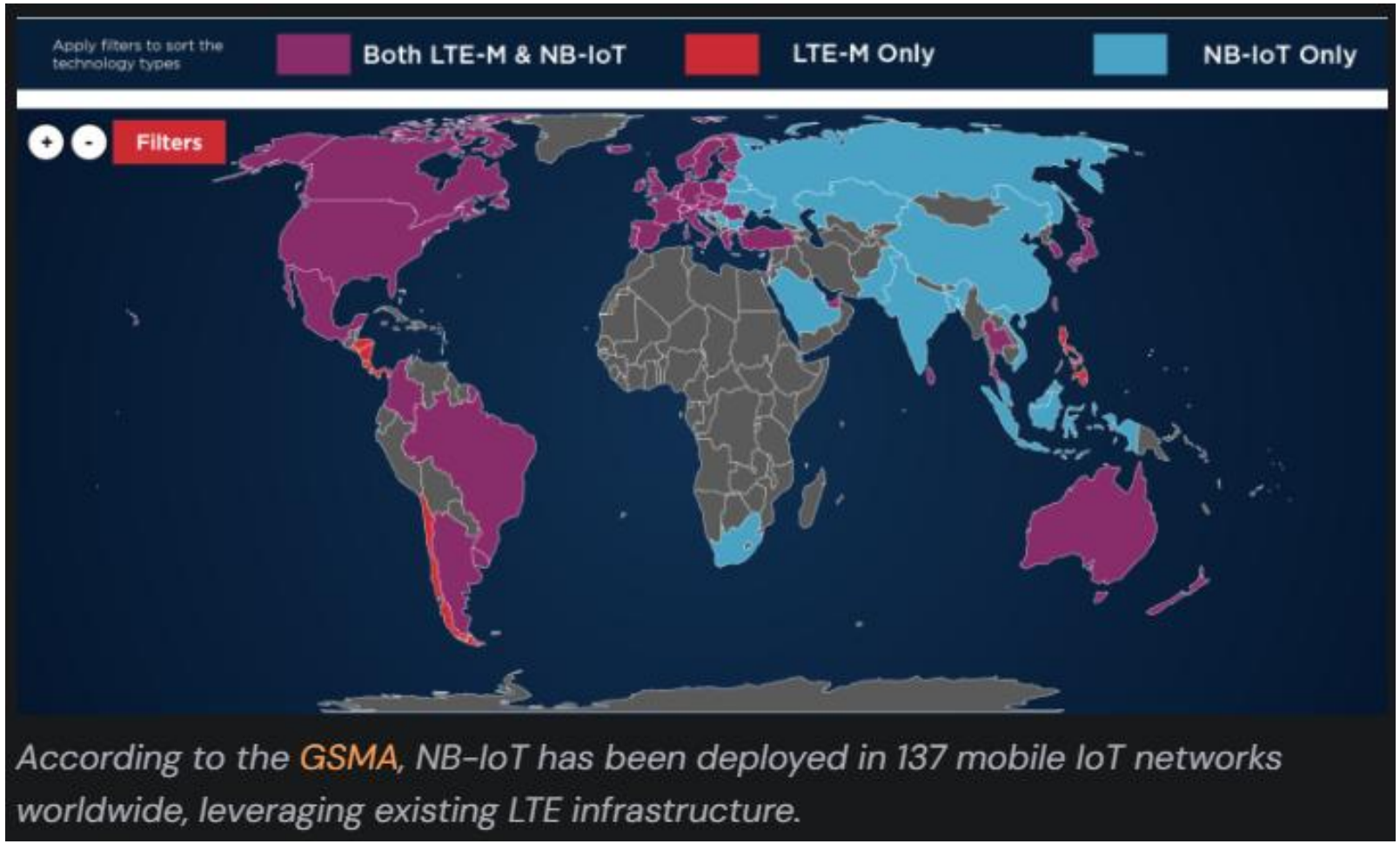
1. 3GPP MTC



IOT Data link layer and Network layer protocols

1. 3GPP MTC

Global distribution map of mobile connectivity

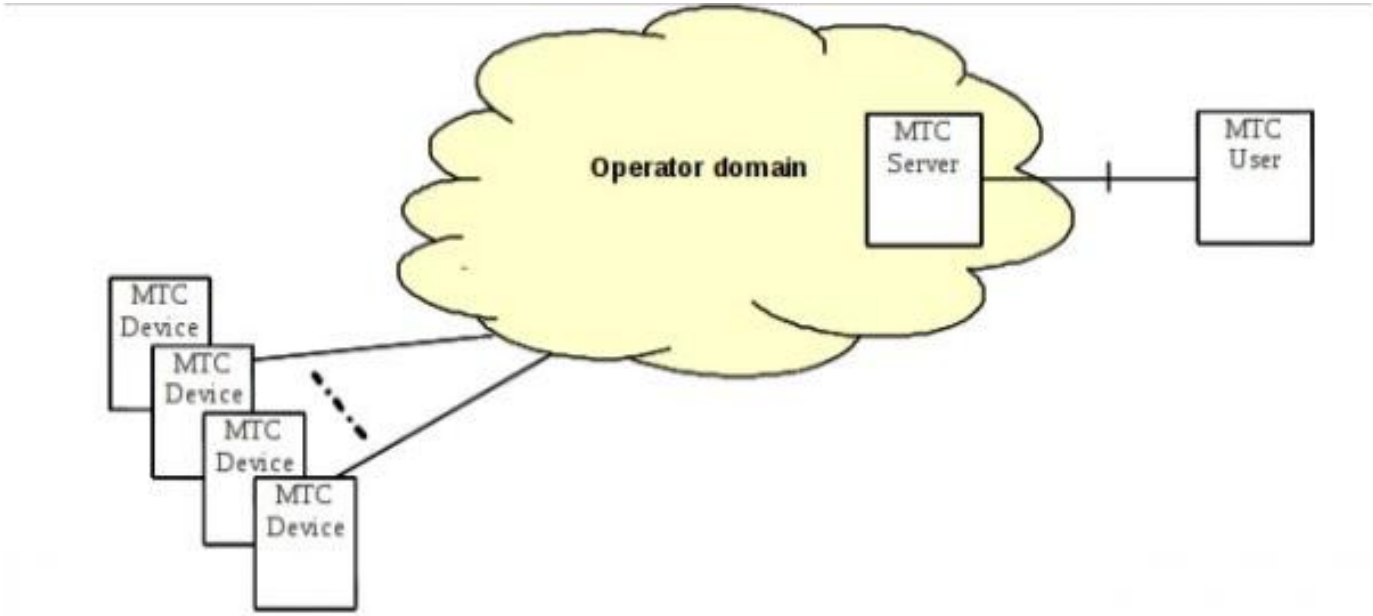


IOT Data link layer and Network layer protocols

1. 3GPP MTC

MTC communication scenarios

1) Communication scenario with MTC devices communicating with MTC server. MTC server is located in the operator domain



3GPP MTC Architecture-MTC server located in the operated domain

IOT Data link layer and Network layer protocols

1. 3GPP MTC

MTC communication scenarios

2) Communication scenario with MTC devices communicating with MTC server. MTC server is located outside the operator domain:

The network operator provides network connectivity to MTC Server(s). This applies to MTC Servers controlled by the network operator or to MTC Servers not controlled by the network operator



3GPP MTC Architecture-MTC server located outside the operated domain

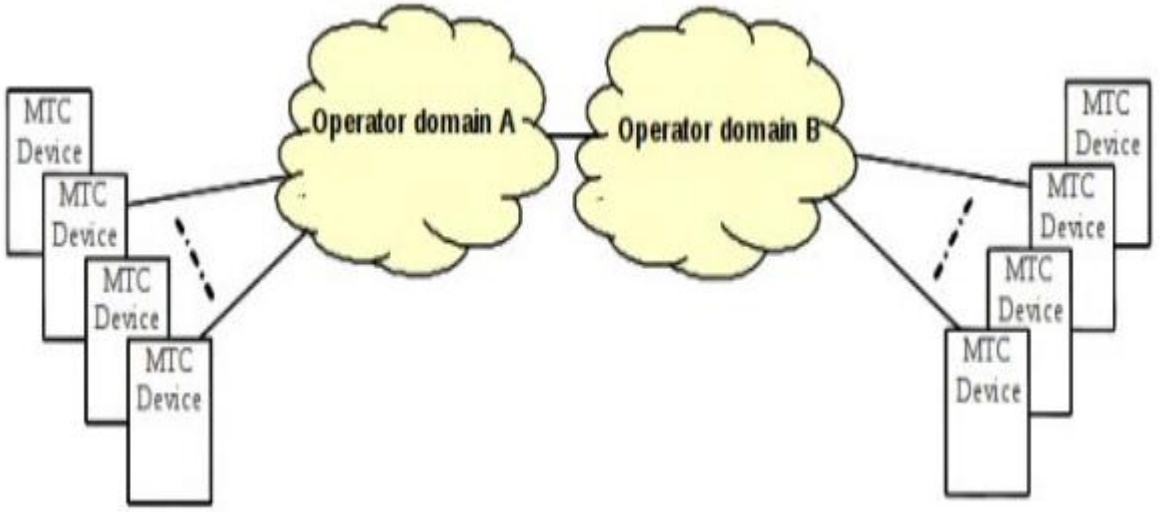
IOT Data link layer and Network layer protocols

1. 3GPP MTC

MTC communication scenarios

3) MTC devices communicating with each other: The communication scenario where the

MTC Devices communicate directly without intermediate MTC Server



IOT Data link layer and Network layer protocols

1. 3GPP MTC

MTC communication scenarios

- MTC Device: A MTC Device is a UE equipped for Machine Type Communication, which communicates through a PLMN with MTC Server(s) and/or other MTC Device(s).
- MTC Feature: MTC Features are network functions to optimise the network for use by M2M applications.
- MTC Group: A MTC Group is a group of MTC Devices that share one or more MTC Features and that belong to the same MTC Subscriber.

IOT Data link layer and Network layer protocols

1. 3GPP MTC

MTC communication scenarios

- MTC Server: A MTC Server is a server, which communicates to the PLMN itself, and to MTC Devices through the PLMN. The MTC Server can also have an interface which can be accessed by the MTC User. The MTC Server can:
- Provides services for other servers (e.g. The MTC Server is a Services Capability Server for an Application Server), and/or Provides services for applications and can host the application (e.g. The MTC Server is an Application Server)

IOT Data link layer and Network layer protocols

1. 3GPP MTC

MTC communication scenarios

- MTC User: A MTC User uses the service provided by the MTC Server.
- MTC Subscriber: A MTC Subscriber is a legal entity having a contractual relationship with the network operator to provide service to one or more MTC Devices



IOT Data link layer and Network layer protocols

1. 3GPP MTC

PHY layer features

- **Narrowband Operation:** LTE-M operates within a restricted 1.4 MHz channel , whereas NB-IoT scales down to 180 kHz.
- **Coverage Enhancement (CE) via Subframe Repetition:** Delivers a 15–20 dB link budget gain over standard LTE to reach deep indoors and underground.
- **Half-Duplex FDD (HD-FDD):** By disabling RX while TX is active, devices remove the need for costly and space-consuming RF duplexers.
- **Lower-Order Modulation:** Uses robust, simplified modulation schemes like BPSK and QPSK.
- **Reduced Transmit Power:** Supports lower maximum output power classes (such as 14 dBm or 20 dBm instead of standard 23 dBm) to avoid peak current spikes on small batteries.

IOT Data link layer and Network layer protocols

1. 3GPP MTC

MAC layer features

- **Power Saving Mode (PSM):** Allows a device to turn off its radio transceiver and enter a deep sleep state for days or weeks.
- **Extended Discontinuous Reception (eDRX):** Lengthens the paging listening cycles up to 40 minutes for LTE-M (and several hours for NB-IoT), letting the device sleep soundly between scheduled downlinks while staying reachable.
- **Early Data Transmission (EDT):** Permits small data payloads to be included directly inside the 4-step Random Access Channel (RACH) handshakes.
- **Repetition Scheduling & CE Modes:** The MAC layer dynamically manages frame scheduling based on Coverage Enhancement levels.

IOT Data link layer and Network layer protocols

1. 3GPP MTC

Hardware Modules

- **Quectel**: BG95-M3, BG96
- **SIMCom**: SIM7080G, SIM7000G
- **Nordic Semiconductor**: nRF9160, nRF9161
- **u-blox**: SARA-R5, SARA-R4
- **Telit Cinterion**: ME910G1, EXS82
- **Sierra Wireless**: AirPrime HL7800



Quectel BG95-M3

IOT Data link layer and Network layer protocols

1. 3GPP MTC



SIM7080G



SIM7000E

IOT Data link layer and Network layer protocols

IEEE 802.11

What is IEEE 802.11 ?





IOT Data link layer and Network layer protocols

IEEE 802.11

Why is IEEE 802.11 ?

- This standard is a key IoT wireless access technology,
- either for connecting endpoints such as fog computing nodes,
- high-data-rate sensors, and
- audio or video analytics devices or
- for deploying Wi-Fi backhaul infrastructures, such as
- outdoor Wi-Fi mesh in smart cities, oil and mining, or other environments.
- However, Wi-Fi lacks sub-GHz support for better signal penetration, low power for battery-powered nodes, and the ability to support a large number of devices.

IOT Data link layer and Network layer protocols

IEEE 802.11

- IEEE 802.11 standard, popularly known as WiFi, lays down the architecture and specifications of wireless LANs (WLANs).
- WiFi or WLAN uses high frequency radio waves for connecting the nodes.

- IEEE 802.11 is a set of standards for wireless LAN technology created by the Institute of Electrical and Electronics Engineers (IEEE)

- The original 802.11 standard was released in 1997



IOT Data link layer and Network layer protocols

IEEE 802.11

There are several standards of IEEE 802.11 WLANs.

The prominent among them are

802.11,

802.11a,

802.11b,

802.11g,

802.11n and

802.11p.



- All the standards use carrier-sense multiple access with collision avoidance (CSMA/CA).
- Also, they have support for both centralized base station based as well as ad-hoc networks

IOT Data link layer and Network layer protocols

IEEE 802.11

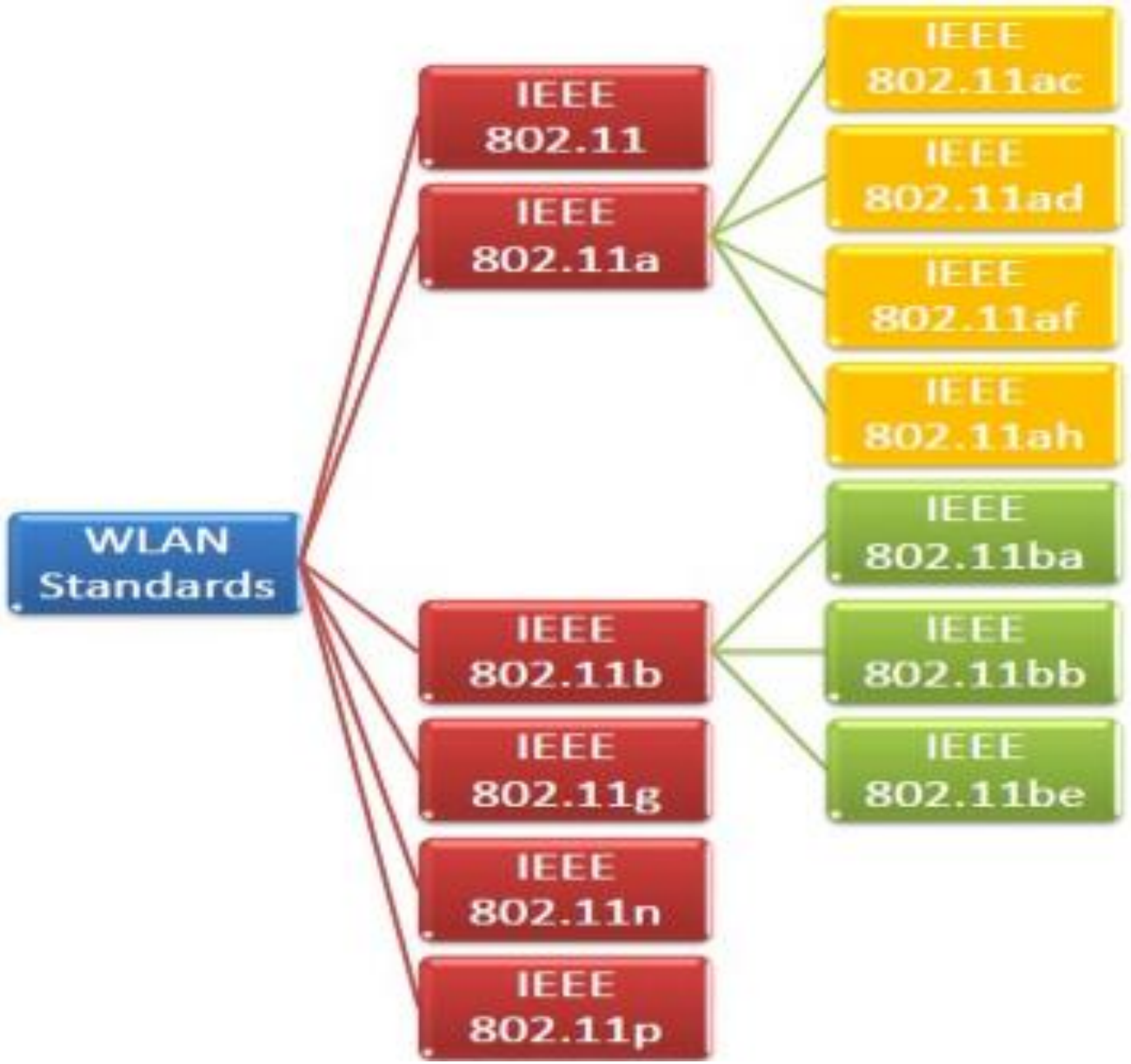


Evolution of Wi-Fi Standards

Wi-Fi Gen	IEEE Standard	Release Date	2.4 GHz	5 GHz	Max Data Rate
Wi-Fi	802.11	1997	Yes	No	2 Mbps
Wi-Fi 1	802.11b	1999	Yes	No	11 Mbps
Wi-Fi 2	802.11a	1999	No	Yes	54 Mbps
Wi-Fi 3	802.11g	2003	Yes	No	54 Mbps
Wi-Fi 4	802.11n	2009	Yes	Yes	600 Mbps
Wi-Fi 5	802.11ac	2013	No	Yes	6.93 Gbps
Wi-Fi 6	802.11ax	2019	Yes	Yes	14 Gbps

IOT Data link layer and Network layer protocols

IEEE 802.11



IOT Data link layer and Network layer protocols

IEEE 802.11

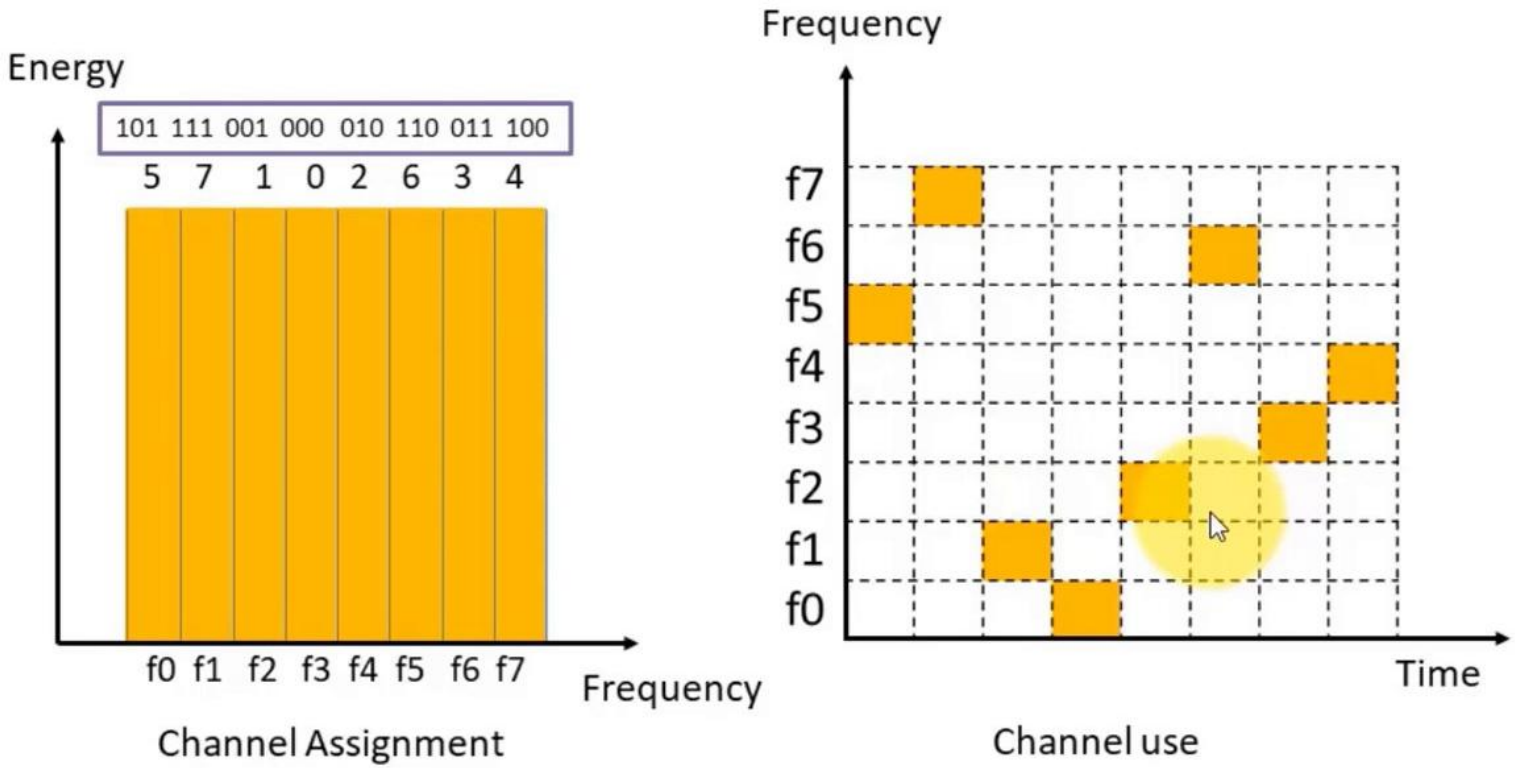
- IEEE 802.11:
- was the original version released in 1997.
- It provided 1 Mbps or 2 Mbps data rate in the 2.4 GHz band and used either frequency-hopping spread spectrum (FHSS) or direct-sequence spread spectrum (DSSS).
- It is obsolete now.



IOT Data link layer and Network layer protocols

IEEE 802.11

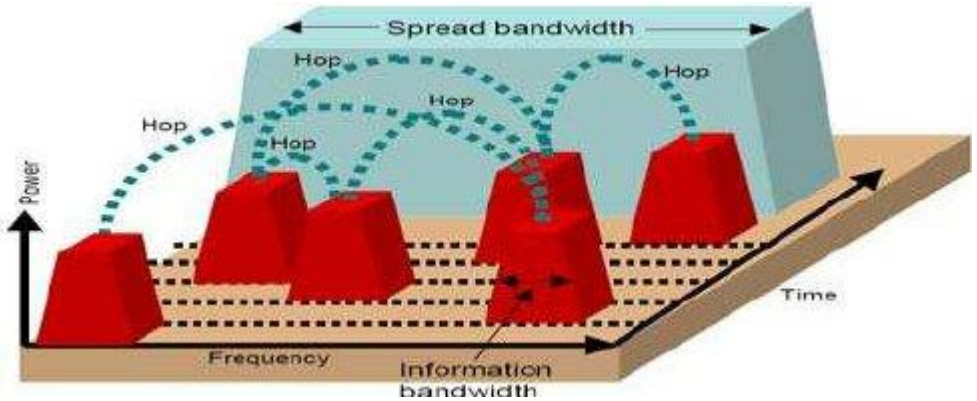
Frequency-hopping Spread Spectrum (FHSS)

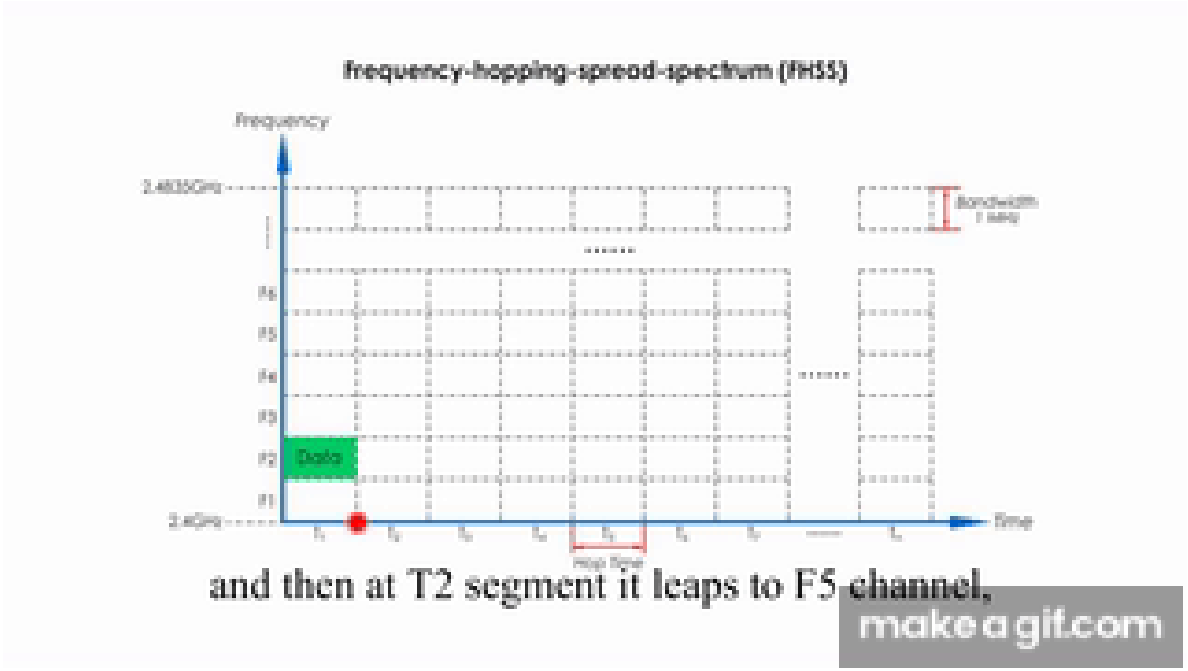


IOT Data link layer and Network layer protocols

IEEE 802.11

Frequency-hopping Spread Spectrum (FHSS)

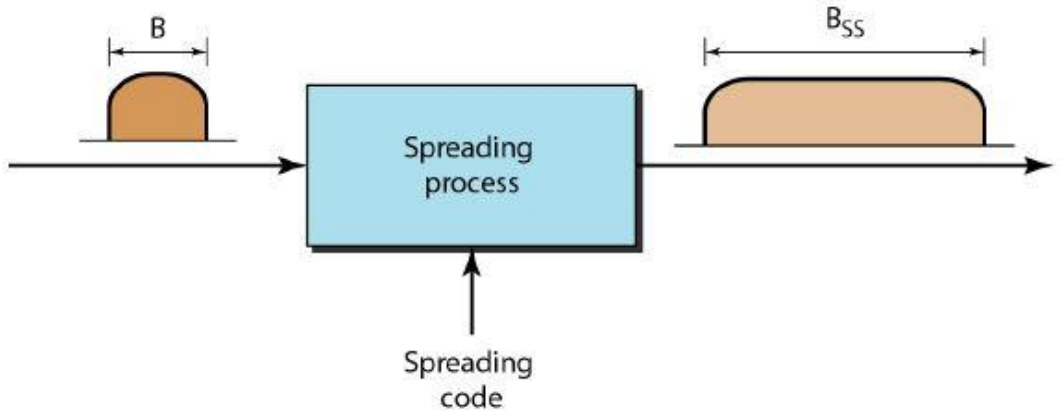


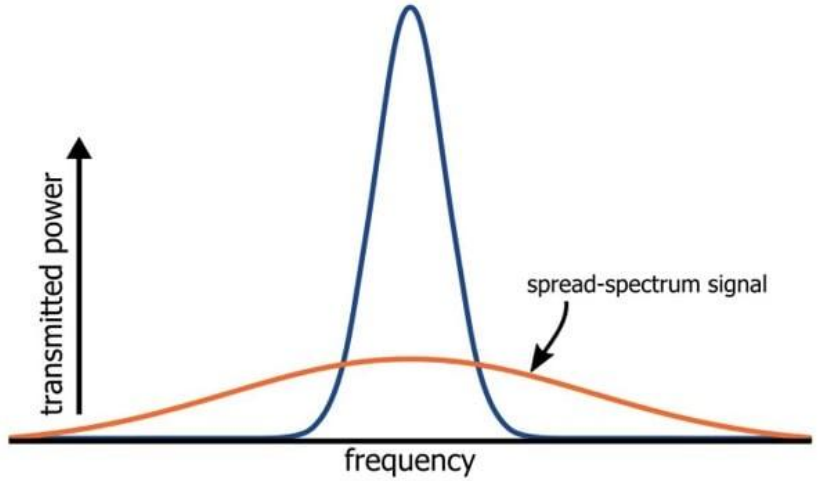


IOT Data link layer and Network layer protocols

IEEE 802.11

Direct-sequence Spread Spectrum (DSSS)





The spreading code is a series of numbers that look random, but are actually a pattern

IOT Data link layer and Network layer protocols

IEEE 802.11

- IEEE 802.11a :
- was published in 1999 as a modification to 802.11
- with orthogonal frequency division multiplexing (OFDM) based air interface in physical layer instead of FHSS or DSSS of 802.11.
- It provides a maximum data rate of 54 Mbps operating in the 5 GHz band.
- Besides it provides error correcting code.
- As 2.4 GHz band is crowded, relatively sparsely used 5 GHz imparts additional advantage to 802.11a.
- Further amendments to 802.11a are 802.11ac, 802.11ad, 802.11af, 802.11ah, 802.11ai, 802.11aj etc

IOT Data link layer and Network layer protocols

IEEE 802.11

- IEEE 802.11b:
- Is a direct extension of the original 802.11 standard that appeared in early 2000.
- It uses the same modulation technique as 802.11, i.e. DSSS and operates in the 2.4 GHz band.
- It has a higher data rate of 11 Mbps as compared to 2 Mbps of 802.11, due to which it was rapidly adopted in wireless LANs.
- However, since 2.4 GHz band is pretty crowded, 802.11b devices faces interference from other devices.
- Further amendments to 802.11b are 802.11ba, 802.11bb, 802.11bc, 802.11bd and 802.11be

IOT Data link layer and Network layer protocols

IEEE 802.11

- IEEE 802.11g :
- was indorsed in 2003.
- It operates in the 2.4 GHz band (as in 802.11b) and provides a average throughput of 22 Mbps.
- It uses OFDM technique (as in 802.11a).
- It is fully backward compatible with 802.11b.
- 802.11g devices also faces interference from other devices operating in 2.4 GHz band.

IOT Data link layer and Network layer protocols

IEEE 802.11

- IEEE 802.11n:
- was approved and published in 2009 that operates on both the 2.4 GHz and the 5 GHz bands.
- It has variable data rate ranging from 54 Mbps to 600 Mbps.
- It provides a marked improvement over previous standards 802.11 by incorporating multiple-input multiple-output antennas (MIMO antennas).

IOT Data link layer and Network layer protocols

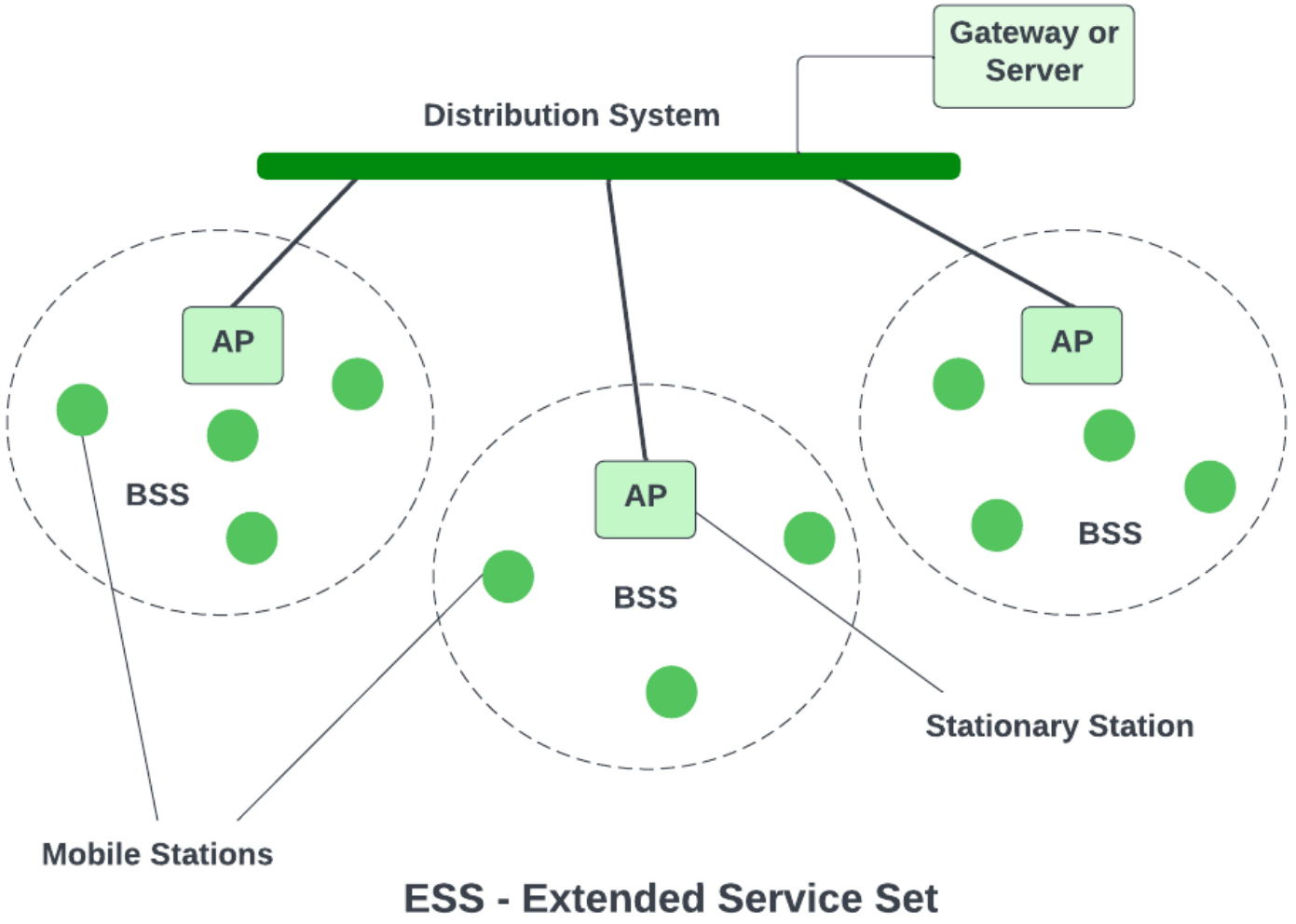
IEEE 802.11

- IEEE 802.11p:
- Is an amendment for including wireless access in vehicular environments (WAVE) to support Intelligent Transportation Systems (ITS).
- They include network communications between vehicles moving at high speed and the environment.
- They have a data rate of 27 Mbps and operate in 5.9 GHz band.

IOT Data link layer and Network layer protocols

IEEE 802.11

IEEE 802.11 protocol architecture:



IOT Data link layer and Network layer protocols

IEEE 802.11

IEEE 802.11 protocol architecture:

- Wireless LANs are those Local Area Networks that use high frequency radio waves instead of cables for connecting the devices in LAN.
- Users connected by WLANs can move around within the area of network coverage.
- Most WLANs are based upon the standard IEEE 802.11 or WiFi.

IOT Data link layer and Network layer protocols

IEEE 802.11

IEEE 802.11 protocol architecture:

- The components of IEEE 802.11 architecture

i. Stations (STA)

ii. Basic Service Set (BSS)

iii. Extended Service Set (ESS)

iv. Distribution System (DS)

IOT Data link layer and Network layer protocols

IEEE 802.11

IEEE 802.11 protocol architecture:

i. Stations (STA):

- Stations comprise all devices and equipments that are connected to the wireless LAN.
- A station can be of two types:
  - Wireless Access Pointz (WAP) – WAPs or simply access points (AP) are generally wireless routers that form the base stations or access.
  - Client. – Clients are workstations, computers, laptops, printers, smart phones, etc.
- Each station has a wireless network interface controller.

IOT Data link layer and Network layer protocols

IEEE 802.11

IEEE 802.11 protocol architecture:

ii. Basic Service Set (BSS)

- A basic service set is a group of stations communicating at physical layer level.
- BSS can be of two categories depending upon mode of operation:
  - Infrastructure BSS – Here, the devices communicate with other devices through access points.
  - Independent BSS – Here, the devices communicate in peer-to-peer basis in an ad hoc manner.

IOT Data link layer and Network layer protocols

IEEE 802.11

IEEE 802.11 protocol architecture:

iii. Extended Service Set (ESS)

- It is a set of all connected BSS.

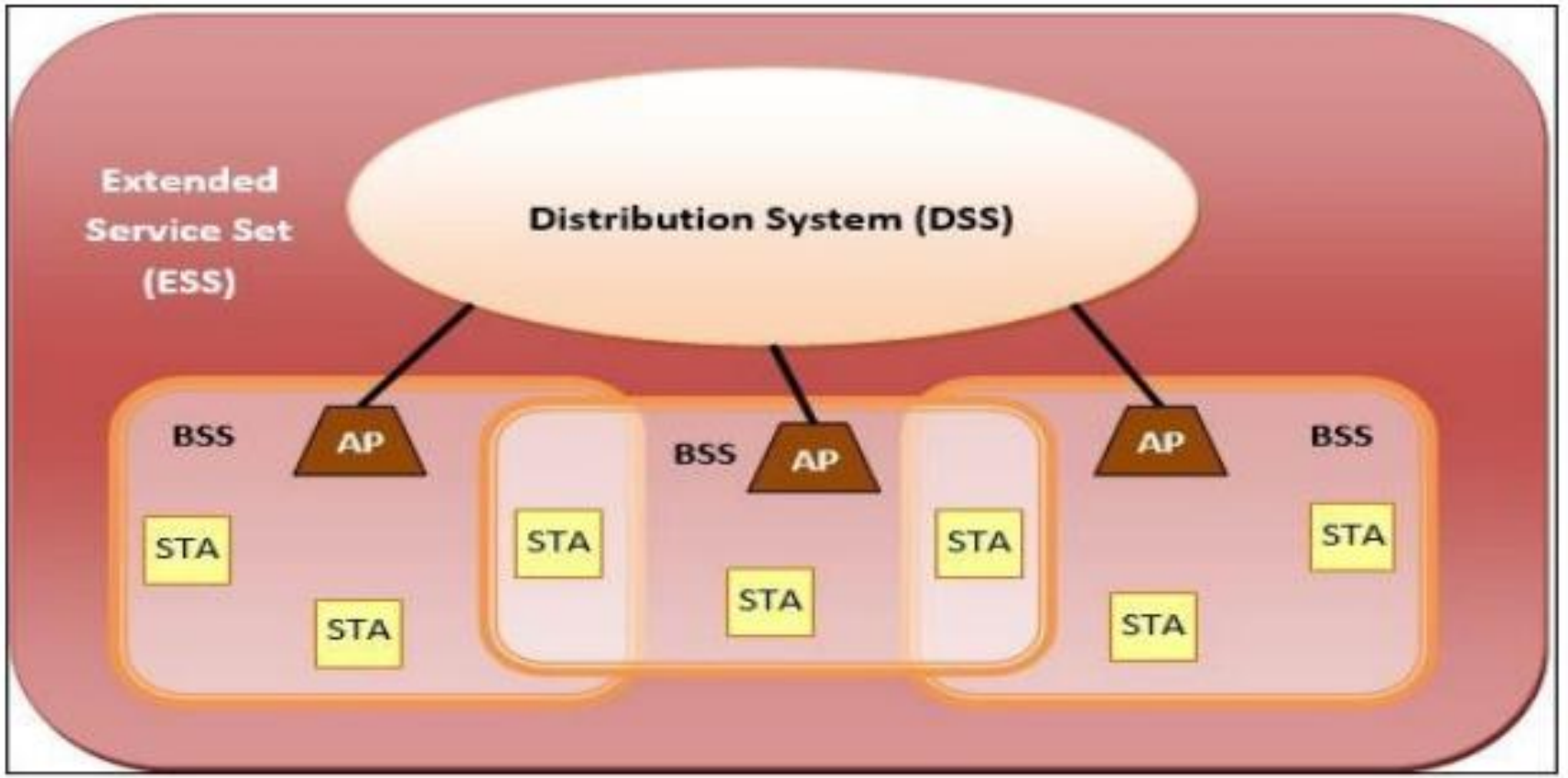
iv. Distribution System (DS)

- It connects access points in ESS.

IOT Data link layer and Network layer protocols

IEEE 802.11

IEEE 802.11 protocol architecture:



IOT Data link layer and Network layer protocols

IEEE 802.11

IEEE 802.11 protocol architecture:

- An ESS can also provide gateway access for wireless users into a wired network.
- Each end station associates itself with one access point.
- If station A, associated with AP-1 wants to send a frame to another station associated with AP- 2, the first sends a frame to its access point (AP-1), which forwards the frame across the distribution system to the access point AP-2.
- AP-2 finally delivers it to the destination station.

IOT Data link layer and Network layer protocols

IEEE 802.11

IEEE 802.11 protocol architecture:

- The technique used for this purpose is known as scanning, which involves the following steps:
- A station sends a probe frame.
- All APs within reach reply with a probe response frame.
- The station selects one of the access points, and sends the AP an Association Request frame.
- The AP replies with an Association Response frame.



IOT Data link layer and Network layer protocols

IEEE 802.11

- Medium Access Control:

  - CSMA CA stands for Carrier Sense Multiple Access – Collison Avoidance.
  - It is a network protocol that operates in the data link layer of the OSI model.
  - It uses carrier sensing to check if the channel is idle before transmitting data.

IOT Data link layer and Network layer protocols

IEEE 802.11

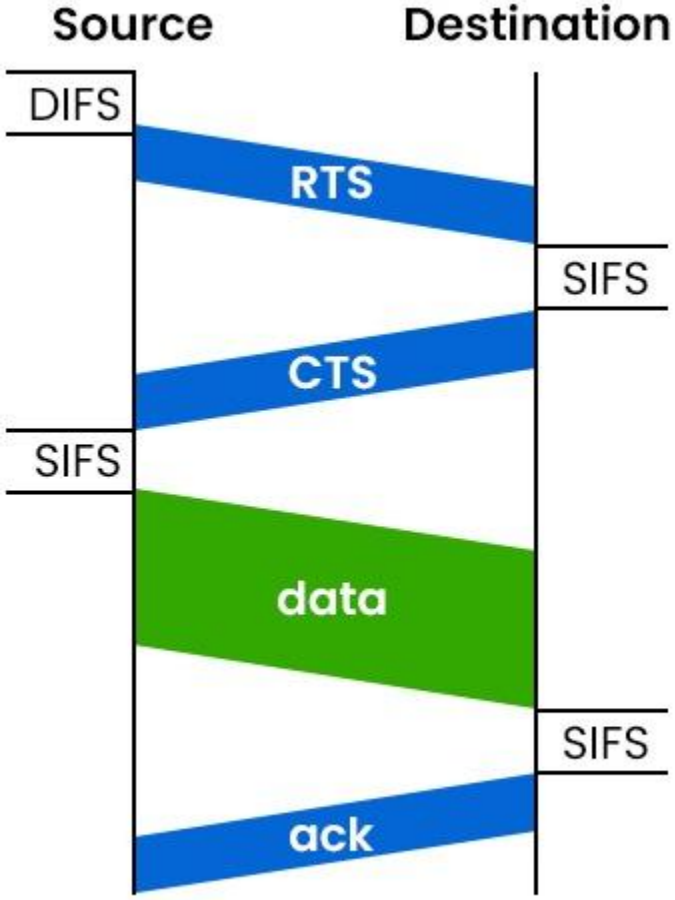
- Medium Access Control:
  - It also uses collision avoidance techniques such as interframe spacing, contention windows, and acknowledgments to reduce the chances of collisions.
  - CSMA/CA is mainly used in wireless networks, where collision detection is difficult or impossible.



IOT Data link layer and Network layer protocols

IEEE 802.11

- Medium Access Control:
  - CSMA CA
    - To prevent a collision, devices rely on signals called "Ready to Send" (RTS) and "Clear to Send" (CTS).
    - A device will scan the airwaves for a signal when it is ready to send.
    - If there isn't already one, it will create a request to transmit (RTS) signal.
    - It then delivers its own data and concludes with a CTS signal, which lets other wireless devices know it's their turn to communicate.



CSMA CA

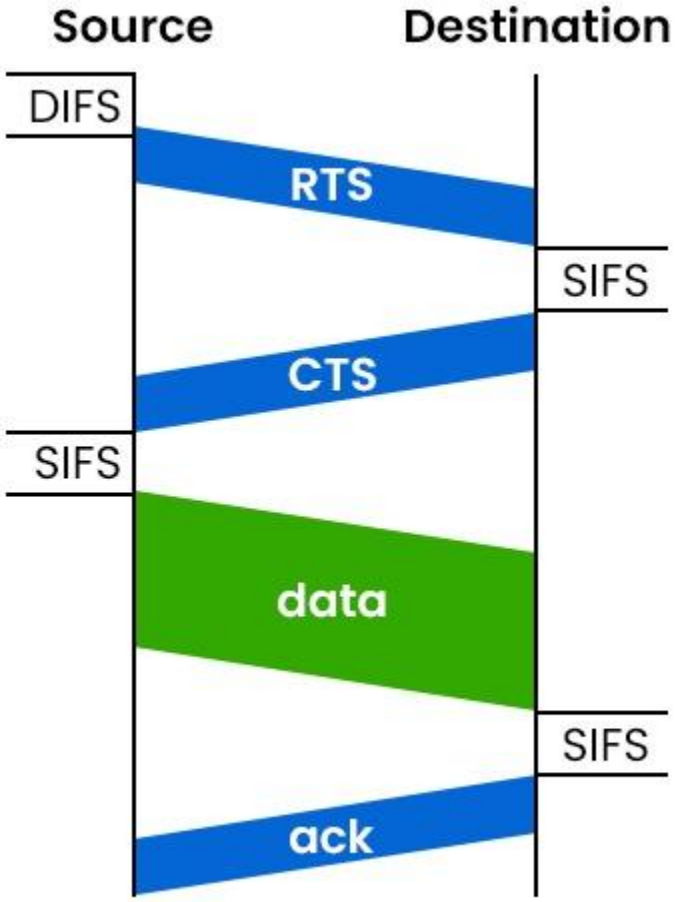
IOT Data link layer and Network layer protocols

IEEE 802.11

- Medium Access Control:
  - CSMA CA

DIFS- distributed inter-frame space- is the time a wireless station waits to send data after checking that the channel is clear

SIFS-Short Interframe Space-SIFS is the smallest IFS and is used for sending an ACK after receiving data.



CSMA CA

IOT Data link layer and Network layer protocols

IEEE 802.11

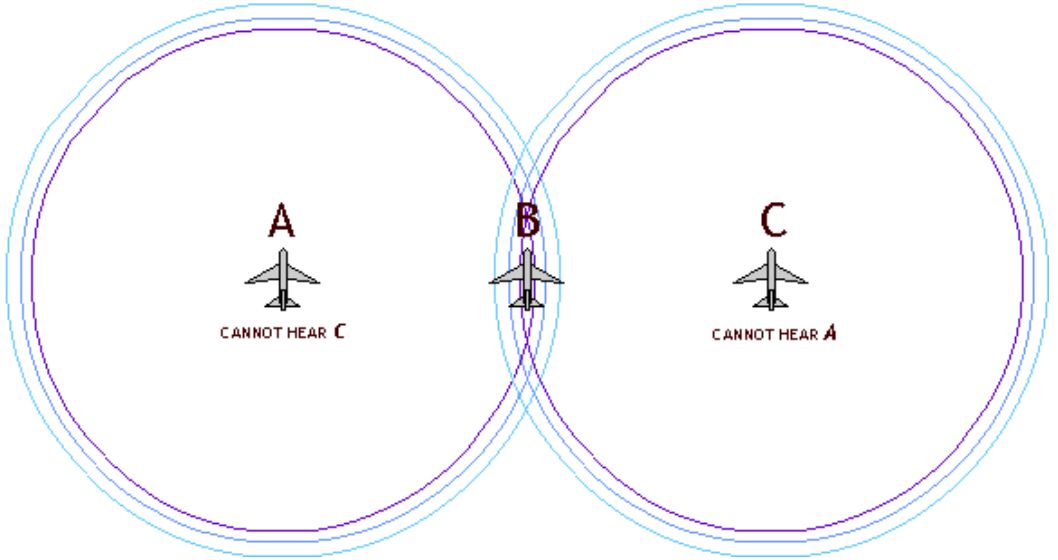


- Medium Access Control:
  - However, the wireless medium presents some unique challenges not present in wired LANs that must be dealt with by the MAC used for IEEE 802.11.
  - Some of the challenges are:
    - The wireless LAN is prone to more interference and is less reliable.
    - The wireless LAN is susceptible to unwanted interception leading to security problems.
    - There are so called hidden station and exposed station problems.

IOT Data link layer and Network layer protocols

IEEE 802.11

- Medium Access Control:
  - The Hidden Station Problem
    - A is transmitting to B
    - If C senses the media, it will not hear anything because it is out of range
    - thus will falsely conclude that no transmission is going on and will start transmit to B.
    - the transmission will interfere at B, wiping out the frame from A
    - C act as a hidden station to A

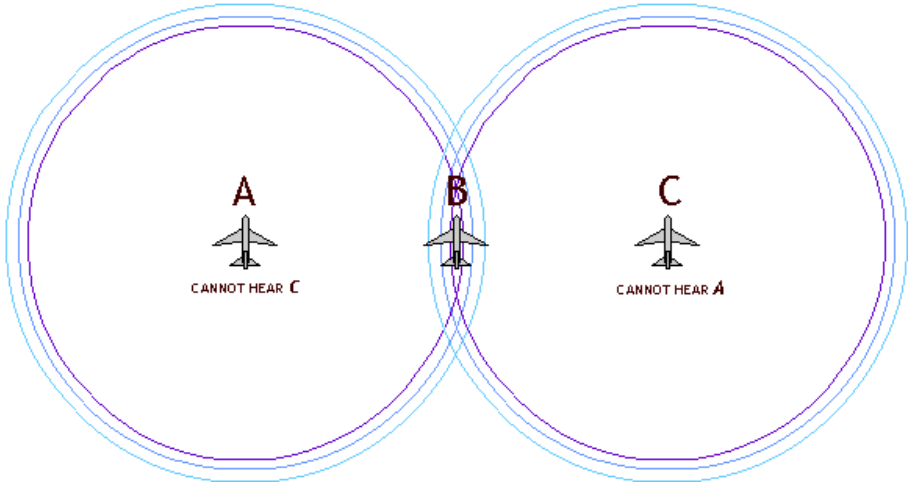


IOT Data link layer and Network layer protocols

IEEE 802.11

- Medium Access Control:

  - The Hidden Station Problem

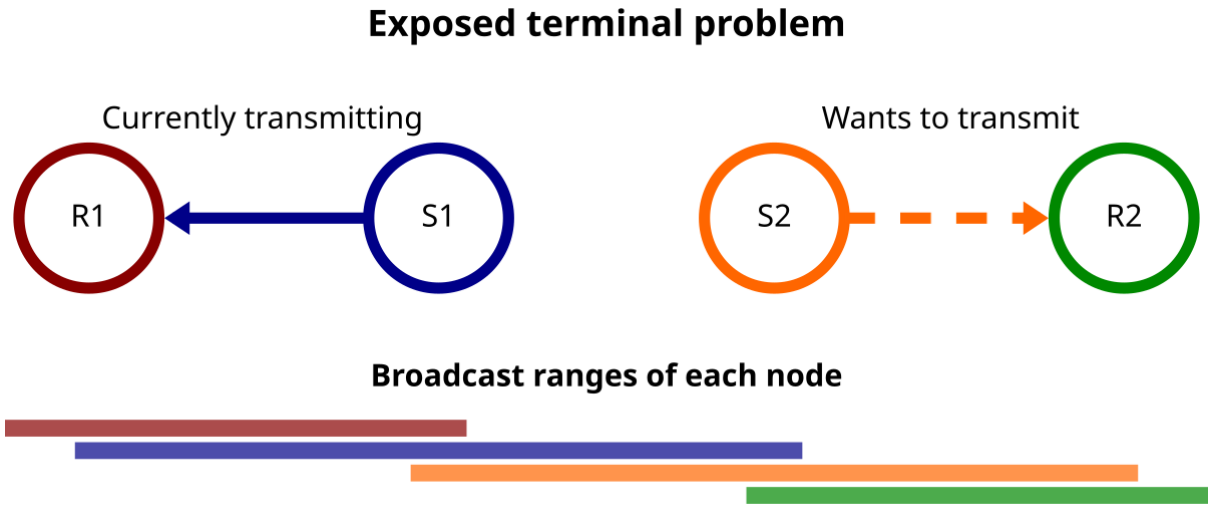


  - The problem of a station not been able to detect a potential competitor for the medium because the competitor is too far away is referred as Hidden Station Problem.

IOT Data link layer and Network layer protocols

IEEE 802.11

- Medium Access Control:
  - Exposed Station problem

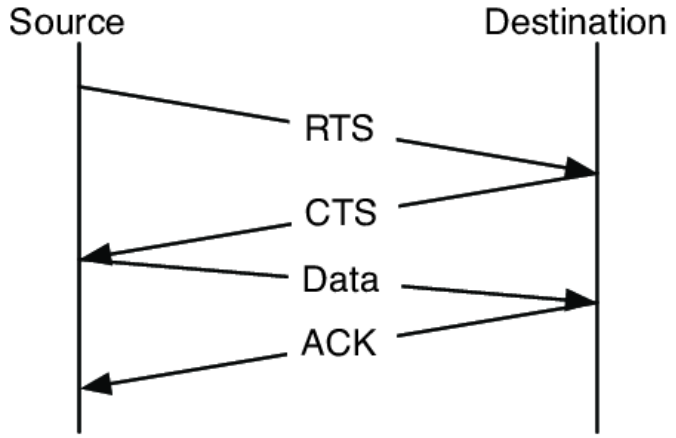


- S1 is transmitting to R1,
- and S2 sense the medium and detects the ongoing transmission between S1 and R1.
- S2 falsely conclude that it cannot transmit to R2, when the fact is that such transmission would cause no problem.
- A transmission could cause a problem only when the destination is in zone between S1 and S2.

IOT Data link layer and Network layer protocols

IEEE 802.11

- Remedy to Hidden Node and Exposed node Problem



•**Use request-to-send/clear-to-send (RTS/CTS) handshaking**:

•This is implemented at the Access Point along with the Carrier sense multiple access with collision avoidance (CSMA/CA) scheme.

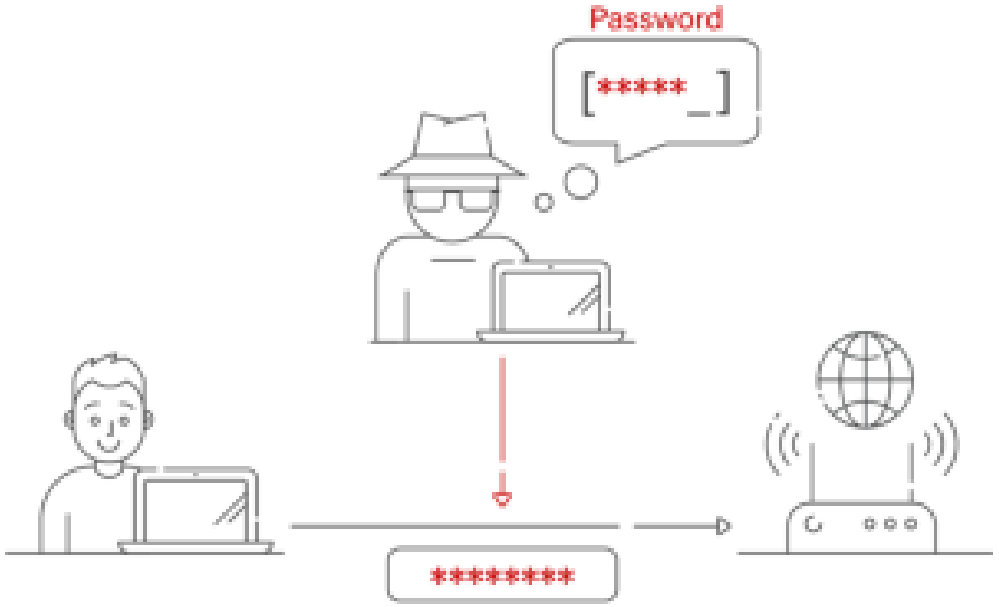
• RTS/CTS handshaking helps to overcome the hidden node problem, but it's not a complete solution.



IOT Data link layer and Network layer protocols

IEEE 802.11

- Security in MAC layer

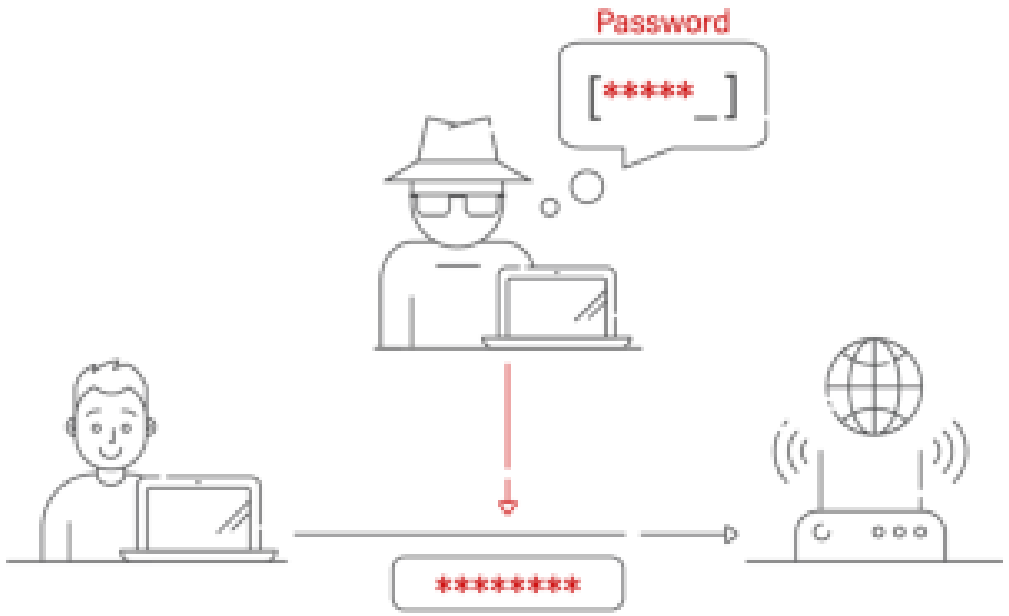


- Wireless LANs are subjected to possible breaches from unwanted monitoring.
- To overcome this problem, IEEE 802.11 specifies an optional MAC layer security system known as Wired Equivalent Privacy (WEP).

IOT Data link layer and Network layer protocols

IEEE 802.11

- Security in MAC layer
  - The objective is to provide a level of privacy to the wireless LAN similar to that enjoyed by wired Ethernets.
  - It is achieved with the help of a 40-bit shared key authentication service.

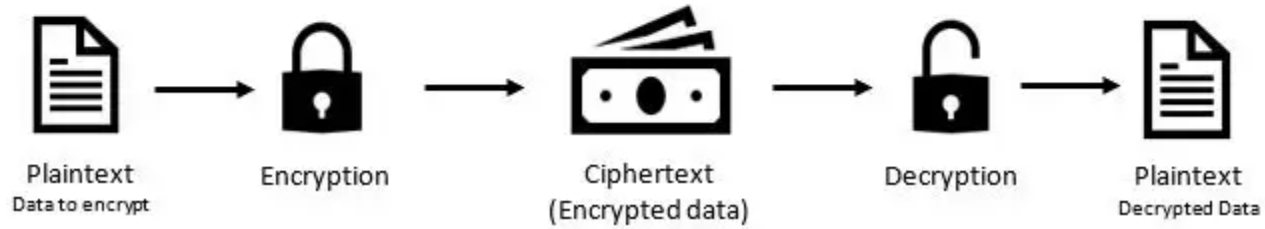


IOT Data link layer and Network layer protocols

IEEE 802.11

- Security in MAC layer
  - By default each BSS supports up to four 40-bit keys that are shared by all the clients in the BSS.
  - Keys unique to a pair of communicating clients and direction of transmission may also be used.
  - Advanced Encryption Standard (AES) for authentication and encryption is recommended as a long term solution.

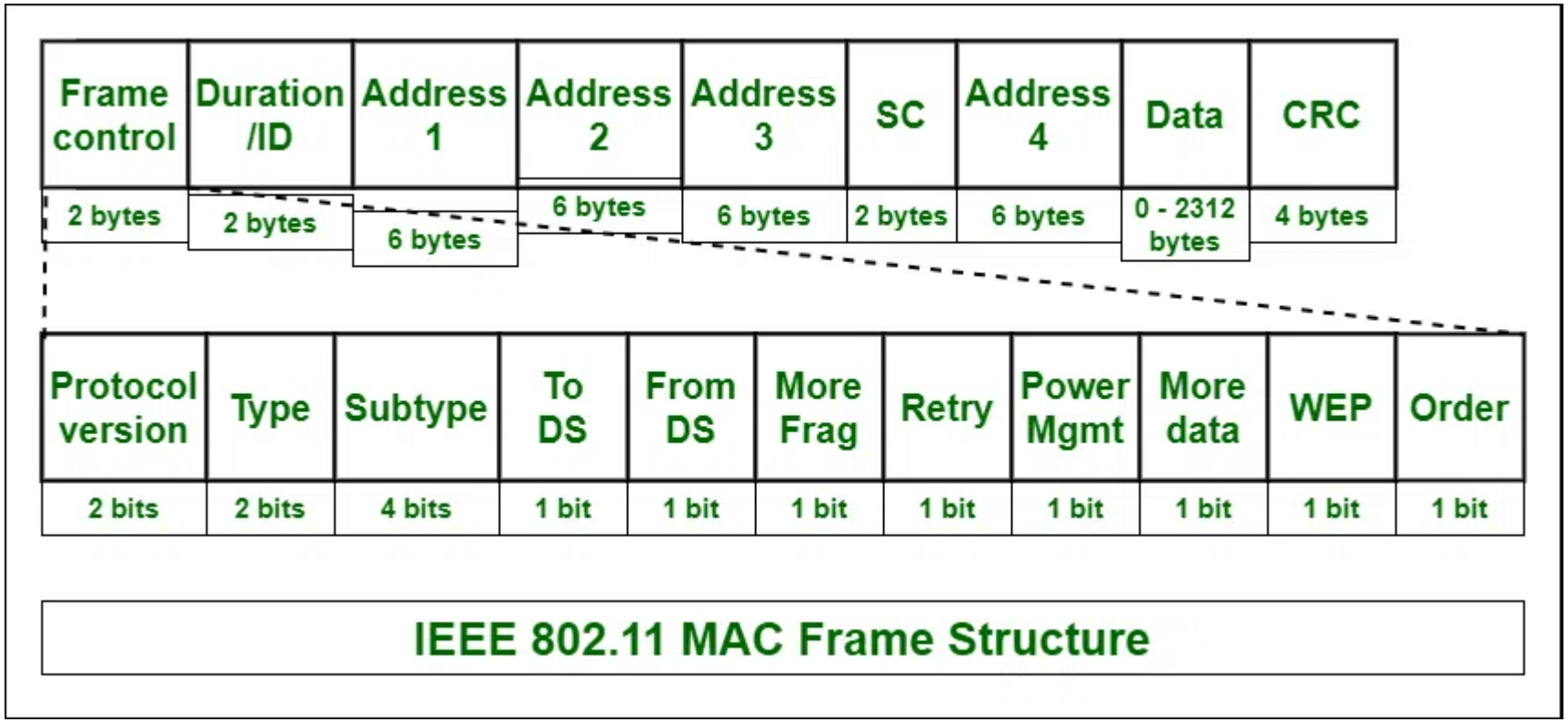




IOT Data link layer and Network layer protocols

IEEE 802.11

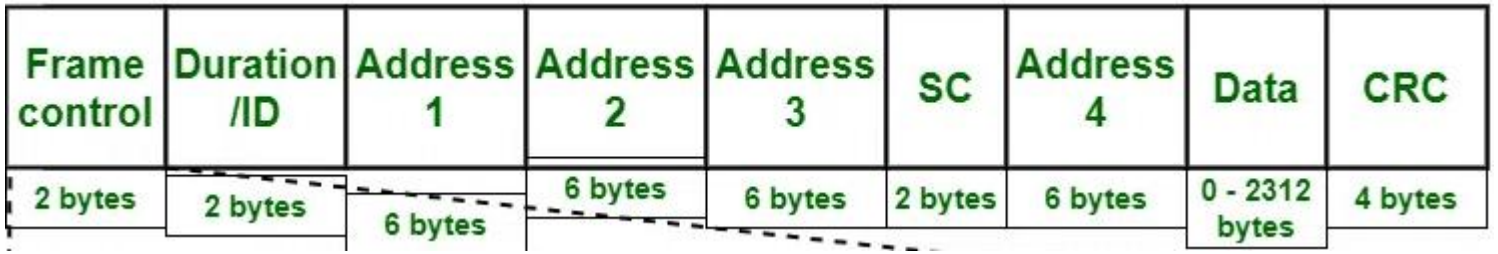
- Frame Format of IEEE 802.11



IOT Data link layer and Network layer protocols

IEEE 802.11

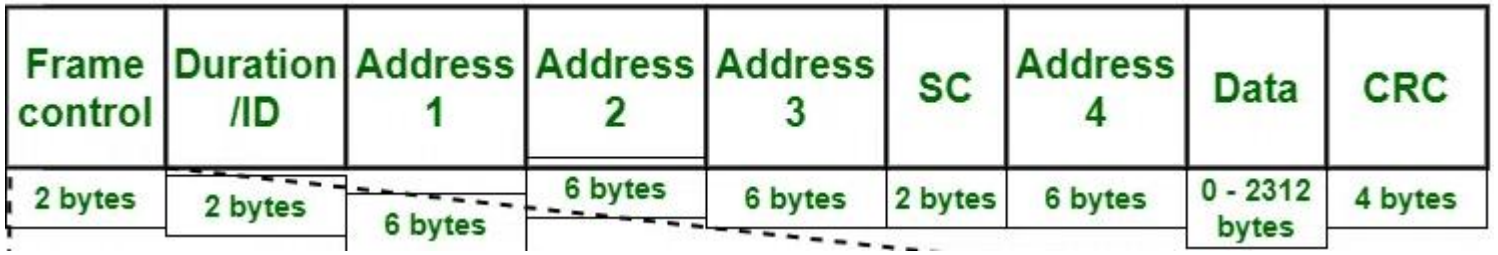
- Frame Format of IEEE 802.11



- The main fields of a frame of wireless LANs as laid down by IEEE 802.11 are
- Frame Control –
  - It is a 2 bytes starting field composed of 11 subfields.
  - It contains control information of the frame.
- Duration –
  - It is a 2-byte field that specifies the time period for which the frame and its acknowledgment occupy the channel.

IOT Data link layer and Network layer protocols

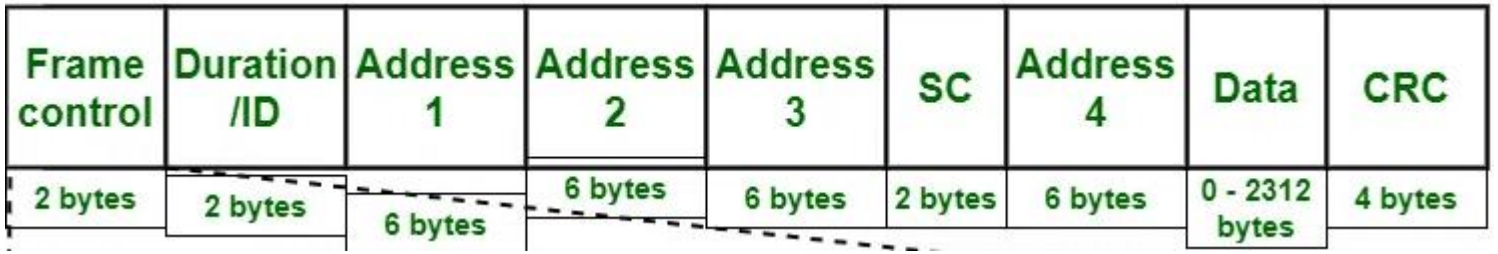
IEEE 802.11



- Frame Format of IEEE 802.11
  - Address fields –
    - There are three 6-byte address fields containing addresses of source, immediate destination, and final endpoint respectively.
  - Sequence –
    - It a 2 bytes field that stores the frame numbers.

IOT Data link layer and Network layer protocols

IEEE 802.11



- Frame Format of IEEE 802.11
  - Data –
    - This is a variable-sized field that carries the data from the upper layers.
    - The maximum size of the data field is 2312 bytes.
  - Check Sequence –
    - It is a 4-byte field containing error detection information.

IOT Data link layer and Network layer protocols

IEEE 802.11

- Frame Format of IEEE 802.11

Protocol version	Type	Subtype	To DS	From DS	More Frag	Retry	Power Mgmt	More data	WEP	Order
2 bits	2 bits	4 bits	1 bit	1 bit	1 bit	1 bit	1 bit	1 bit	1 bit	1 bit

Frame Control(FC) –

It is 2 bytes long field which defines type of frame and some control information. Various fields presentation FC are:

1. Version: It is a 2 bit long field which indicates the current protocol version which is fixed to be 0 for now.

2. Type: It is a 2 bit long field which determines the function of frame i.e management (00), control (01) or data (10). The value 11 is reserved.

IOT Data link layer and Network layer protocols

IEEE 802.11

- Frame Format of IEEE 802.11

Protocol version	Type	Subtype	To DS	From DS	More Frag	Retry	Power Mgmt	More data	WEP	Order
2 bits	2 bits	4 bits	1 bit	1 bit	1 bit	1 bit	1 bit	1 bit	1 bit	1 bit

Frame Control(FC) –

3. Subtype: It is a 4 bit long field which indicates sub-type of the frame like 0000 for association request, 1000 for beacon.

4. To DS: It is a 1 bit long field which when set indicates that destination frame is for DS(distribution system).

5. From DS: It is a 1 bit long field which when set indicates frame coming from DS.

IOT Data link layer and Network layer protocols

IEEE 802.11

- Frame Format of IEEE 802.11

Protocol version	Type	Subtype	To DS	From DS	More Frag	Retry	Power Mgmt	More data	WEP	Order
2 bits	2 bits	4 bits	1 bit	1 bit	1 bit	1 bit	1 bit	1 bit	1 bit	1 bit

Frame Control(FC) –

6. More frag (More fragments): It is 1 bit long field which when set to 1 means frame is followed by other fragments.

7. Retry: It is 1-bit long field, if the current frame is a retransmission of an earlier frame, this bit is set to 1.

IOT Data link layer and Network layer protocols

IEEE 802.11

- Frame Format of IEEE 802.11

Protocol version	Type	Subtype	To DS	From DS	More Frag	Retry	Power Mgmt	More data	WEP	Order
2 bits	2 bits	4 bits	1 bit	1 bit	1 bit	1 bit	1 bit	1 bit	1 bit	1 bit

Frame Control(FC) –

8. Power Mgmt (Power management): It is 1-bit long field that indicates the mode of a station after successful transmission of a frame. Set to 1 the field indicates that the station goes into power-save mode. If the field is set to 0, the station stays active.

IOT Data link layer and Network layer protocols

IEEE 802.11

- Frame Format of IEEE 802.11

Protocol version	Type	Subtype	To DS	From DS	More Frag	Retry	Power Mgmt	More data	WEP	Order
2 bits	2 bits	4 bits	1 bit	1 bit	1 bit	1 bit	1 bit	1 bit	1 bit	1 bit

Frame Control(FC) –

9. More data: It is 1-bit long field that is used to indicate receiver that a sender has more data to send than the current frame.

This can be used by an access point to indicate to a station in power-save mode that more packets are buffered or it can be used by a station to indicate to an access point after being polled that more polling is necessary as the station has more data ready to transmit.

IOT Data link layer and Network layer protocols

IEEE 802.11

- Frame Format of IEEE 802.11

Protocol version	Type	Subtype	To DS	From DS	More Frag	Retry	Power Mgmt	More data	WEP	Order
2 bits	2 bits	4 bits	1 bit	1 bit	1 bit	1 bit	1 bit	1 bit	1 bit	1 bit

Frame Control(FC) –

10. WEP: It is 1 bit long field which indicates that the standard security mechanism of 802.11 is applied.

11. Order: It is 1 bit long field, if this bit is set to 1 the received frames must be processed in strict order.

IOT Data link layer and Network layer protocols

IEEE 802.11



ESP-01 ESP8266 Serial WiFi Transceiver Module

IOT Data link layer and Network layer protocols

IEEE 802.15

What is



wireless specialty networks (WSNs)-

WSNs include

- wireless personal area networks (WPANs),
- Bluetooth,
- mesh networks, and
- body area networks.

IOT Data link layer and Network layer protocols

IEEE 802.15

- IEEE 802.15.1: WPAN / Bluetooth
- IEEE 802.15.2: Coexistence
- IEEE 802.15.3: High Rate WPAN
- IEEE 802.15.4: Low Rate WPAN
- IEEE 802.15.5: Mesh Networking
- IEEE 802.15.6: Body Area Networks
- IEEE 802.15.7: Visible Light Communication
- IEEE P802.15.8: Peer Aware Communications
- IEEE P802.15.9: Key Management Protocol
- IEEE P802.15.10: Layer 2 Routing
- IEEE 802.15.13: Multi-Gigabit/s Optical Wireless Communications

IOT Data link layer and Network layer protocols

IEEE 802.15

IEEE 802.15.1



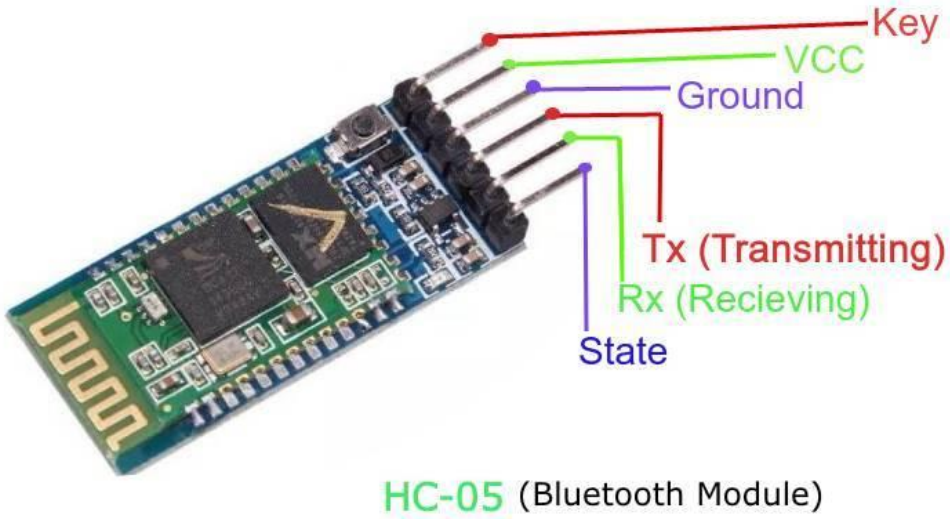
- **IEEE definition of WPAN**

- Wireless personal area networks (WPANs) are used to convey information over short distances among a private, intimate group of participant devices.

IOT Data link layer and Network layer protocols

IEEE 802.15

IEEE 802.15.1



HC-05 (Bluetooth Module)

- **WPAN**
- Unlike a wireless local area network (WLAN), a connection made through a WPAN involves little or no infrastructure or direct connectivity to the world outside the link.
- This allows small, power-efficient, inexpensive solutions to be implemented for a wide range of devices.

IOT Data link layer and Network layer protocols

IEEE 802.15



- **IEEE 802.15.1: WPAN / Bluetooth**

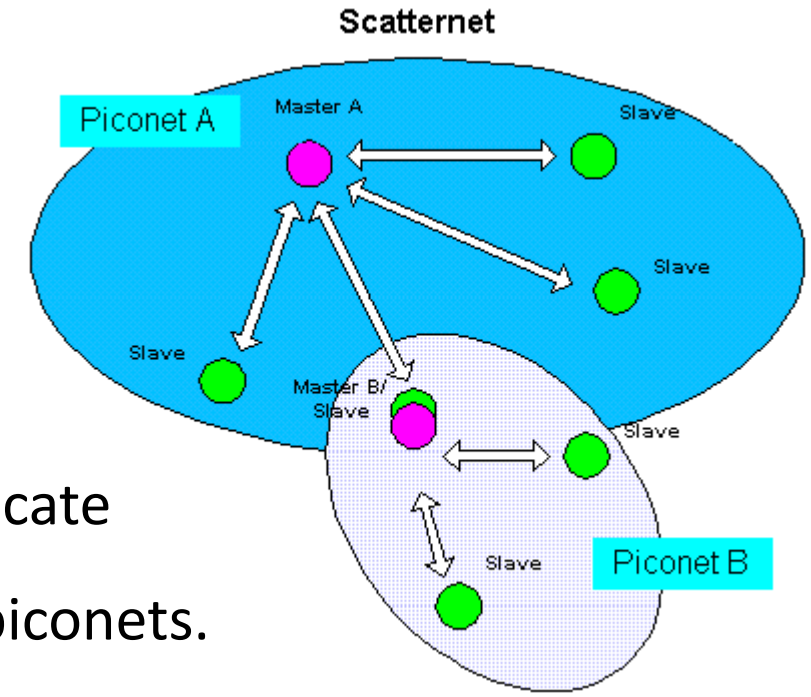
- A widely used WPAN technology is known as Bluetooth (version 1.2 or version 2.0)
- The IEEE 802.15.1 standard specifies the architecture and operation of Bluetooth devices, but only as far as physical layer and media access control (MAC) layer operation is concerned (the core system architecture).
- Higher protocol layers and applications defined in usage profiles are standardized by the Bluetooth SIG.

•In 1998, Ericsson, IBM, Intel, Nokia and Toshiba formed the "Bluetooth Special Interest Group" to establish standards for short-range radio.

IOT Data link layer and Network layer protocols

IEEE 802.15

- **IEEE 802.15.1: WPAN / Bluetooth**



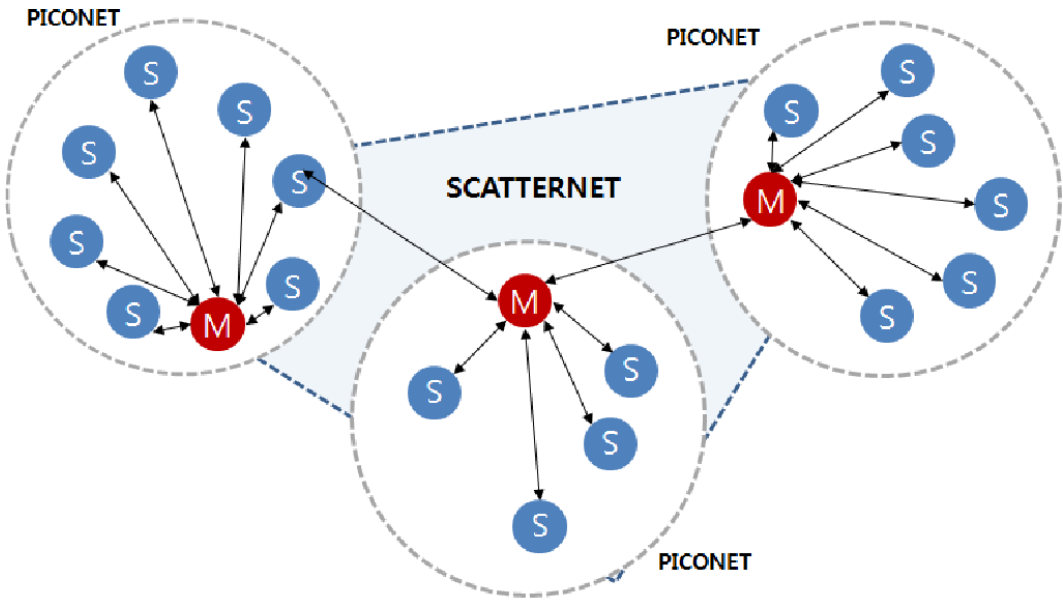
- Piconets
- Bluetooth enabled electronic devices connect and communicate wirelessly through short-range, ad hoc networks known as piconets.
- Piconets are established dynamically and automatically as Bluetooth enabled devices enter and leave radio proximity.

Up to 8 devices in one piconet (1 master and 7 slave devices). Max range 10 m.

IOT Data link layer and Network layer protocols

IEEE 802.15

- **IEEE 802.15.1: WPAN / Bluetooth**



- Scatternet
- It is a network which connects multiple piconets using Bluetooth and it acts as a master and another type of piconet acts as a slave.
- It has more than 6 devices that can be connected.
- Multiple devices are active, so there is an effective use of channel bandwidth.
- It is a connection of multiple piconets therefore it is applicable for devices belonging to large areas.

IOT Data link layer and Network layer protocols

IEEE 802.15

- **IEEE 802.15.1: WPAN / Bluetooth**

Specifications

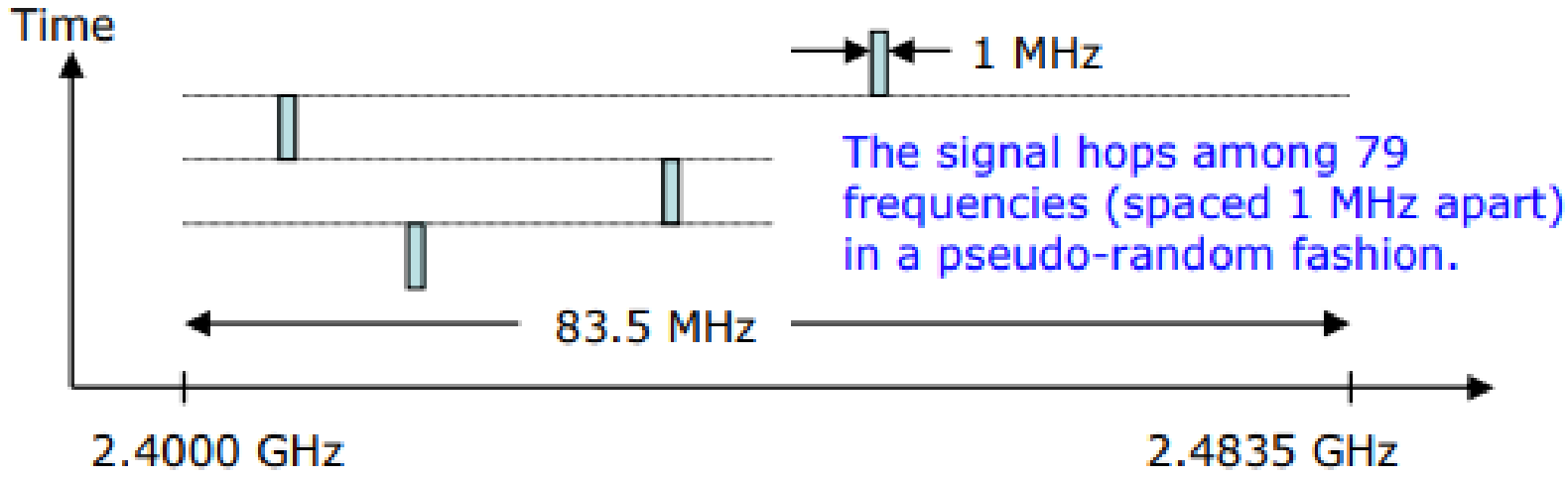
Topology	Up to 7 simultaneous links
Modulation	Gaussian filtered FSK
RF bandwidth	220 kHz (-3 dB), 1 MHz (-20 dB)
RF band	2.4 GHz ISM frequency band
RF carriers	79 (23 as reduced option)
Carrier spacing	1 MHz
Access method	FHSS-TDD-TDMA
Freq. hop rate	1600 hops/s

IOT Data link layer and Network layer protocols

IEEE 802.15

- **IEEE 802.15.1: WPAN / Bluetooth**

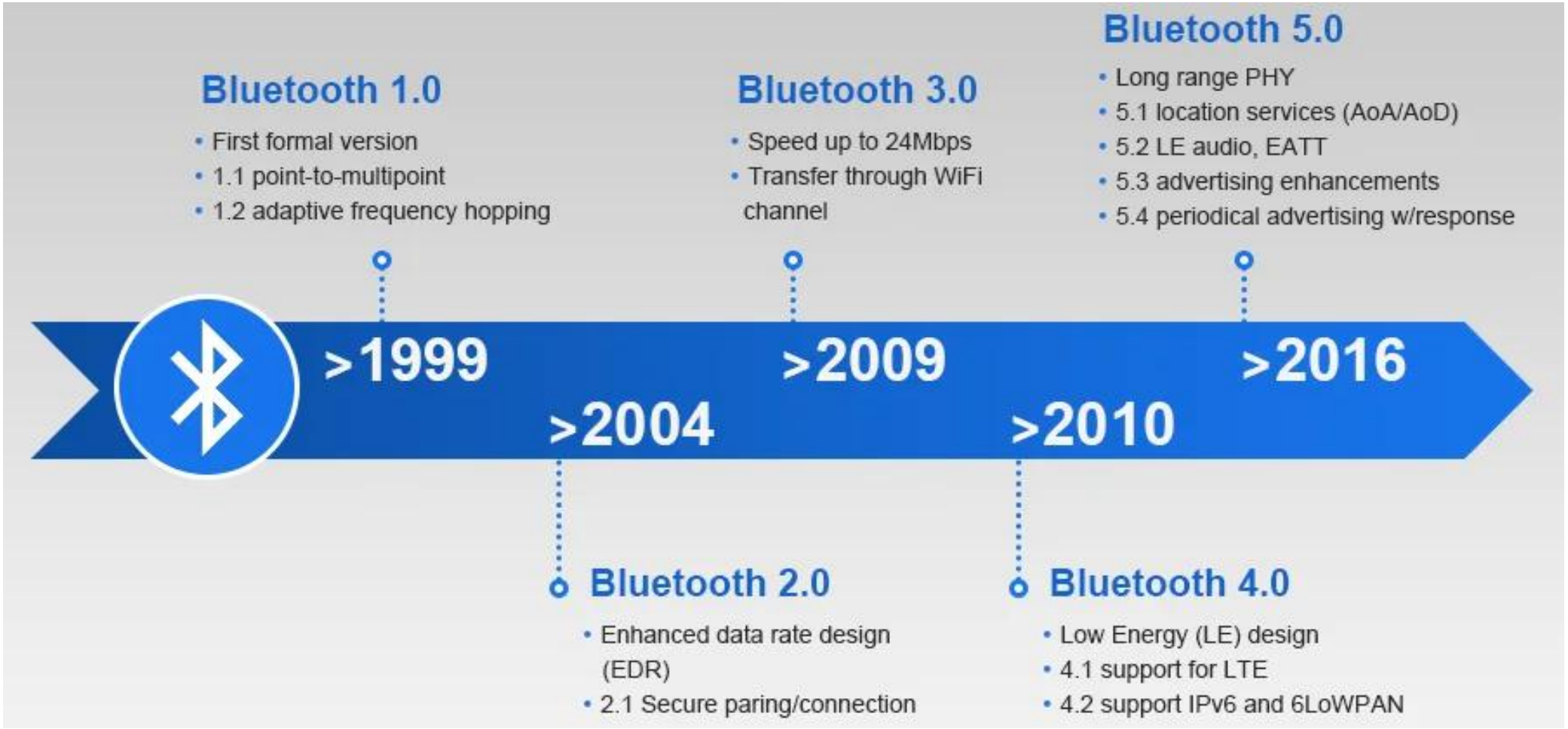
Bluetooth technology operates in the 2.4 GHz ISM band, using a spread spectrum, frequency hopping, full-duplex signal at a nominal rate of 1600 hops/second.



IOT Data link layer and Network layer protocols

IEEE 802.15

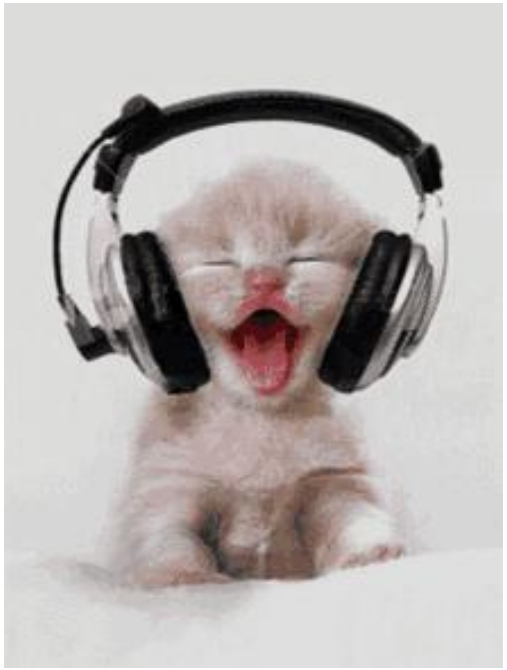
- **IEEE 802.15.1: WPAN / Bluetooth**



IOT Data link layer and Network layer protocols

IEEE 802.15

- **IEEE 802.15.1: WPAN / Bluetooth**
  - **Types**
  - **Classic Bluetooth versions: Bluetooth 1.0**



- **Bluetooth low energy versions: Bluetooth 4.0 – 5.4**



IOT Data link layer and Network layer protocols

IEEE 802.15

- **IEEE 802.15.2: Coexistence**
  - IEEE 802.15.2 is a recommended practice that describes how wireless personal area networks (WPANs) and wireless local area networks (WLANs) can coexist
  - The standard's goal is to enable WPAN devices to operate alongside other wireless devices in unlicensed frequency bands

The standard was published in 2003, reaffirmed in 2009, and withdrawn in 2018

IEEE 802.15.3 — Overview

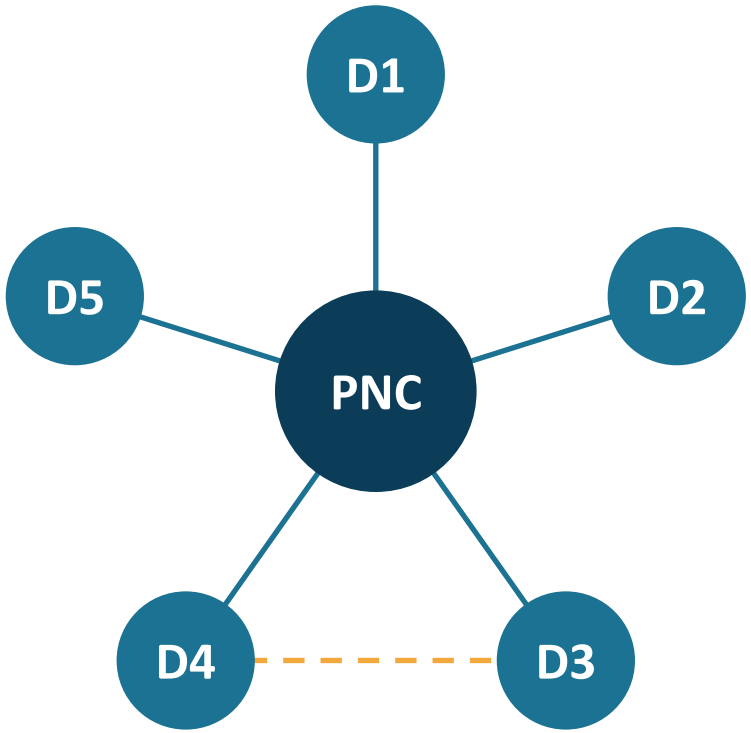
11–55	~10 m	2003	2.4 GHz
Mbps data rate	operating range	year ratified	ISM band

Why 802.15.3 exists

- Bluetooth (802.15.1) was too slow for multimedia; Wi-Fi (802.11) consumed too much power for handheld devices.
- 802.15.3 fills the gap: high throughput plus low power, aimed at streaming audio/video between nearby consumer devices.
- Forms self-organizing, ad hoc networks called piconets with no fixed infrastructure required.

802.15.3 Network Architecture — The Piconet

- A piconet is an independent, ad hoc network of at least two devices (DEVs) in radio proximity.
- One device becomes the Piconet Coordinator (PNC), providing synchronization, QoS, power management, and access control.
- The PNC allocates channel time; once allocated, DEVs can talk directly to each other, peer-to-peer.
- Any DEV can become the PNC — the role is negotiated, not fixed hardware.



Solid = PNC control links Dashed = direct peer-to-peer data link

802.15.3 Superframe Structure

- Channel time in a piconet is organized into repeating superframes controlled by the PNC.
- Beacon: carries timing and control information used to synchronize all DEVs.
- CAP (Contention Access Period): command and asynchronous data, accessed via CSMA/CA.
- CTAP (Channel Time Allocation Period): time-division slots (MCTAs / CTAs) guaranteed to specific DEVs — supports QoS.



802.15.3 Physical Layer

Spectrum & channels

- Operates in the unlicensed 2.4 GHz ISM band (2400–2483.5 MHz).
- Defines 5 channels, each 15 MHz wide; a piconet uses one channel at a time.
- Single physical layer shared by all devices in the piconet.

Modulation & data rates

QPSK	11 Mbps
DQPSK	22 Mbps
16-QAM	33 Mbps
32-QAM	44 Mbps
64-QAM	55 Mbps

Higher-order modulation trades range for throughput — 64-QAM delivers the fastest 55 Mbps link but needs the strongest signal.

802.15.3 MAC Layer Features

1 Quality of Service

Guaranteed Time Slots (GTS) in the CTAP reserve bandwidth for streaming audio/video.

2 Security

Optional AES-128 based encryption and authentication protects piconet data.

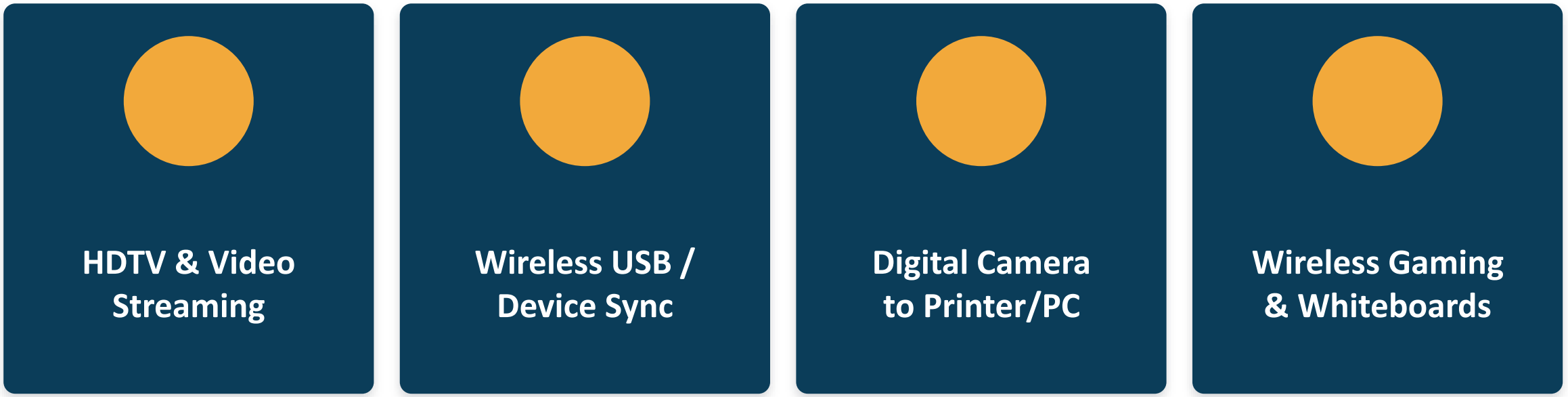
3 Power Management

DEVs use sleep and active modes negotiated with the PNC to save energy.

4 Ad Hoc Formation

Piconets form and dissolve dynamically with no fixed infrastructure.

802.15.3 Applications



Common thread: all require sustained high throughput over a short range — exactly what 802.15.3's TDMA-based CTAP guarantees.

IEEE 802.15.4 — Overview

20–250	10–100 m	2003	Years
kbps data rate	operating range	year ratified	battery lifetime

Design philosophy

- Optimized for low cost, low power, and low data rate — the opposite trade-off from 802.15.3.
- The MAC and PHY foundation for ZigBee, 6LoWPAN, Thread, and WirelessHART.
- Devices should run for months or years on small batteries, so the protocol favors idle/sleep time.

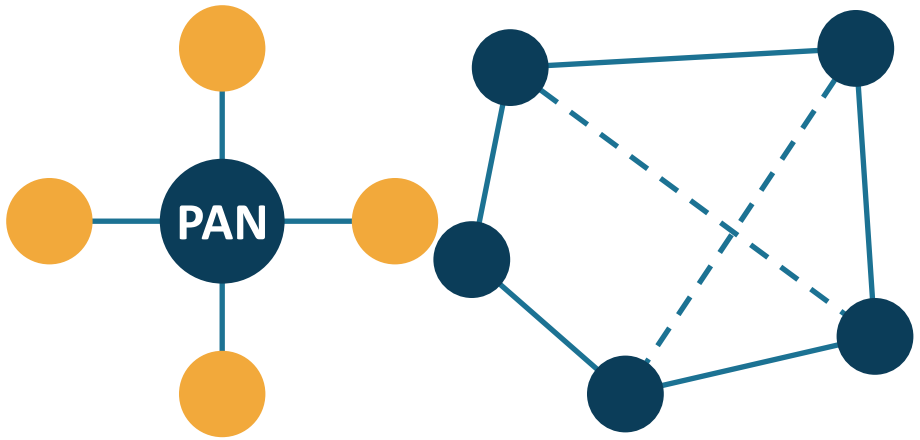
Device Types & Network Topologies

Device types

- FFD (Full Function Device) — can act as coordinator, router, or end device; full protocol.
- RFD (Reduced Function Device) — simple end device, e.g. a sensor; talks to one FFD only.

Topologies

- Star — all RFDs communicate through one central PAN coordinator.
- Peer-to-peer / Mesh — any FFD can talk to any FFD in range, enabling multi-hop routing.
- Cluster-tree — a hybrid, hierarchical combination of star clusters linked in a tree.



A PAN Coordinator (an FFD) is mandatory in every 802.15.4 network — it starts the network and assigns its PAN ID.

802.15.4 Physical Layer — Frequency Bands

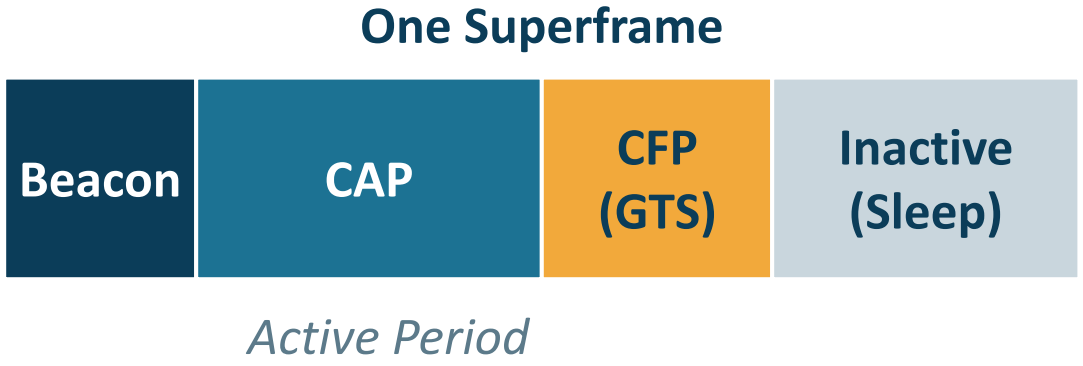
Band	Region	Channels	Rate	Modulation
868 MHz	Europe	1	20 kbps	BPSK
915 MHz	Americas	10	40 kbps	BPSK
2.4 GHz	Global (ISM)	16	250 kbps	O-QPSK

Key takeaway

- The 2.4 GHz band offers the most channels and the highest rate, so it is used worldwide (this is what ZigBee runs on).
- 868/915 MHz sub-GHz bands trade data rate for longer range and better wall penetration.

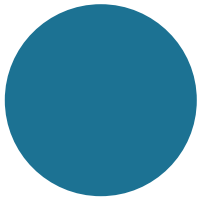
802.15.4 MAC Layer — Channel Access

- Beacon-enabled mode: coordinator sends periodic beacons; devices synchronize to an active/inactive superframe.
- Non-beacon mode: devices use unslotted CSMA/CA and transmit whenever the channel is free — simpler, but less deterministic.
- CFP (Contention-Free Period) offers Guaranteed Time Slots (GTS) for latency-sensitive devices.
- The Inactive Period lets the coordinator and devices sleep, which is the main source of multi-year battery life.

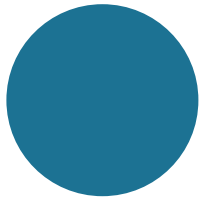


Superframe duration is set by two parameters: Beacon Order (BO) and Superframe Order (SO), both configured by the coordinator.

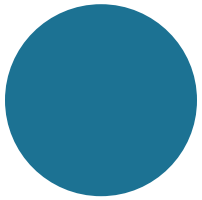
802.15.4 Applications



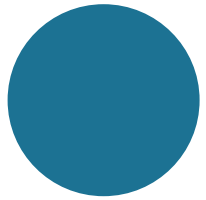
Home Automation (ZigBee)



Industrial Sensing (WirelessHART)



Smart Metering / Smart Grid



Healthcare & Wearable Monitors

Common thread: sparse, small data payloads sent infrequently by battery-powered devices — the exact profile 802.15.4 is optimized for.

COMPARATIVE ANALYSIS

802.15.3 vs 802.15.4 at a Glance

Parameter	IEEE 802.15.3	IEEE 802.15.4
Data rate	11 – 55 Mbps	20 – 250 kbps
Range	~10 m	10 – 100 m
Power use	Moderate	Ultra-low
Topology	Piconet (star + peer)	Star / Mesh / Cluster-tree
Target use	Multimedia streaming	Sensing & control (IoT)
Coordinator	PNC	PAN Coordinator (FFD)

The IEEE 802.15 Working Group

Wireless Personal Area Networks (WPANs): short-range, low-power, low-cost connectivity

S **Scope**	Standards for short-range wireless links among devices carried by, worn by, or near a person
C **Core**	802.15.4 defines the base PHY/MAC; task groups 5–13 extend it for new applications
F **Focus of this lecture**	Nine amendments and recommended practices: mesh, body networks, optics, coexistence and more

Standards Covered Today

15.5 | 15.6 | 15.7 | 15.8 | 15.9 | 15.10 | 15.11 | 15.12 | 15.13

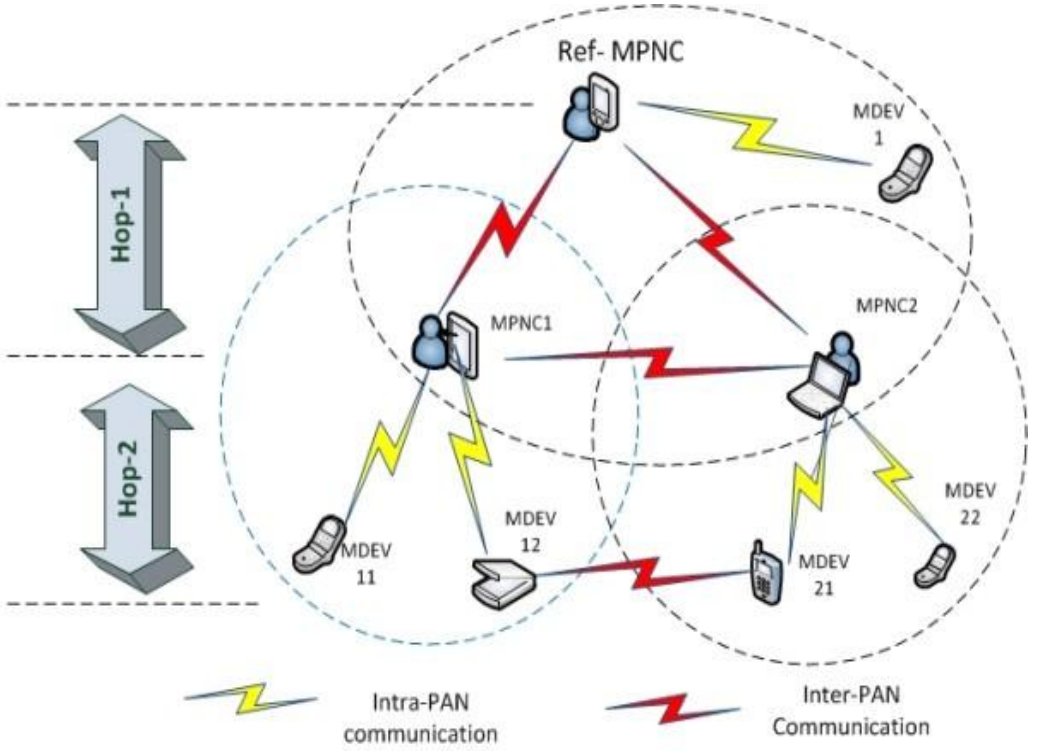
Each amendment solves a distinct connectivity problem beyond the base 802.15.4 standard.

IOT Data link layer and Network layer protocols

IEEE 802.15

- **IEEE 802.15.5: Mesh Networking**

- This standard is composed of two parts:
  - low-rate WPAN mesh and
  - high-rate WPAN mesh networks.
- The features include
- network initialization,
- addressing, and
- multi-hop unicasting.





802.15.5

Mesh Networking

“Let small networks talk over long distances.”

Approved 2009

Purpose

A recommended practice enabling mesh topology capability on top of 802.15.4 (low-rate) and 802.15.3 (high-rate) WPANs, extending effective network coverage.

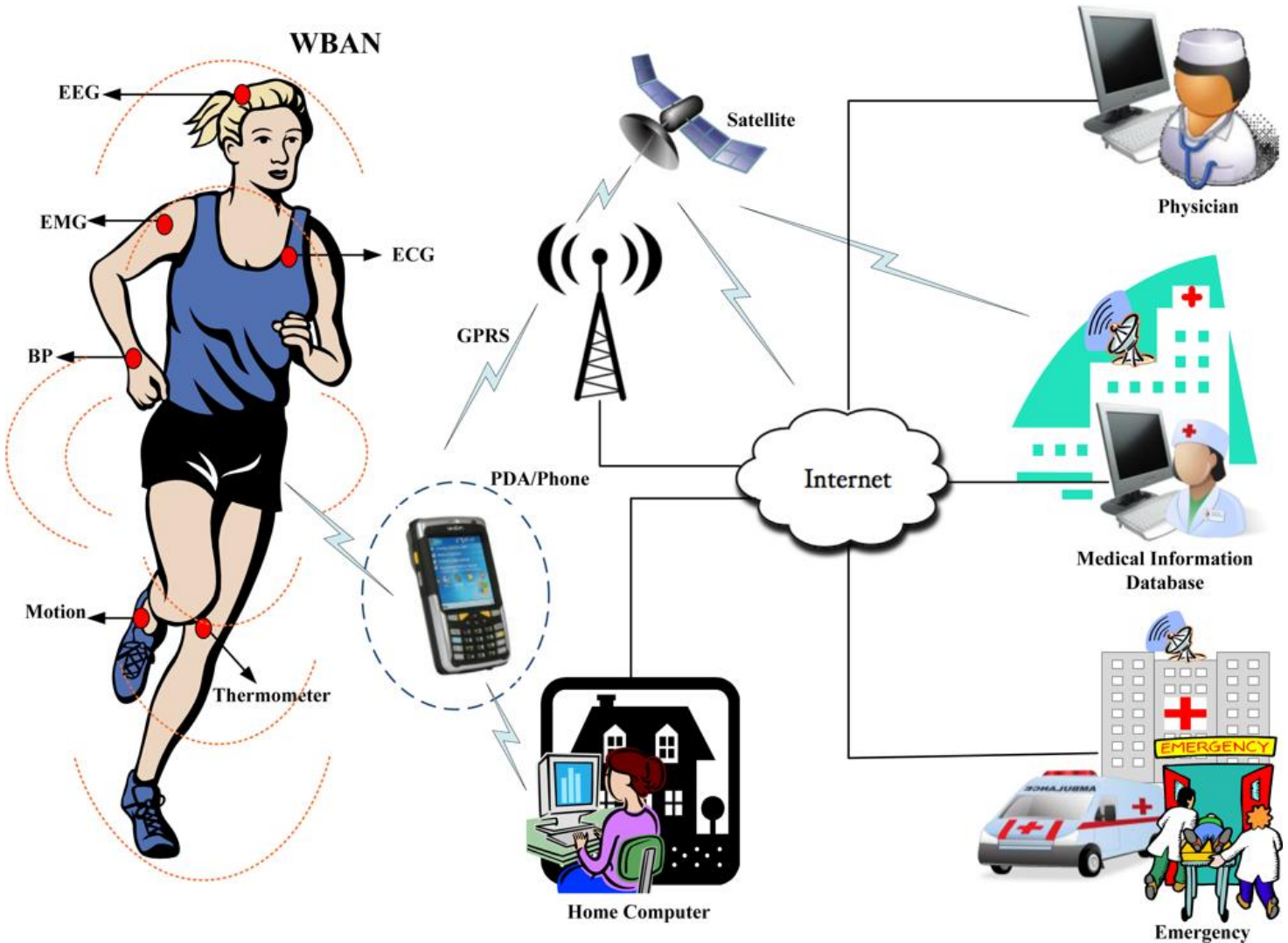
Base layer	Built above 802.15.4 / 802.15.3 MAC
Key feature	Multi-hop routing, self-healing topology
Benefit	Extends range without extra infrastructure
Typical use	Industrial monitoring, large sensor fields

Think of it as giving short-range WPAN radios the reach of a multi-hop mesh network.

IOT Data link layer and Network layer protocols

IEEE 802.15

- **IEEE 802.15.6: Body Area Networks**





802.15.6

Body Area Networks

“Connectivity in, on, and around the human body.”

Approved 2012

IEEE 802.15 WPAN Family

Purpose

Defines a low-power, short-range, highly reliable wireless standard for devices operating in, on, or around the human body, for medical and consumer use.

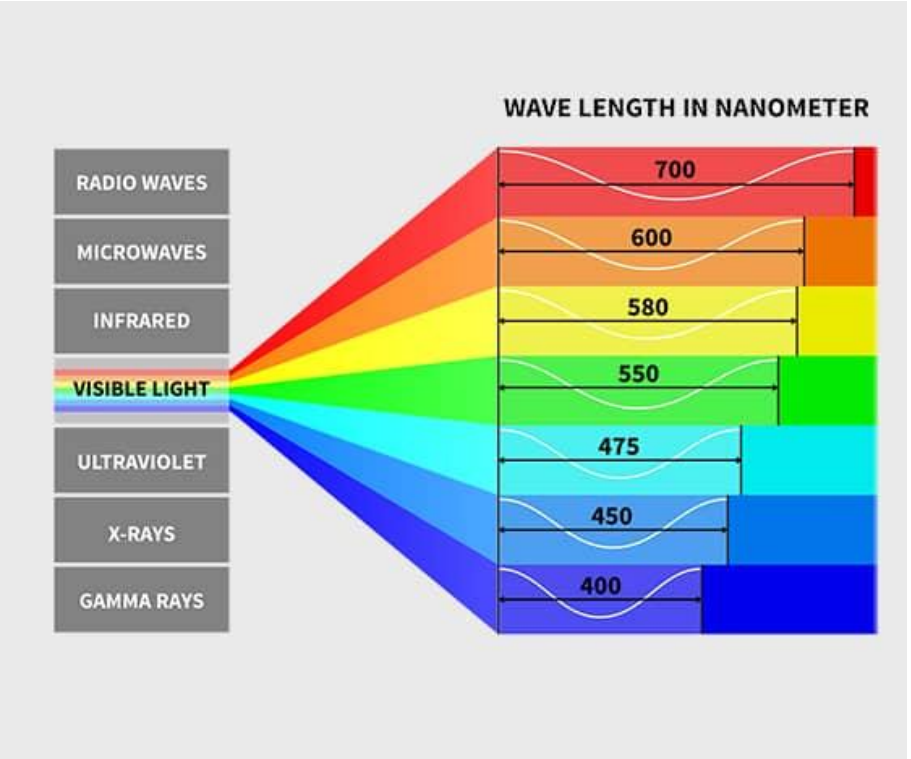
Frequency	NB, UWB, and Human Body Comm. (HBC) modes
Data rate	Up to ~10 Mbps depending on PHY
Priority	Extremely low power, high reliability, safety
Applications	ECG/EEG monitors, fitness bands, implants

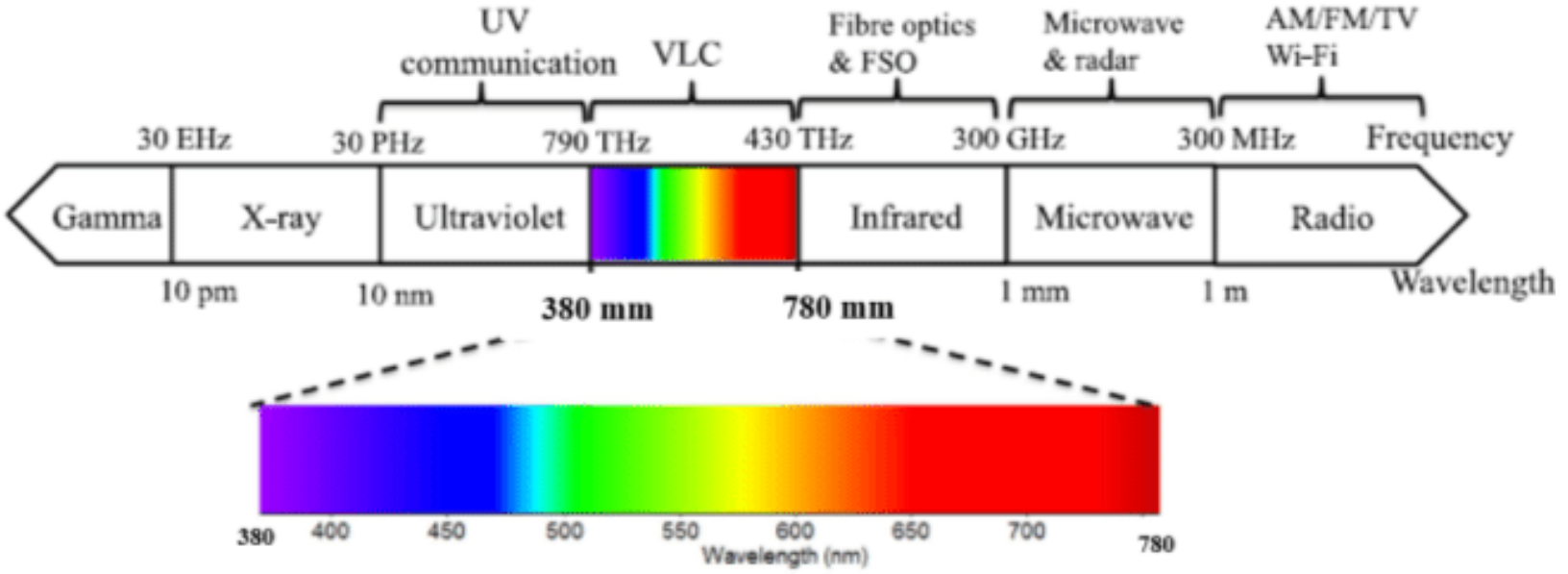
Strict SAR (radiation absorption) limits make 802.15.6 unique among WPAN standards.

IOT Data link layer and Network layer protocols

IEEE 802.15

- **IEEE 802.15.7: Visible Light Communication**

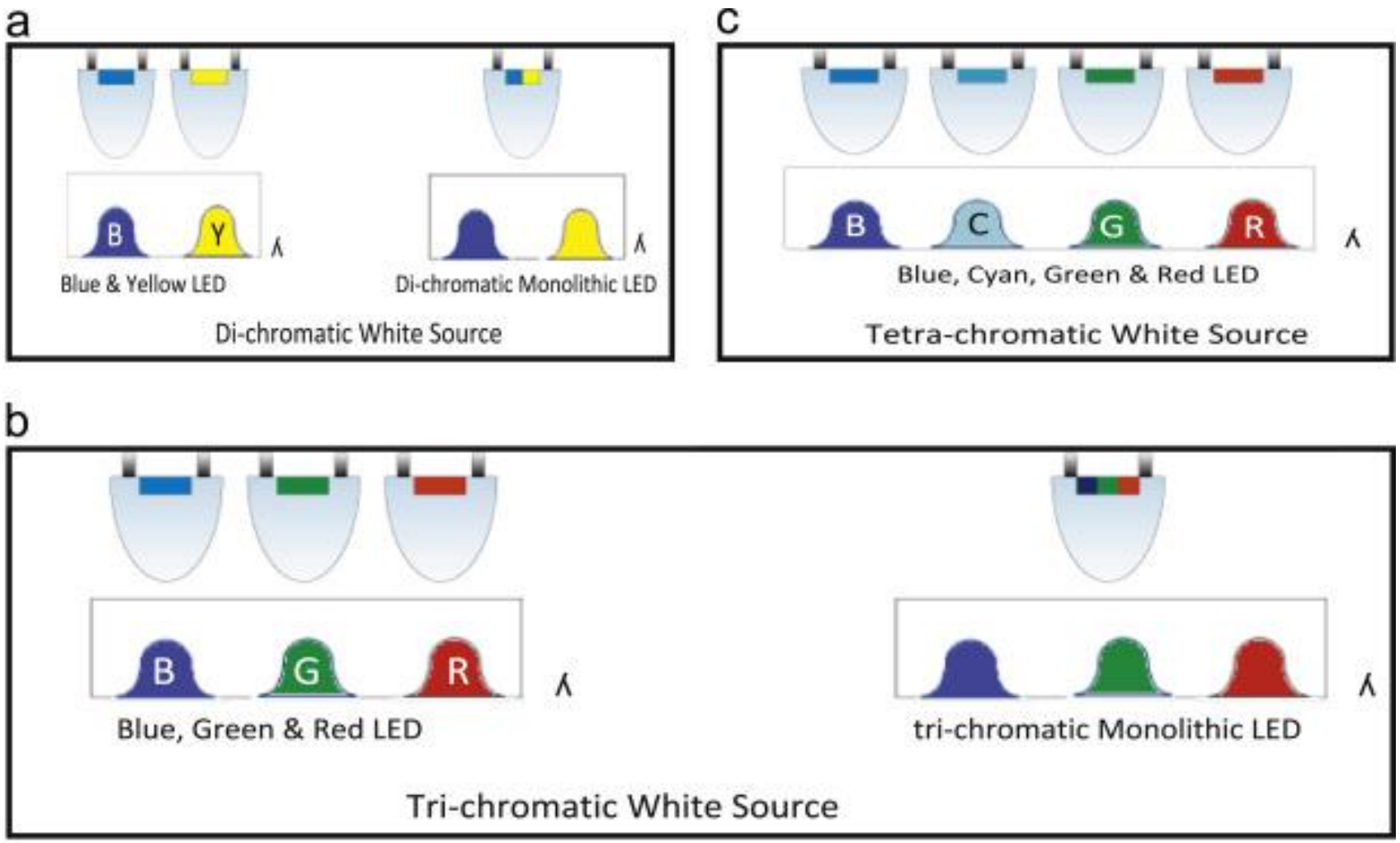




IOT Data link layer and Network layer protocols

IEEE 802.15

- **IEEE 802.15.7: Visible Light Communication**





802.15.7

Visible Light Communication

“Data travels on the light you already switched on.”

Approved 2011 (revised 2018)

IEEE 802.15 WPAN Family

Purpose

Specifies PHY and MAC layers for short-range optical wireless communication using visible light (LEDs) as the medium — combining illumination with data transfer.

Spectrum	Visible light, 400–800 THz
Data rate	Kbps up to hundreds of Mbps (2018 rev.)
Advantage	No RF interference; usable where RF is restricted
Applications	Indoor positioning, hospitals, aircraft cabins

Often the starting point for what is popularly called ‘Li-Fi.’



802.15.8

Peer Aware Communications

“Devices find and talk to each other — no network needed.”

Standard development (PAC)

IEEE 802.15 WPAN Family

Purpose

Defines infrastructure-less, peer-to-peer communication that lets a large number of autonomous devices discover and interact directly and dynamically.

Topology	Fully distributed, no access point required
Scale	Designed for dense, large device populations
Key feature	Autonomous peer discovery and session setup
Applications	Social proximity apps, IoT device discovery

PAC targets scenarios where relying on fixed infrastructure is impractical.



802.15.9

Key Management Protocol

“A common language for exchanging security keys.”

Approved 2016

IEEE 802.15 WPAN Family

Purpose

A recommended practice defining a transport layer for Key Management Protocol (KMP) datagrams over the 802.15.4 MAC, independent of the specific KMP used.

Role	Carries key-management messages, not a cipher itself
Flexibility	Works with multiple KMPs (e.g., 4-way handshake, HIP)
Layer	Sits directly above the 802.15.4 MAC layer
Benefit	Standardizes secure key exchange across vendors

802.15.9 is about how keys move, not which cryptographic algorithm is used.

802.15.10

Layer 2 Mesh Routing

“Efficient path-finding inside a mesh, at Layer 2.”

Approved 2017

IEEE 802.15 WPAN Family

Purpose

A recommended practice for routing at the MAC (Layer 2) level in an 802.15.4-based mesh network, so routing occurs before the network layer is involved.

Layer	Data-link layer (Layer 2) routing
Key feature	Route discovery, forwarding tables, mesh addressing
Benefit	Lower overhead than network-layer mesh routing
Complements	Works alongside 802.15.5 mesh capability

Distinguishes routing responsibility: 15.5 enables mesh capability, 15.10 defines how routes are found.

802.15.13

Multi-Gb/s Optical Wireless

“Optical links fast enough for cable-free desktops.”

Approved 2023

IEEE 802.15 WPAN Family

Purpose

Defines PHY and MAC for multi-gigabit-per-second optical wireless communication (OWC), targeting high-throughput indoor links using infrared/visible light.

Data rate	Multi-Gbps (up to ~10 Gbps class links)
Spectrum	Infrared and visible light bands
Advantage	Huge unlicensed optical bandwidth, no RF interference
Applications	Wireless docking, AR/VR, high-speed indoor links

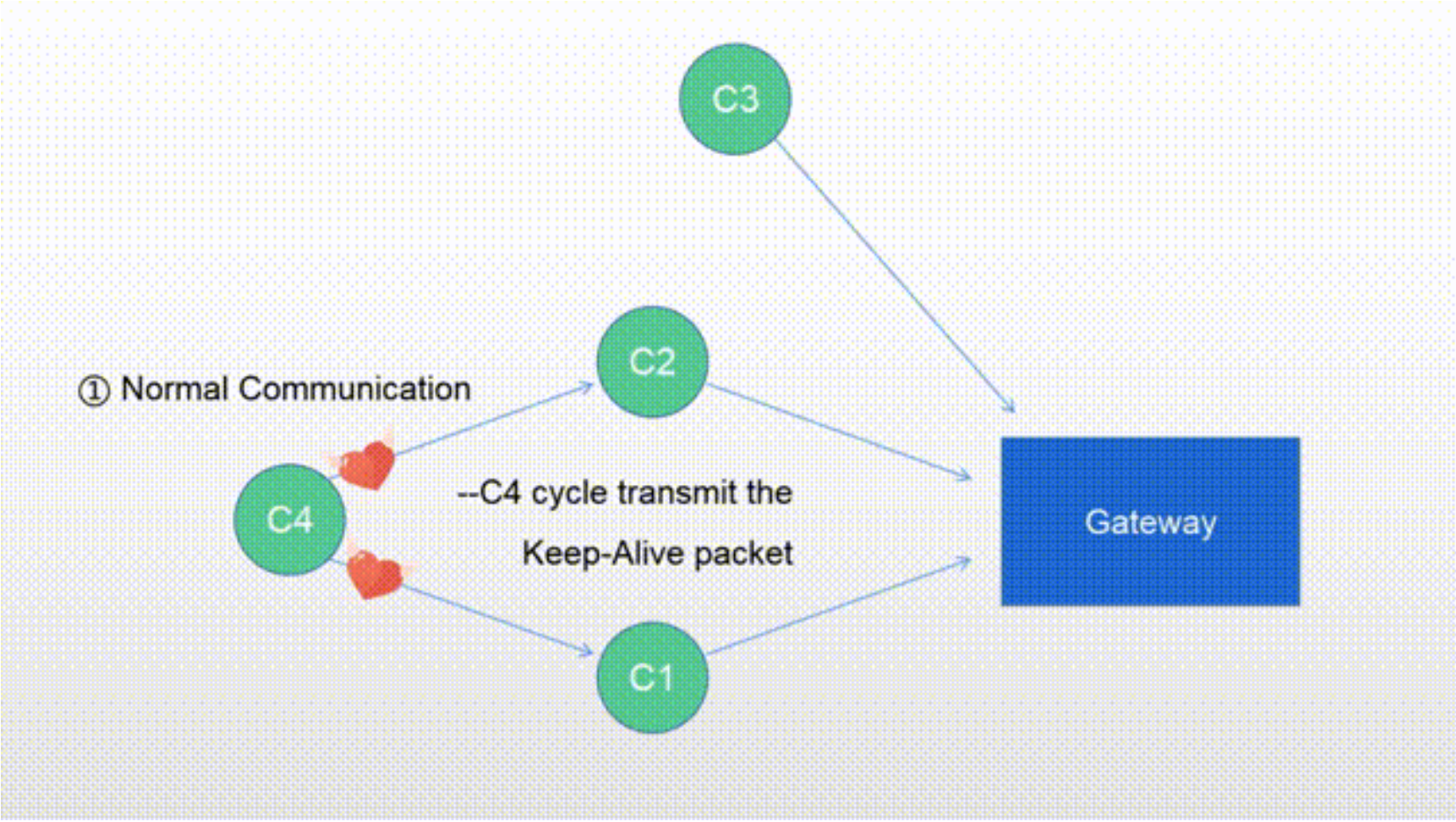
The successor vision to 802.15.7 — optics engineered specifically for gigabit-class throughput.

Standards at a Glance

Standard	Domain	Primary Contribution
802.15.5	Mesh	Mesh topology capability for 15.4 / 15.3
802.15.6	Body Area	Ultra-low-power on/in-body communication
802.15.7	Optical (VLC)	Visible-light data + illumination
802.15.8	Peer-to-Peer	Infrastructure-less peer discovery
802.15.9	Security	Transport for key management protocols
802.15.10	Mesh Routing	Layer-2 route discovery & forwarding
802.15.11	Coexistence	Fast session transfer between radios
802.15.12	Interface	Common upper-layer interface (ULI)
802.15.13	Optical (OWC)	Multi-Gbps optical wireless links

IOT Data link layer and Network layer protocols

Wireless HART





LECTURE OUTLINE

What We Will Cover

1. **From Wired HART to Wireless**
   How the protocol evolved for mesh radio

2. **Network Architecture**
   Gateway, Network Manager, field devices, handhelds

3. **The Protocol Stack**
   Physical, data-link, network & application layers

4. **Security & Reliability**
   Defense-in-depth encryption and self-healing mesh

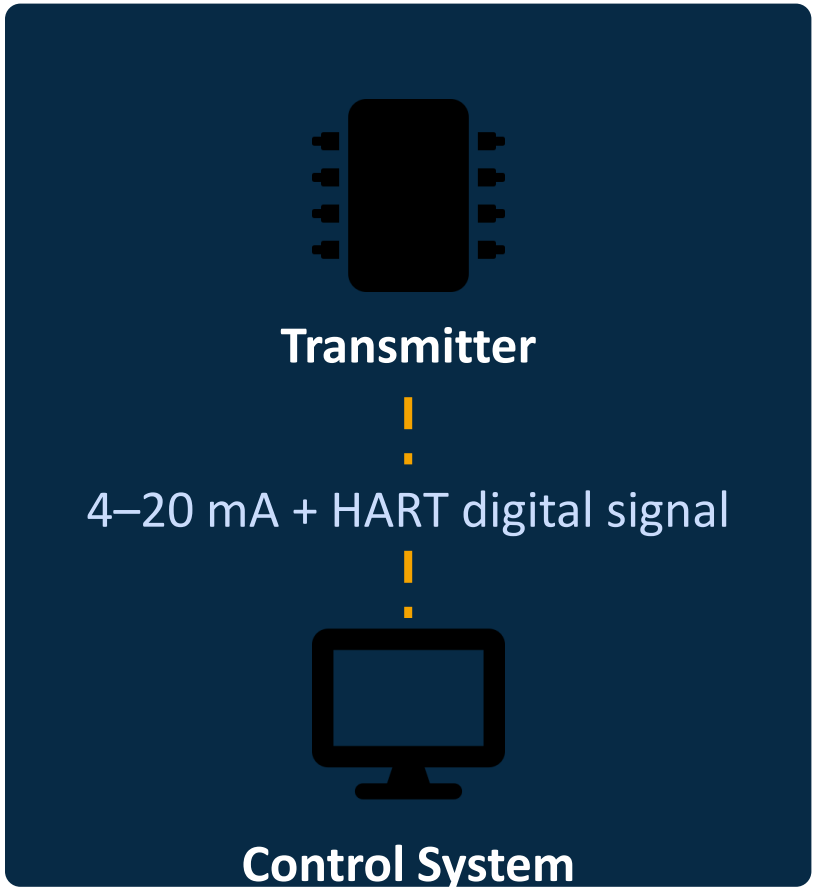
5. **Real-World Applications**
   Where industry deploys WirelessHART today

6. **Challenges Ahead**
   Open problems and practical limitations

Recap: The Wired HART Protocol

HART (Highway Addressable Remote Transducer) has connected process instruments since the 1980s.

- **Hybrid signal:** Digital data riding on the classic 4–20 mA analog loop
- **Two-way communication:** Read process values while writing configuration commands
- **Huge installed base:** Tens of millions of HART field devices worldwide
- **Limitation:** Every device still needs a dedicated pair of wires



DEFINITION

What Is WirelessHART?

The first open, interoperable wireless standard purpose-built for process measurement and control.



Standardized 2007
Published as a secure, self-organizing wireless mesh network technology



IEC 62591
Ratified as an international standard by the IEC

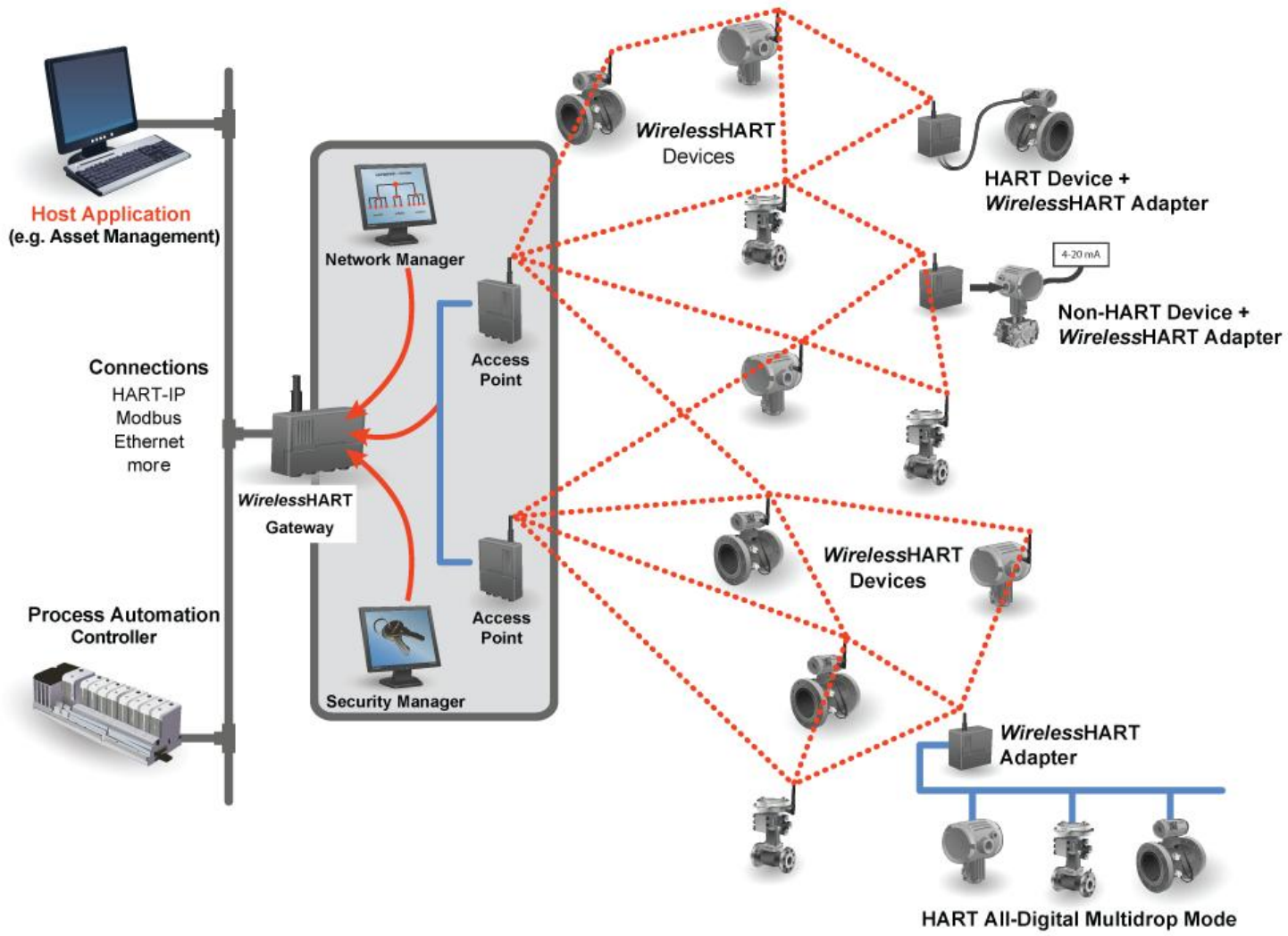


Built on 802.15.4
Uses the IEEE 802.15.4 radio at 2.4 GHz as its physical layer



Backward Compatible
Carries the same HART commands, data & device descriptions

Wireless HART- Architecture



*Wireless*HART Mesh Network

Wireless HART

Three WirelessHART types



WirelessHART gateway

access point radio

+

network manager

+

gateway

connects to the host system



WirelessHART instrument

radio

+

measurement

+

power

wireless measurement



WirelessHART adapter

radio

+

legacy device connectivity

connects wired instruments to WirelessHART

SYSTEM DESIGN

Network Architecture

WIRELESS MESH OF FIELD DEVICES



Field devices relay data hop-by-hop to the Gateway

Gateway
Bridges the wireless mesh to the host system

Network Manager
Builds routes & communication schedules

Security Manager
Generates and manages encryption keys

Handheld
Configures & commissions field devices on-site

ARCHITECTURE

The WirelessHART Protocol Stack

Application Layer	Familiar HART commands, data & device descriptions
Network Layer	Self-organizing, self-healing mesh routing & end-to-end security
Data-Link Layer	TDMA scheduling, channel hopping & channel blacklisting
Physical Layer	IEEE 802.15.4 radio, 2.4 GHz ISM band, O-QPSK



Why layered?

Each layer solves one problem — radio transmission, timing, routing, or meaning — so the whole stack stays reliable and secure.

LAYER 1

Physical Layer

2.4 GHz

Global ISM band — license-free worldwide

15 channels

IEEE 802.15.4 radio channels available for use

250 kbps

Over-the-air raw data rate

O-QPSK

Robust digital modulation scheme

+10 dBm

Typical maximum transmit power

Direct-Sequence

Spread-spectrum coding resists interference

LAYER 2

Data-Link Layer: TDMA & Channel Hopping



Each 10 ms time slot is reserved for one device-to-device transmission — no collisions

Fixed 10 ms slots

Time-division multiple access removes collisions on air

Channel hopping

Every transmission jumps to a new frequency channel

Channel blacklisting

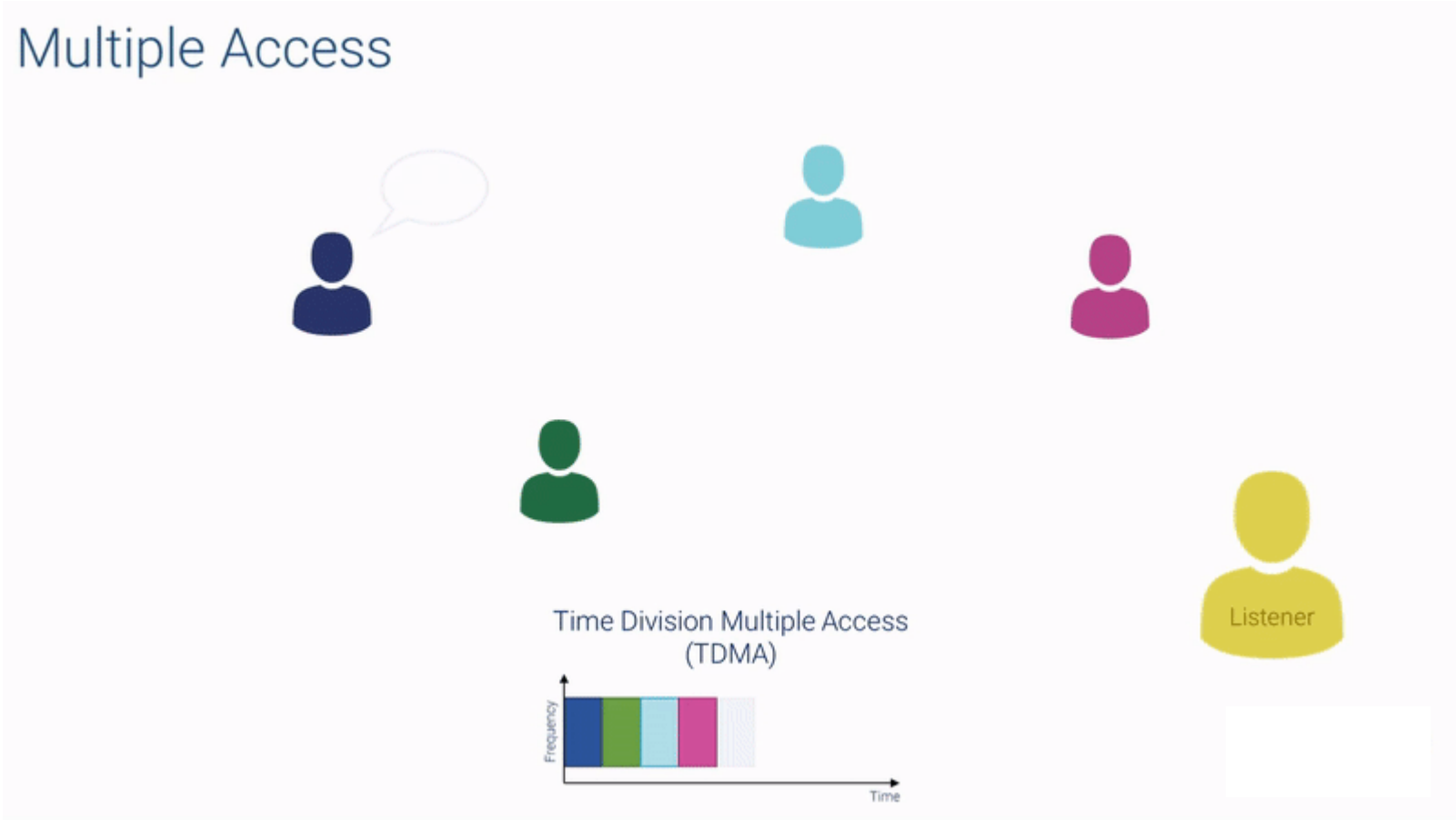
Noisy or interfered channels are avoided automatically

Deterministic delivery

Guarantees predictable, reliable message timing

Wireless HART

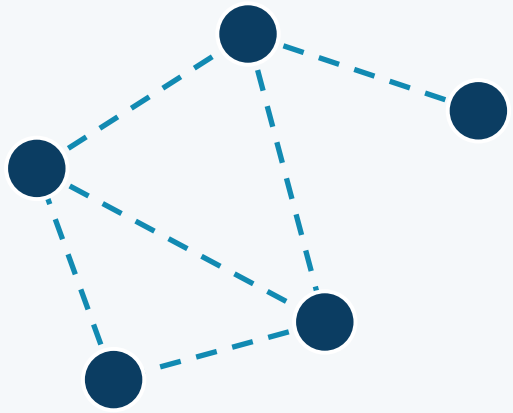
TDMA



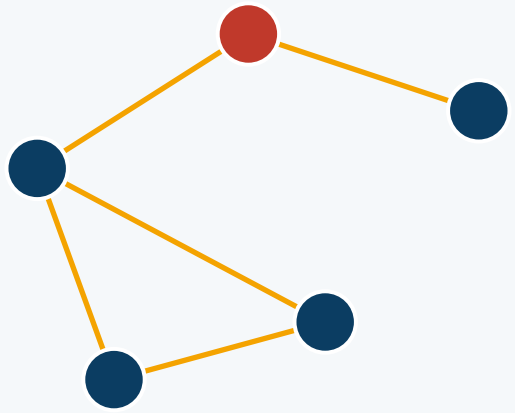
LAYER 3

Network Layer: A Self-Healing Mesh

Normal Operation



Link Fails → Route Reforms



- **Self-organizing:** New devices automatically join and build routes
- **Self-healing:** Traffic reroutes instantly around failed or blocked links
- **Path redundancy:** Multiple alternate paths keep data flowing reliably

PROTECTION

Defense-in-Depth Security



AES-128 Encryption

Every message is encrypted end-to-end across the network



Join-Key Authentication

Devices must prove identity before joining the mesh



Message Integrity Codes

Detects any tampering or corruption in transit



Rotating Session Keys

Fresh keys per session limit exposure from any single key

BENEFITS

Key Advantages

No new wiring

Eliminates cable trenching & conduit runs

Self-organizing

Devices join & route with no manual setup

Secure by design

Encryption protects plant data end-to-end

High reliability

Mesh redundancy tops 99% data reliability

Backward compatible

Reuses existing HART commands & data

Fast to deploy

New measurement points added in hours

IN PRACTICE

Applications Across Industry



Oil & Gas

Tank farms, wellheads & remote pipeline monitoring



Chemical Plants

Corrosion, vibration & tank-level monitoring



Power Generation

Boiler, turbine & emissions instrumentation



Water & Wastewater

Distributed monitoring across treatment facilities

OPEN ISSUES

Challenges & Limitations

Battery lifetime

Field devices run on batteries needing periodic replacement

Network latency

Mesh hopping adds delay versus a direct wired link

RF interference

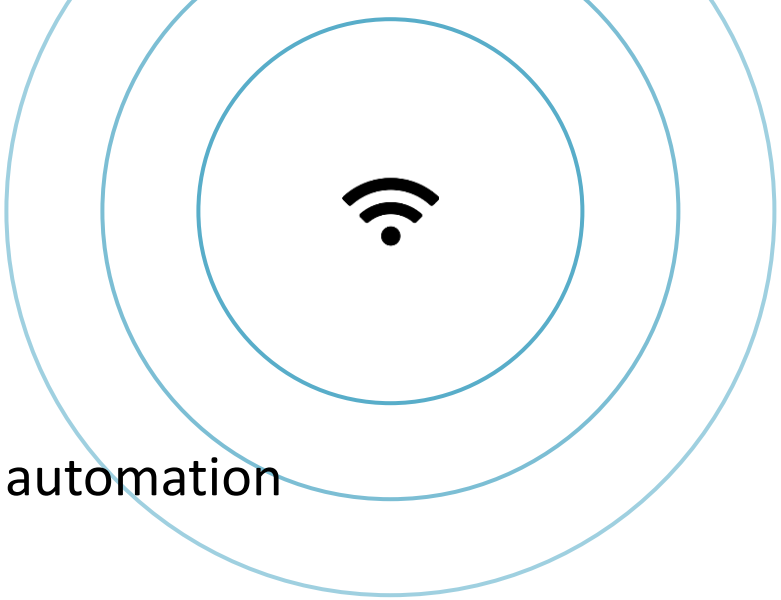
Dense plants with metal structures can weaken signals

Setup complexity

Initial commissioning & scheduling requires trained staff

SUMMARY

Conclusion



- An open, IEC 62591 standard purpose-built for wireless process automation
- Self-organizing, self-healing mesh with deterministic TDMA scheduling
- Defense-in-depth security keeps plant data encrypted end-to-end
- Fully backward compatible with the huge installed base of HART devices
- Reduces cost & time to instrument hard-to-wire locations

IOT Data link layer and Network layer protocols

Z-Wave

A Low-Power Wireless Protocol for the Connected Home



Agenda

01 What Is Z-Wave?

02 History & Evolution

03 How the Mesh Network Works

04 Technical Specifications

05 Protocol Architecture

06 Applications & Devices

07 Z-Wave vs. Zigbee vs. Wi-Fi

08 Security & the Road Ahead

What Is Z-Wave?

Z-Wave is a low-power, sub-1 GHz radio frequency (RF) protocol built specifically for home automation — letting sensors, switches, and controllers talk to each other reliably.

- Purpose-built wireless standard exclusively for smart-home / IoT control
- Devices form a self-healing mesh — no single point of failure
- Extremely low power draw; ideal for battery-run sensors

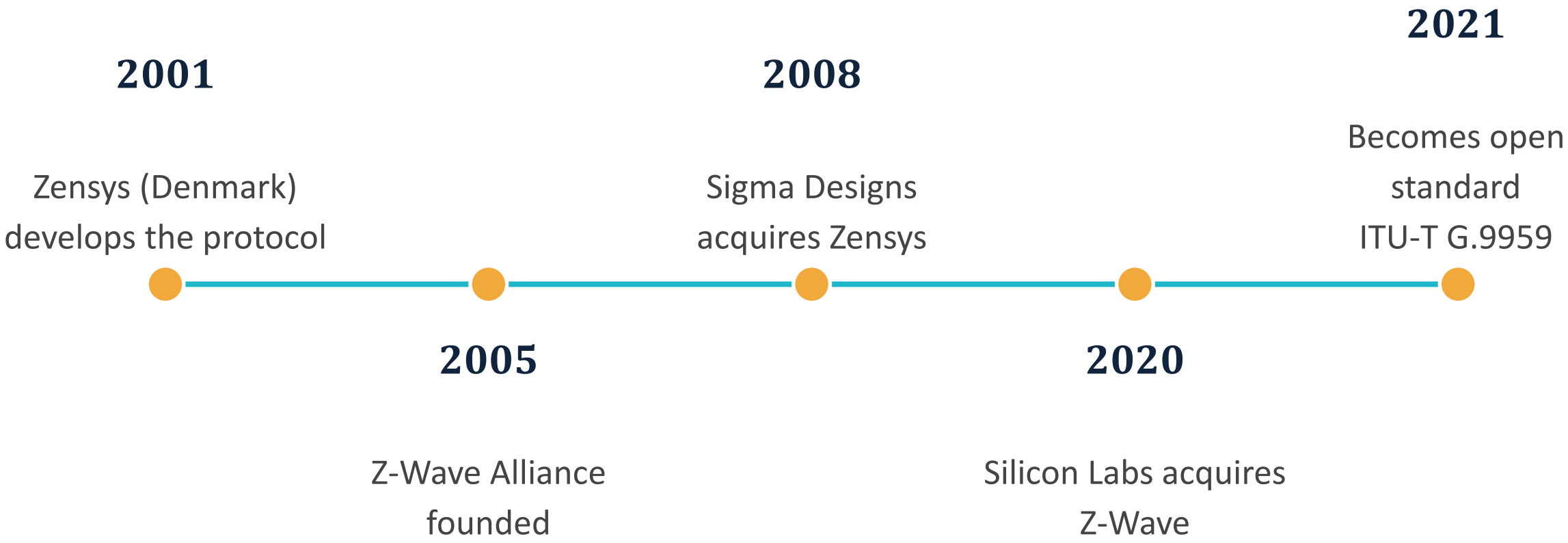


Governed by the Z-Wave Alliance — an open standard managed by hundreds of member companies.

Over 4,400 certified interoperable products worldwide, from over 700 manufacturers.

BACKGROUND

History & Evolution

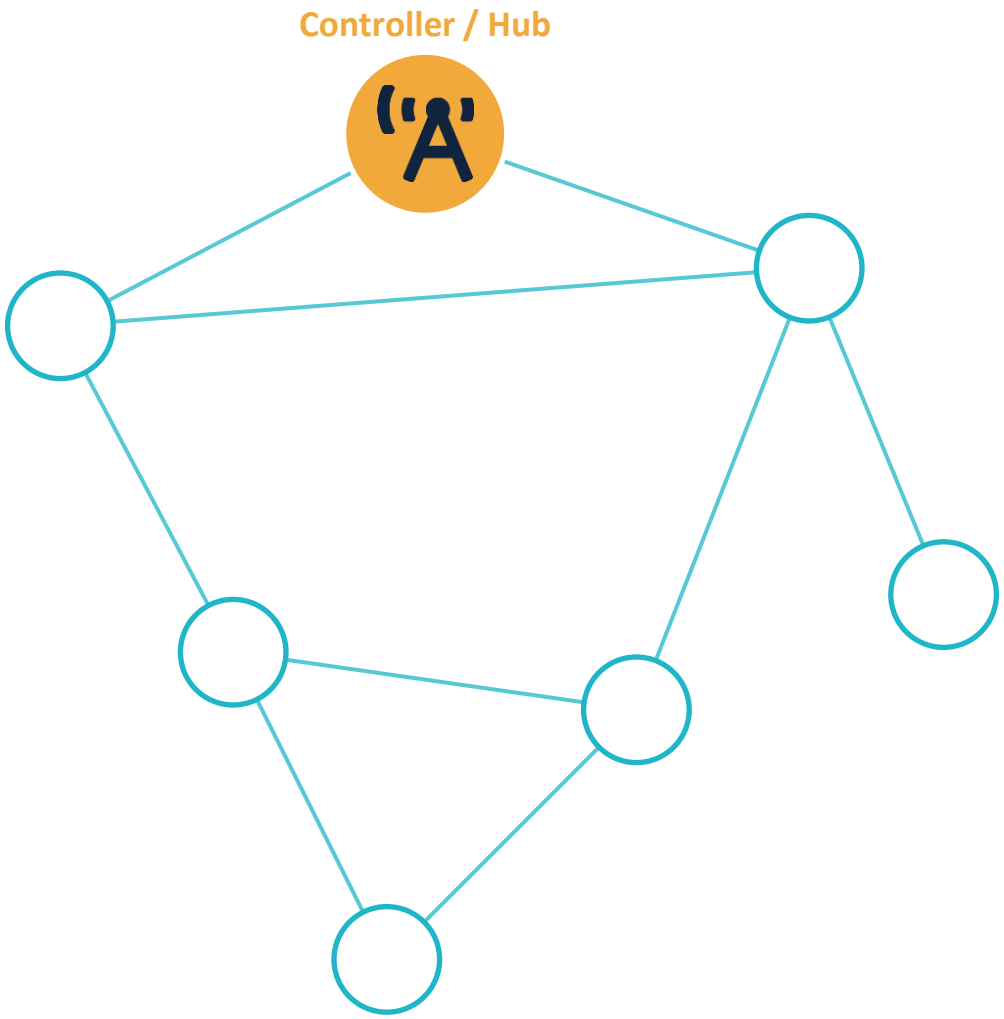


HOW IT WORKS

The Mesh Network

Every mains-powered Z-Wave device acts as a repeater. A signal hops node-to-node until it reaches its target — so the network gets stronger and more reliable as you add devices.

- Controller (hub) sends the command
- Message hops via nearby repeater nodes
- Up to 4 hops to reach the destination
- Network 'self-heals' if a path is blocked



UNDER THE HOOD

Technical Specifications

Frequency Band

868 / 908 / 920 MHz (region-dependent, sub-1 GHz)-865.2MHz in India

Range

~15–30 m per hop
Indoors

Data Rate

9.6 / 40 / 100 kbps

Network Size

Up to 232 nodes per network

Hop Limit

Up to 4 hops source to destination

Z-Wave Long Range

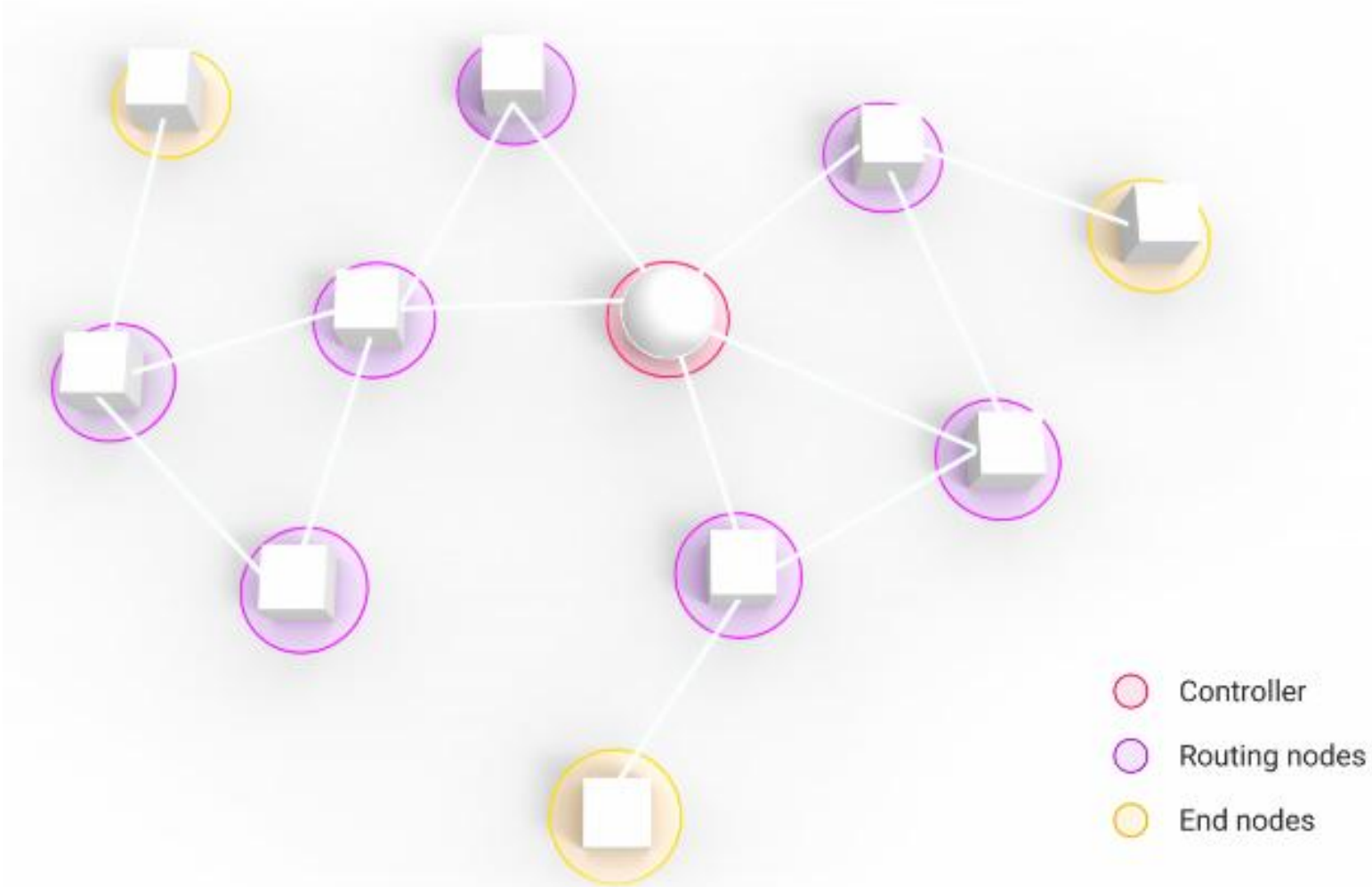
Up to 4,000 nodes, ~1.6 km line-of-sight

Protocol Architecture

Application Layer	Command classes — device behaviors (dimmer, lock, sensor…)
Network Layer	Routing, mesh path discovery, node addressing
Transport Layer	Frame delivery, acknowledgements, retries
MAC Layer	Channel access, collision avoidance
PHY Layer	Radio transmission — sub-1 GHz FSK/GFSK modulation

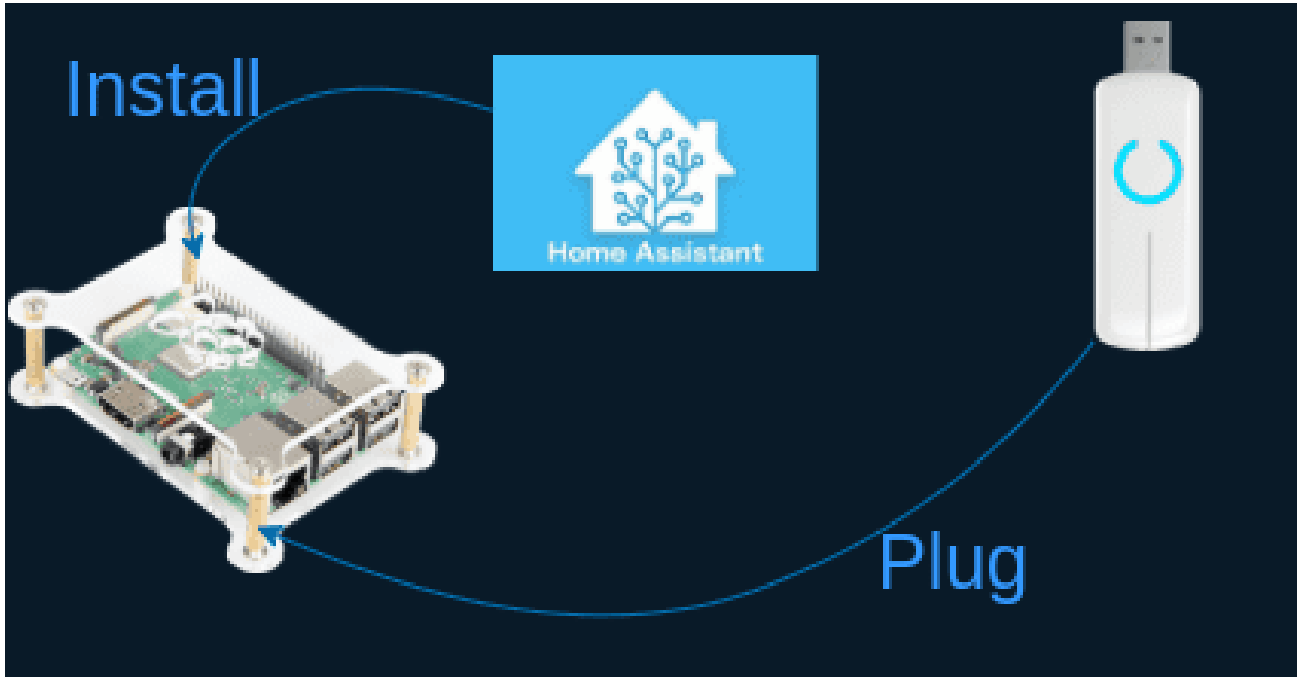
IOT Data link layer and Network layer protocols

Z-Wave



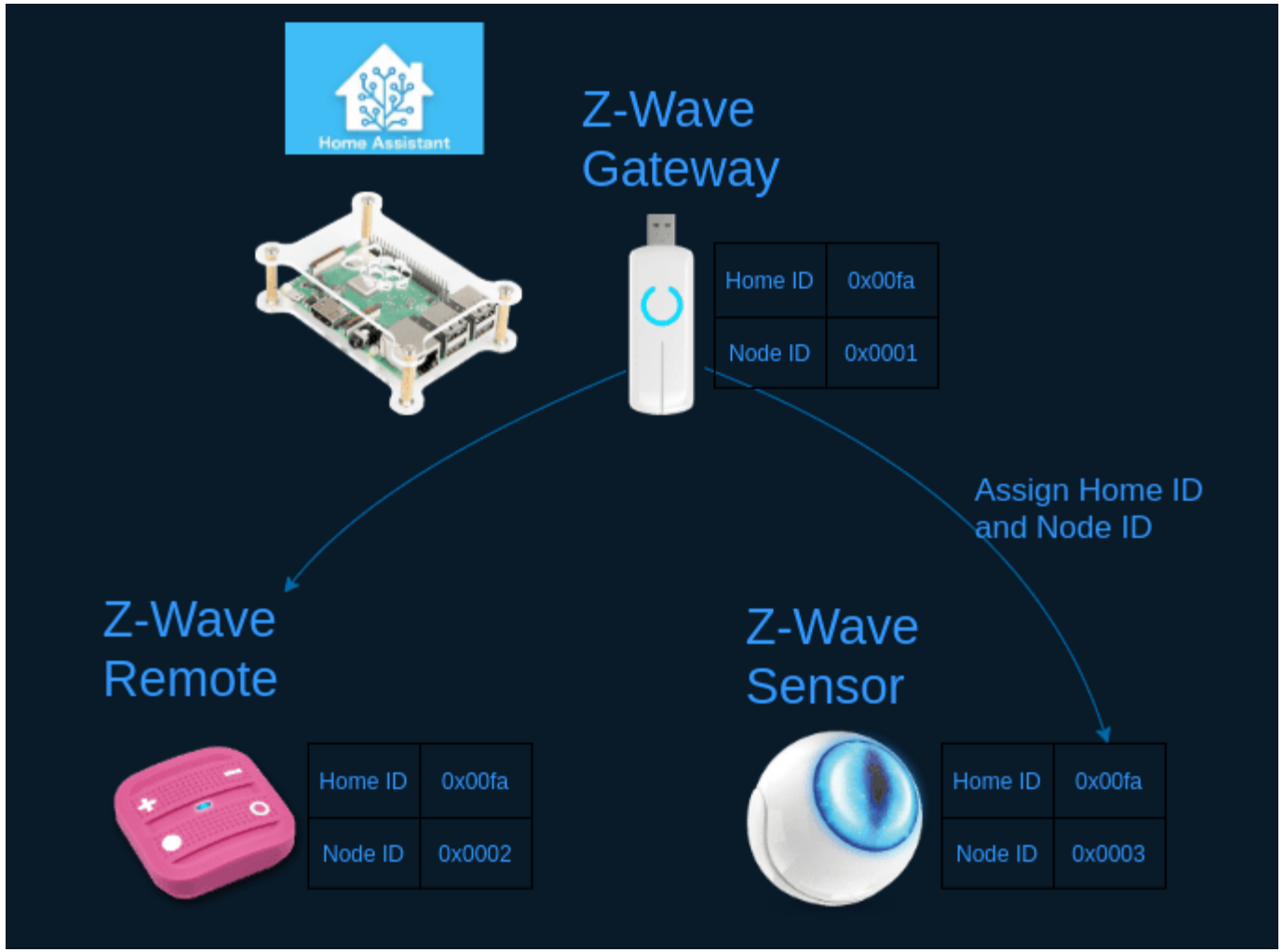
IOT Data link layer and Network layer protocols

Z-Wave



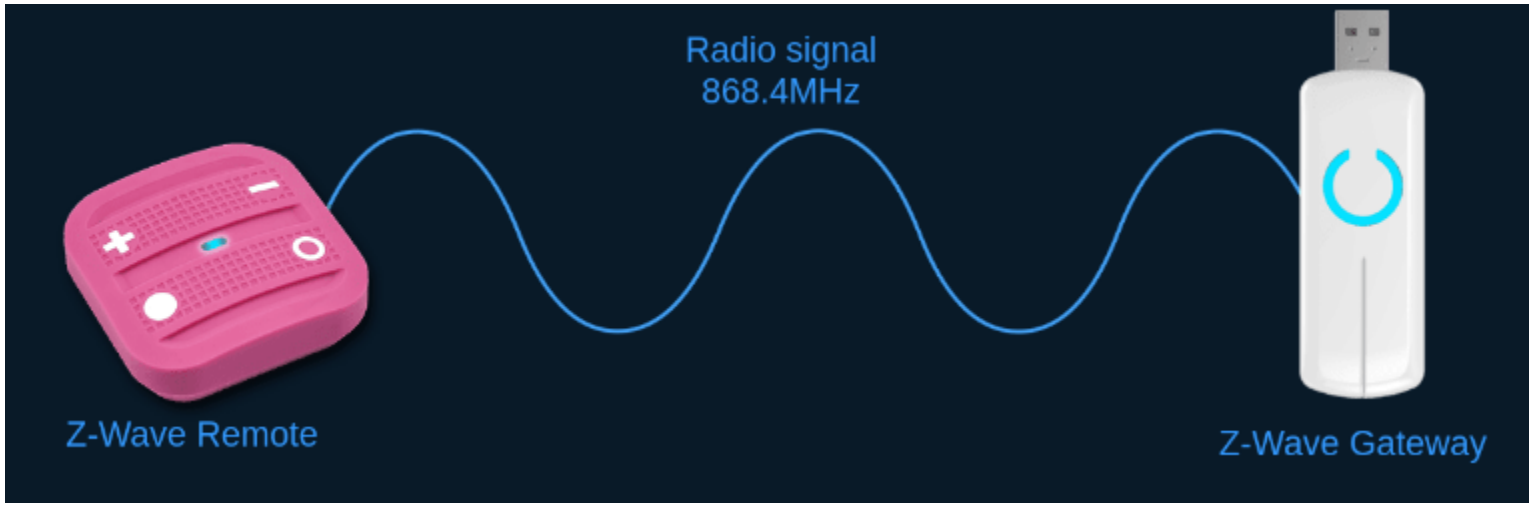
IOT Data link layer and Network layer protocols

Z-Wave



IOT Data link layer and Network layer protocols

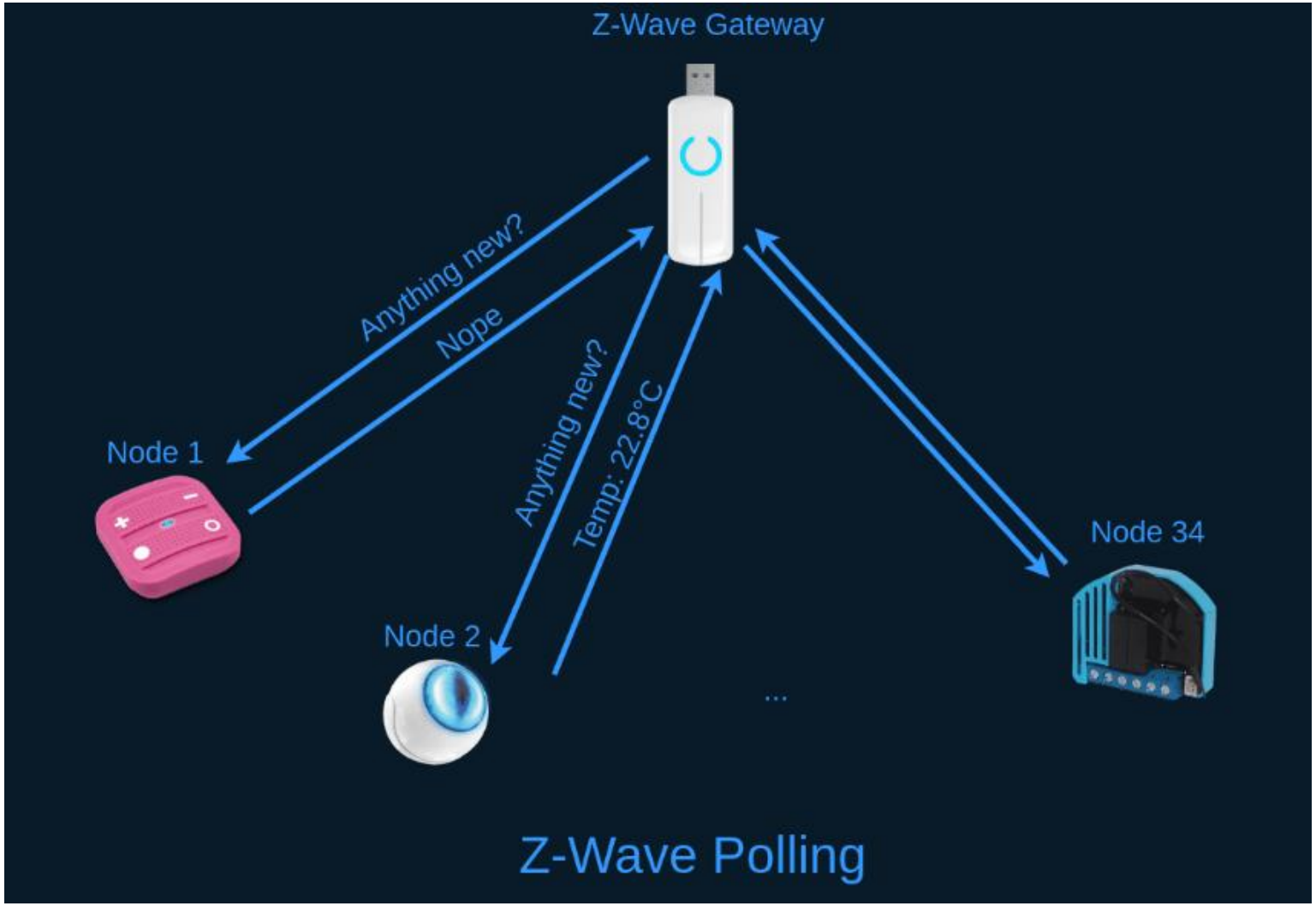
Z-Wave



IOT Data link layer and Network layer protocols

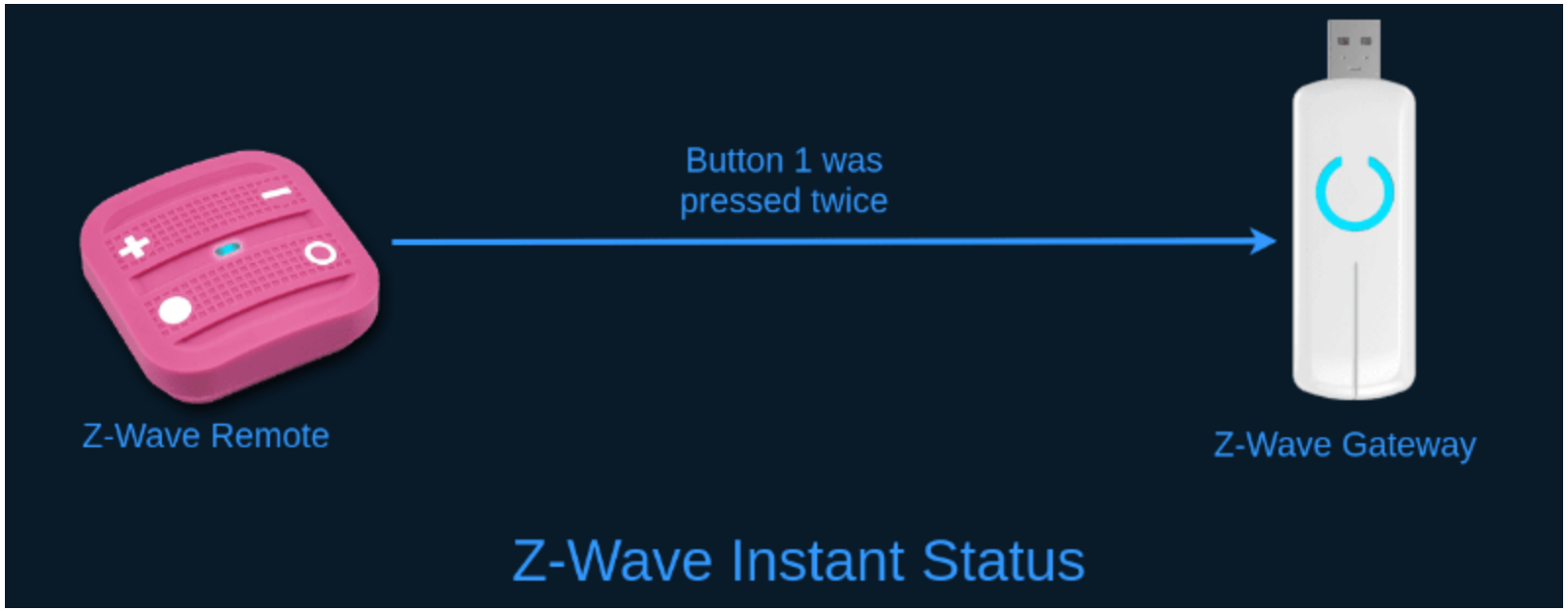
Z-Wave

Polling



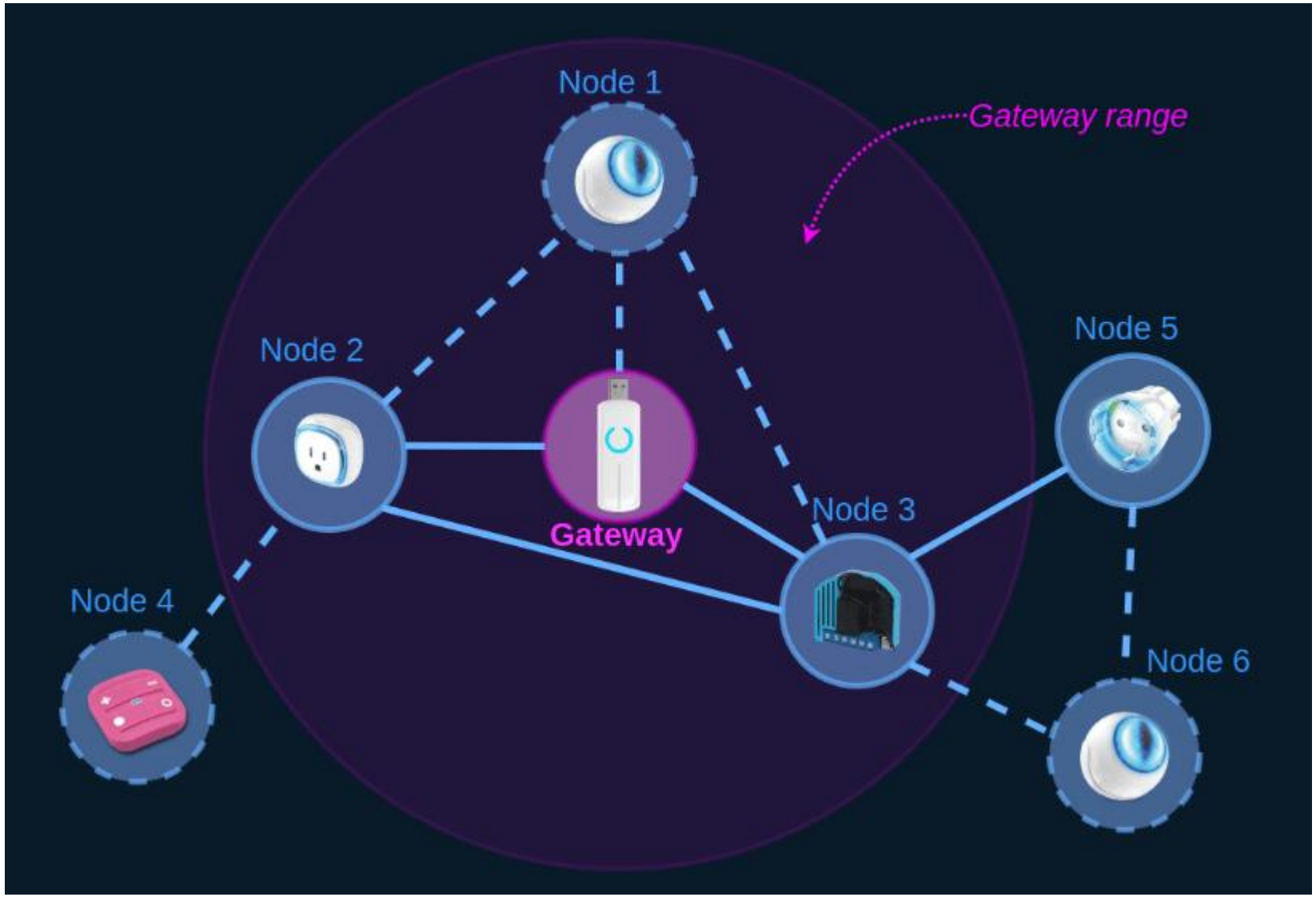
IOT Data link layer and Network layer protocols

Z-Wave



IOT Data link layer and Network layer protocols

Z-Wave



REAL-WORLD USE

Applications & Devices



Smart Lighting



Thermostats



Door Locks



Security Cameras



Smart Plugs



Garage Doors



Water/Leak Sensors



Motorized Blinds

Z-Wave vs. Zigbee vs. Wi-Fi

	Z-Wave	Zigbee	Wi-Fi
Frequency	Sub-1 GHz	2.4 GHz	2.4 / 5 GHz
Range/Hop	~30-100 m	~10-20 m	~50 m
Power Use	Very Low	Very Low	High
Max Nodes	232 (4,000 LR)	65,000+	Dozens
Interference	Low (avoids Wi-Fi band)	Shares band with Wi-Fi	N/A
Interoperability	Mandatory certification	Varies by vendor	N/A

Advantages & Challenges

Advantages

- Mandatory interoperability certification
- Very low power — years of battery life
- No Wi-Fi channel congestion
- Reliable self-healing mesh

Challenges

- Lower max throughput than Wi-Fi
- Needs a dedicated hub/controller
- Sub-1 GHz radios cost more to design
- Regional frequency variants

STAYING PROTECTED

Security: The S2 Framework

Since 2017, every new Z-Wave certified device must support Security 2 (S2) — closing gaps found in the earlier S0 framework.



AES-128 Encryption

All communication between nodes is fully encrypted end-to-end.



Secure Key Exchange

Elliptic-Curve Diffie-Hellman (ECDH) exchanges keys without exposing them over the air.



QR / PIN Pairing

Devices join the network using a scanned code — no default or guessable keys.

LOOKING AHEAD

The Road Ahead

Z-Wave Long Range (Z-Wave LR)

Up to 4,000 devices per network and ~1.6 km range on a single hub — no mesh hopping required.

Matter & Multi-Protocol Bridges

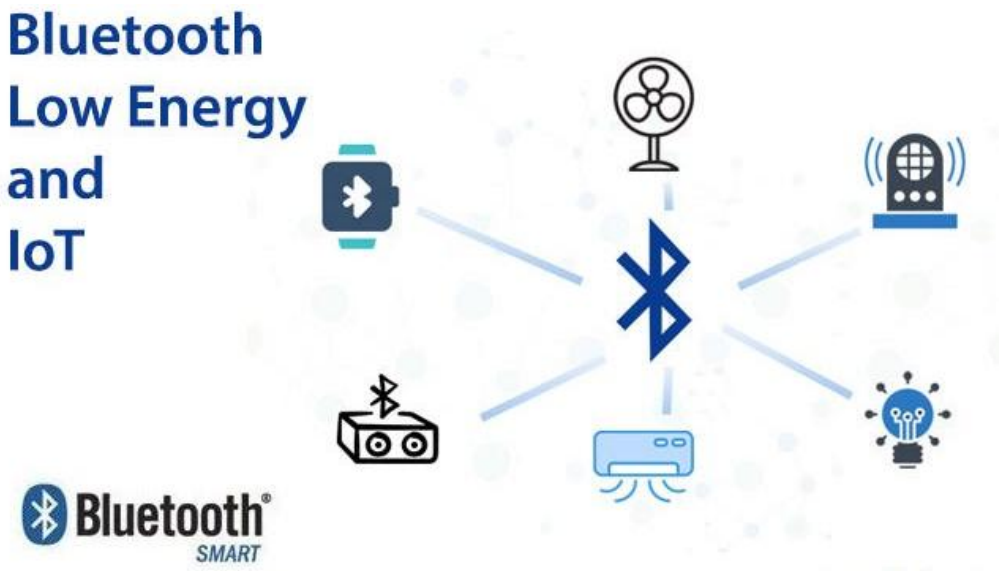
Z-Wave/Matter bridges let ecosystems interoperate, keeping Z-Wave's mesh reliability alongside newer standards.

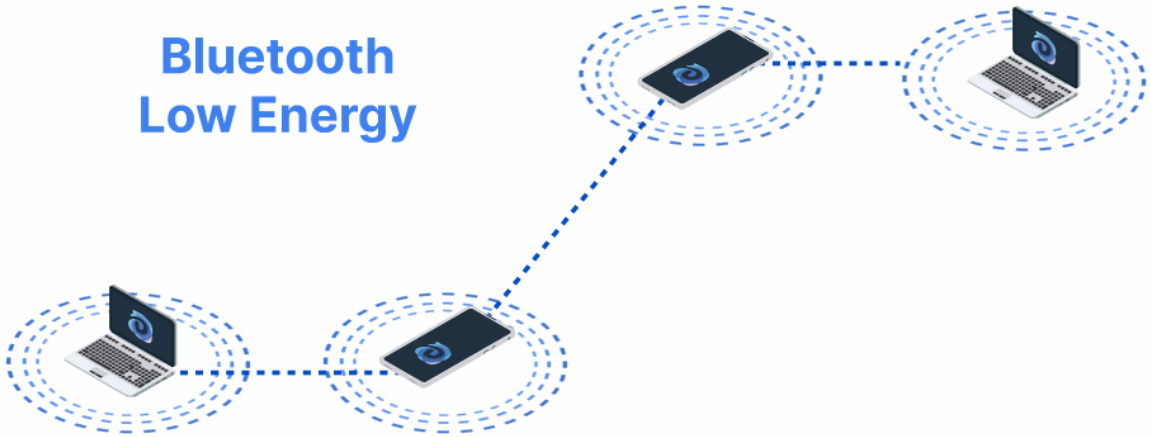
Even Lower Power Silicon

Newer chipsets from Silicon Labs push battery life further for coin-cell sensors.

IOT Data link layer and Network layer protocols

Bluetooth Low Energy





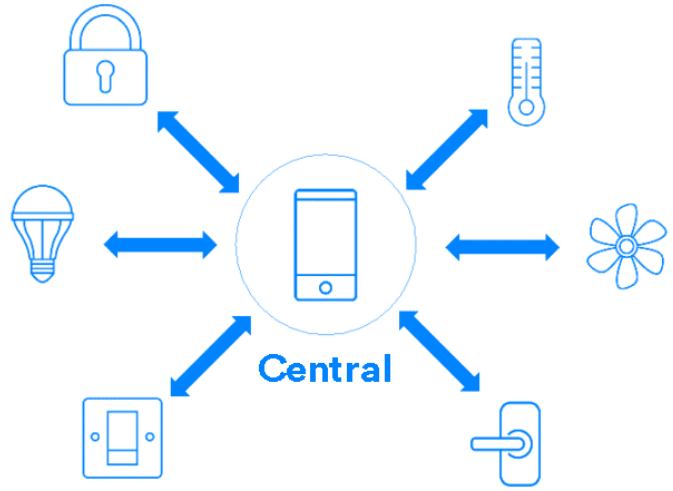


IOT Data link layer and Network layer protocols

Bluetooth Low Energy

1. Introduction to Bluetooth Low Energy
2. BLE vs. Bluetooth Classic
3. BLE Protocol Stack Architecture
4. Key Concepts: GAP & GATT
5. Advertising and Connection Process
6. BLE 5.x Features and Enhancements
7. IoT Applications and Use Cases
8. Summary and Key Takeaways

Bluetooth Low Energy



What is Bluetooth Low Energy?

Definition and Overview

- **Bluetooth Low Energy (BLE)** is a wireless personal area network technology designed for low-power operation
- Introduced with Bluetooth 4.0 specification
- Also known as **Bluetooth Smart**
- Operating in the **2.4 GHz ISM band** (Industrial, Scientific, Medical)
- Purpose-built for IoT applications requiring **long battery life**

Key Design Goals:

- Minimal power consumption
- Low cost
- Interoperability with existing Bluetooth devices
- Simple device discovery and connection

Bluetooth Low Energy

BLE vs Bluetooth Classic- Comparison of Technologies

Feature	BLE	Bluetooth Classic
Channels	40 channels (2 MHz spacing)	79 channels (1 MHz spacing)
Data Rate	125 Kb/s - 2 Mb/s	1 - 3 Mb/s
Power Consumption	~0.01x to 0.5x of Classic	Reference (1x)
Topology	Point-to-Point, Broadcast, Mesh	Point-to-Point (Piconet)
Use Cases	IoT sensors, wearables	Audio streaming, file transfer

BLE sacrifices data rate for **ultra-low power consumption**, making it ideal for battery-powered IoT devices

Bluetooth Low Energy

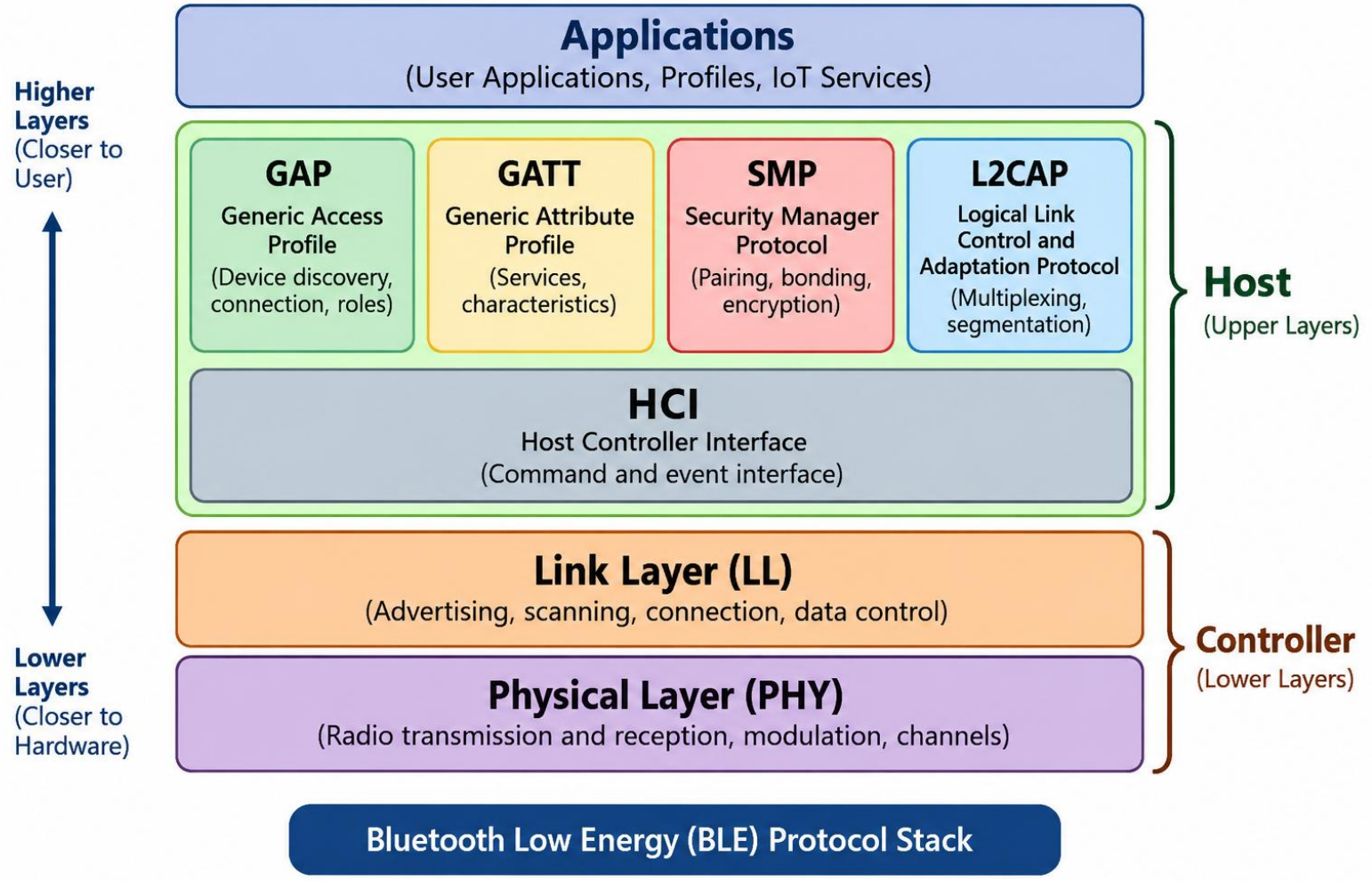
BLE Protocol Stack Overview-Two Main Sections: Controller and Host

- **Controller**:

  PHY and Link Layer (radio hardware)

- **Host**:

  Upper layers (GAP, GATT, SMP, L2CAP)



BLE protocol architecture

Think of BLE as a layered system: radio → controller → host → application

Application / Profiles	Your IoT service: temperature, heart rate, lighting
GAP	Device roles, advertising, scanning & connections
GATT	Services + characteristics: application data structure
SMP	Pairing, bonding, key generation & link encryption
ATT	Attribute operations: read, write, notify, indicate
L2CAP	Multiplexing and transport adaptation
HCI	Interface between host and controller
Link Layer	Advertising, scanning, connections, channel selection
PHY	2.4 GHz radio; LE 1M / 2M / Coded PHY

Bluetooth SIG BLE Primer describes the BLE stack from PHY through GAP, with security and host/controller layers.

Bluetooth Low Energy

Physical Layer (PHY)

Frequency Band: 2.4 GHz ISM (2402 - 2480 MHz)

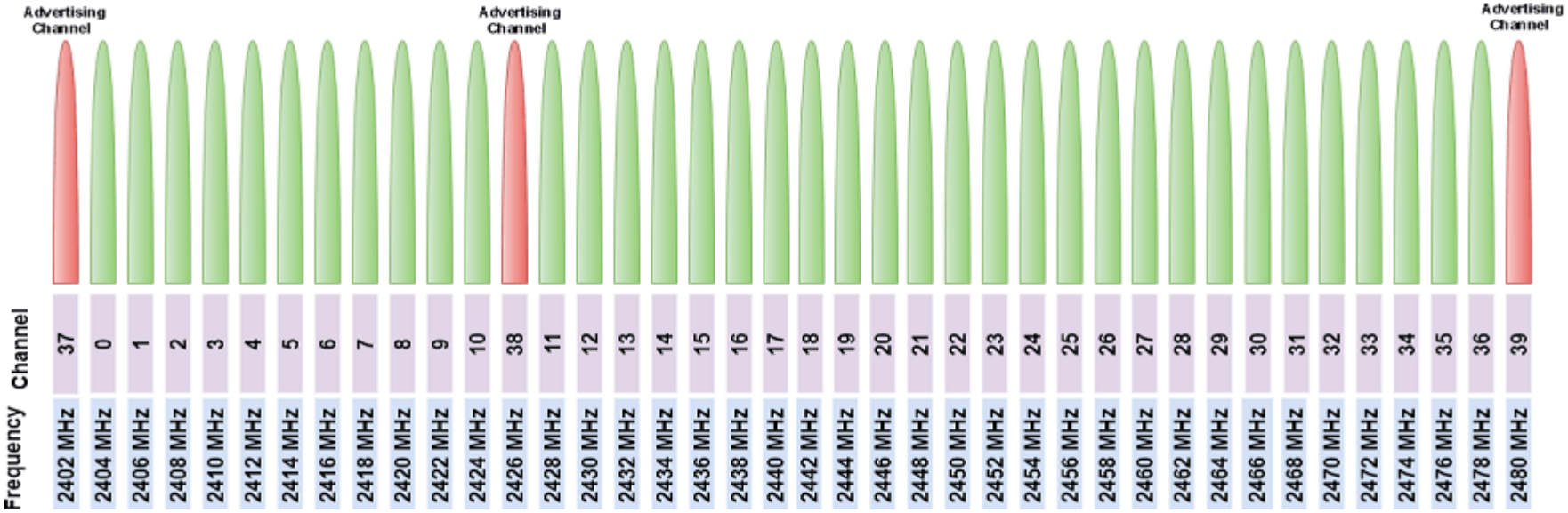
40 RF Channels:

- **3 Advertising Channels** (37, 38, 39) - for device discovery
- **37 Data Channels** - for data transmission after connection

Modulation: GFSK (Gaussian Frequency-Shift Keying)

Data Rates (BLE 5.x):

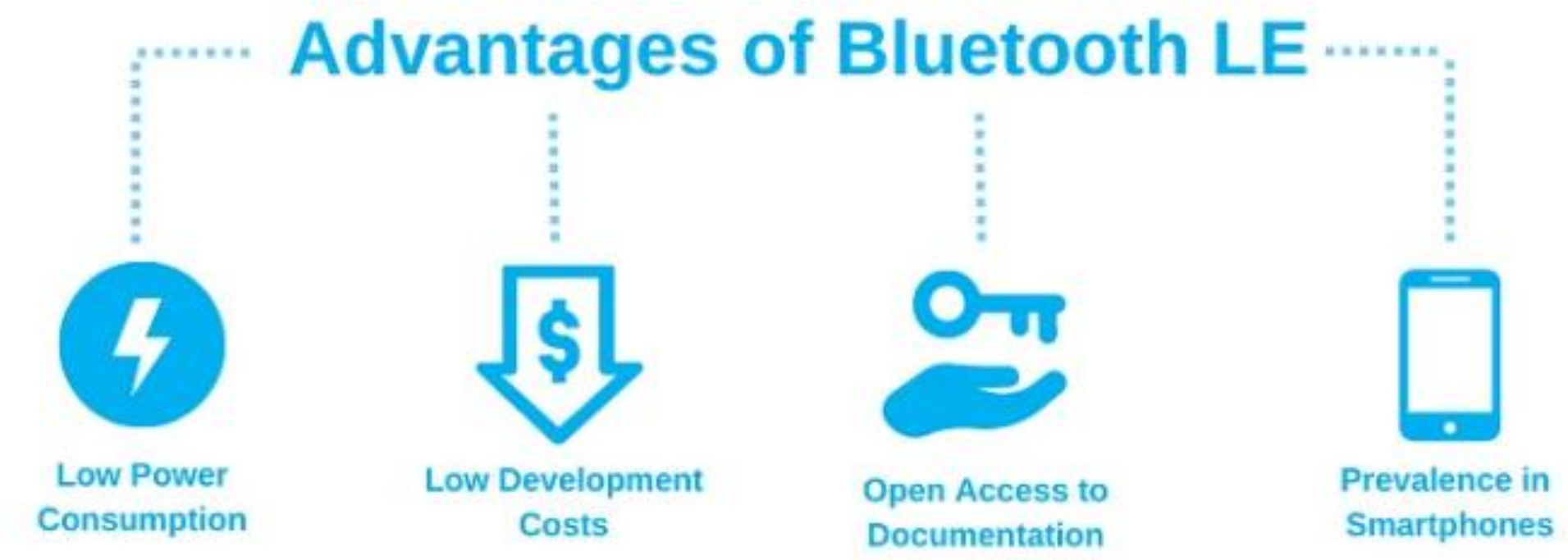
- LE 1M PHY: 1 Mb/s
- LE 2M PHY: 2 Mb/s
- LE Coded PHY (S=2): 500 Kb/s
- LE Coded PHY (S=8): 125 Kb/s



IOT Data link layer and Network layer protocols

Bluetooth Low Energy

Advantages of BLE



IOT Data link layer and Network layer protocols

Bluetooth Low Energy

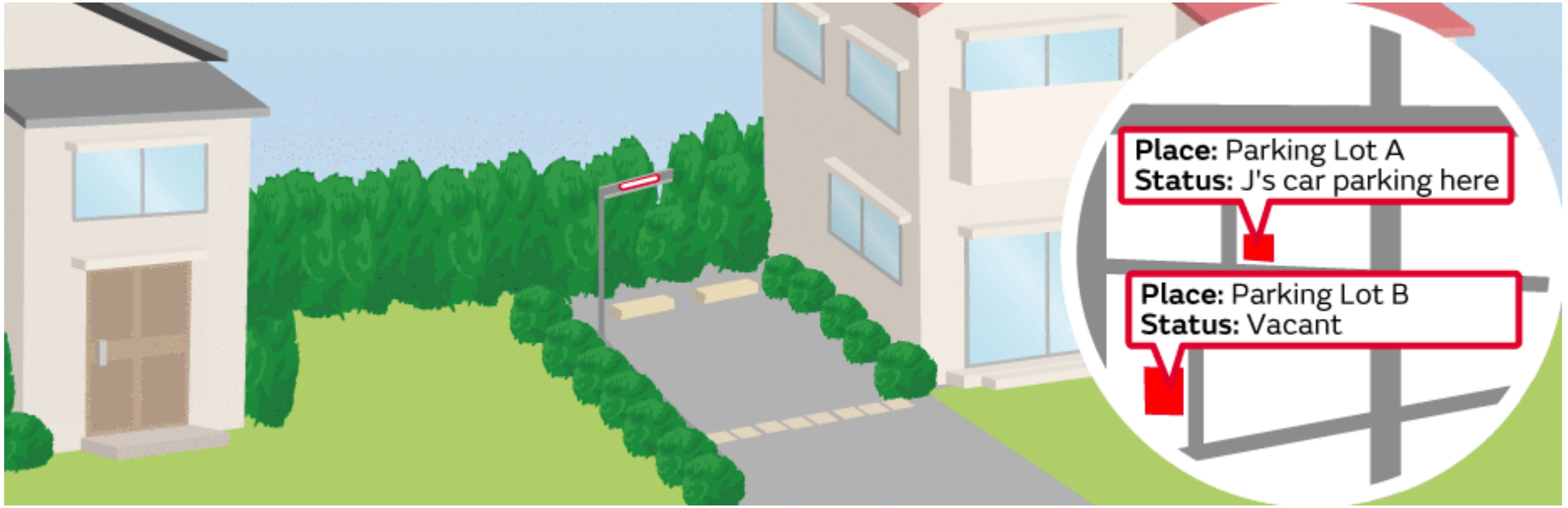
Limitations of BLE



IOT Data link layer and Network layer protocols

Bluetooth Low Energy

Applications of BLE

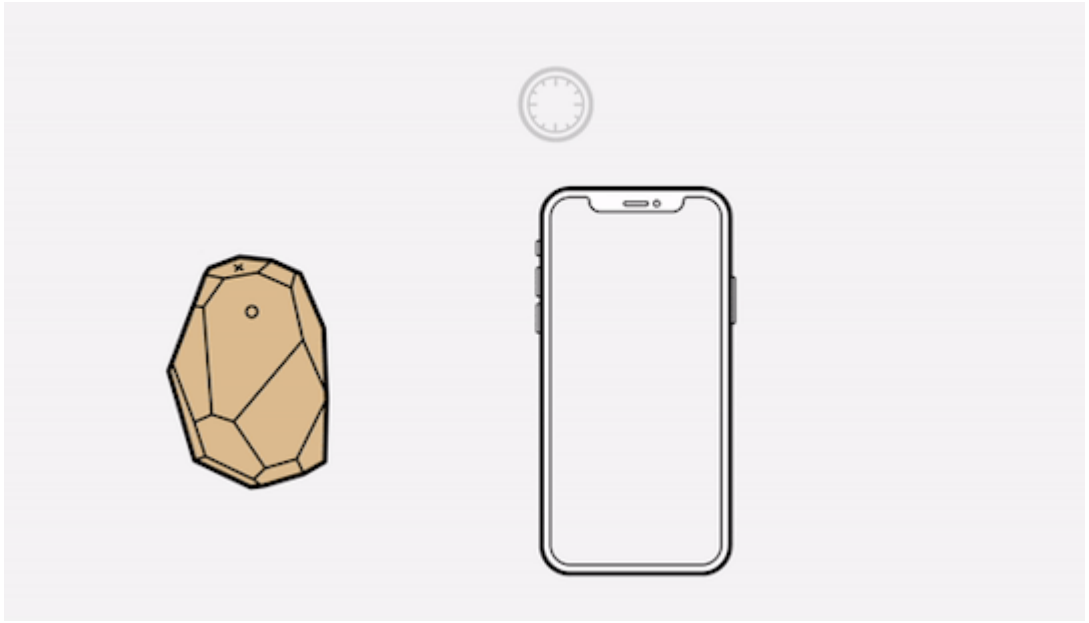


Parking Information

IOT Data link layer and Network layer protocols

Bluetooth Low Energy

Applications of BLE

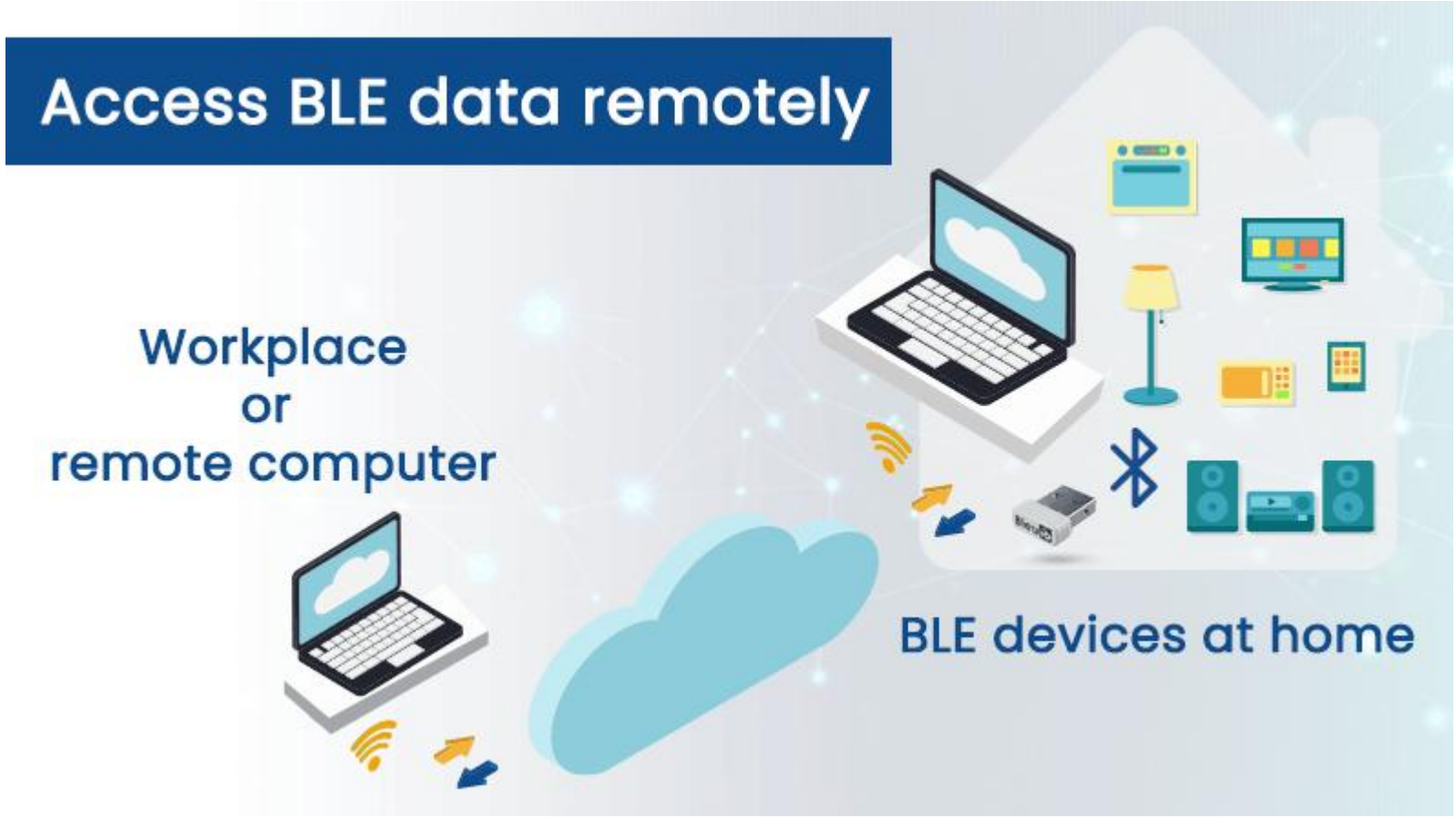


Google Eddystone beacon service

IOT Data link layer and Network layer protocols

Bluetooth Low Energy

Applications of BLE



IOT Data link layer and Network layer protocols

Bluetooth Low Energy

Applications of BLE



- **Wearables and Healthcare**: Fitness trackers, medical monitoring devices.
- **Smart Home**: Door locks, thermostats, lighting control.
- **Beacons**: Location tracking and proximity-based marketing.
- **Industrial IoT**: Asset tracking, predictive maintenance.
- **Agriculture**: Environmental monitoring sensors.

IOT Data link layer and Network layer protocols

Zigbee

•**Zigbee** is a wireless communication protocol based on the IEEE 802.15.4 standard, designed for low-power, low-data-rate, and short-range communication.

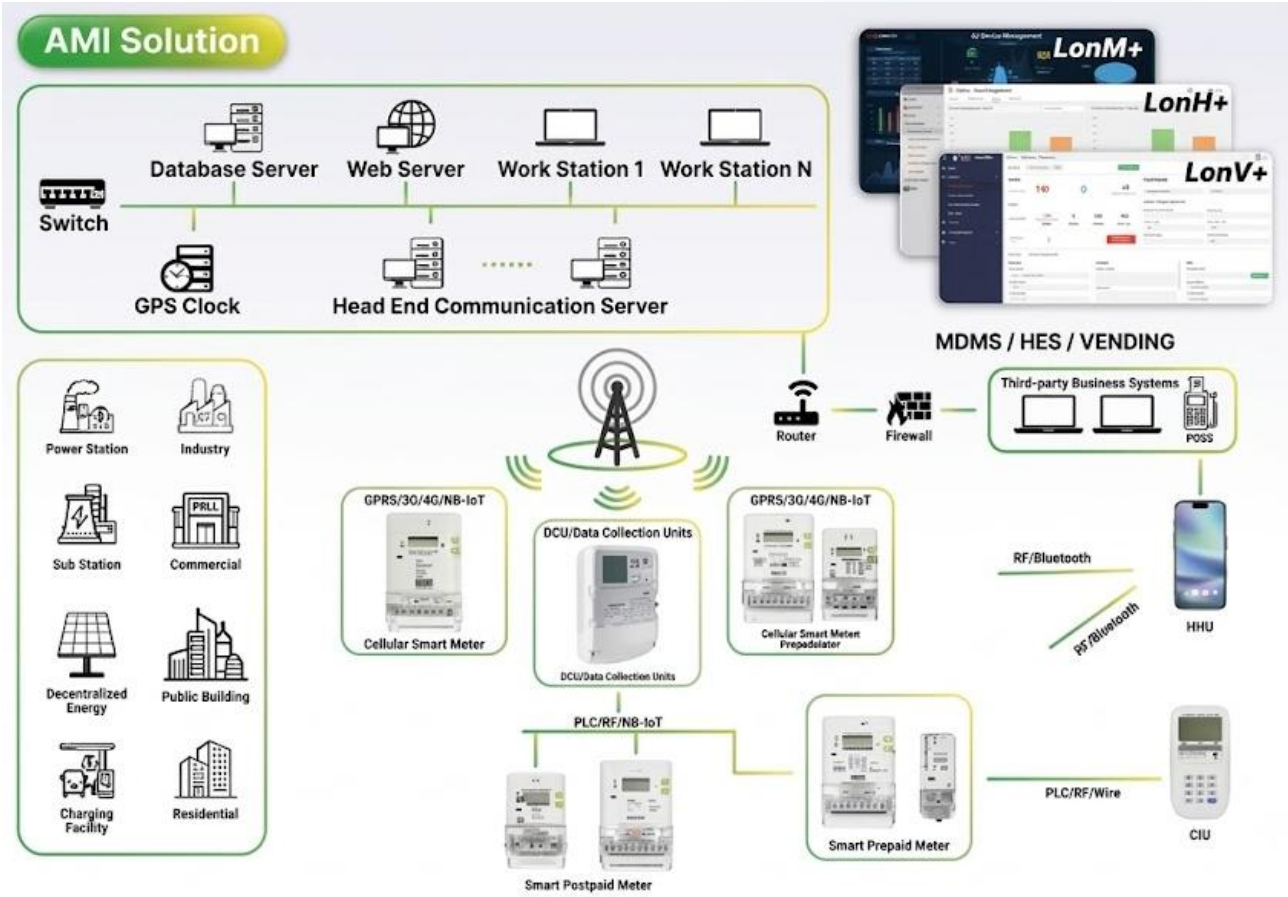
•It is widely used in IoT applications for creating efficient and scalable wireless networks.



Introduction to Zigbee SE

- **Global IoT Standard:** Designed specifically for Home Area Networks (HAN) connecting utilities and consumers.
- **Smart Grid Integration:** Enables real-time energy monitoring, dynamic pricing, and automated load control.
- **Energy Efficiency:** Empowers consumers to reduce peak demand and lower overall utility expenditure.
- **Interoperability:** Ensures seamless communication across diverse vendors and utility assets.

Automated Metering Infrastructure Solution



IOT Data link layer and Network layer protocols

Zigbee Smart Energy

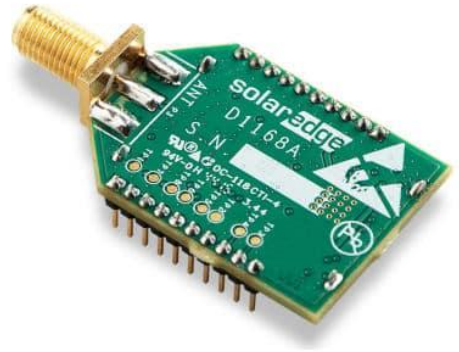


Technical Specifications:

- **Frequency Bands**:
  - **2.4 GHz** (Global availability, 16 channels).
  - **868 MHz** (Europe, 1 channel).
  - **915 MHz** (North America, 10 channels).
- **Data Rate**:
  - Up to **250 kbps** at 2.4 GHz.
  - **40 kbps** at 915 MHz.
  - **20 kbps** at 868 MHz.
- **Range**:
  - Typical range: **10–100 meters** (depending on environmental conditions and obstacles).
  - Mesh networking extends range by relaying data through intermediate devices.

IOT Data link layer and Network layer protocols

Zigbee Smart Energy

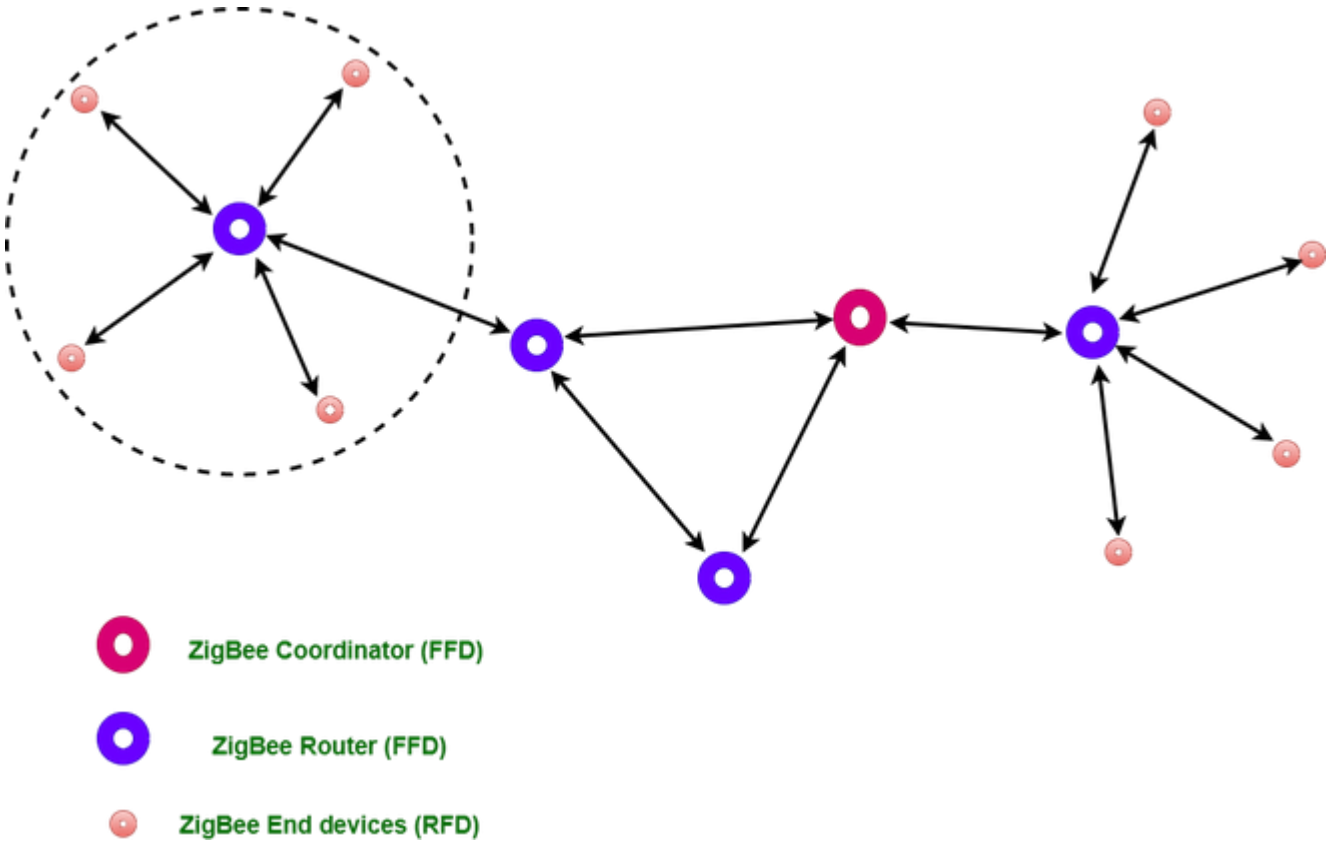


Technical Specifications:

- **Network Topology**:
  - Supports **star**, **tree**, and **mesh** topologies.
  - Can connect up to **65,536 devices** in a single network.
- **Power Consumption**:
  - Ultra-low power, enabling devices to operate on coin-cell batteries for **several years**.
- **Encryption and Security**:
  - **AES-128** encryption for secure data transmission.
  - Key management ensures secure device pairing.
- **Latency**:
  - Low latency, typically **30 ms** for data communication.

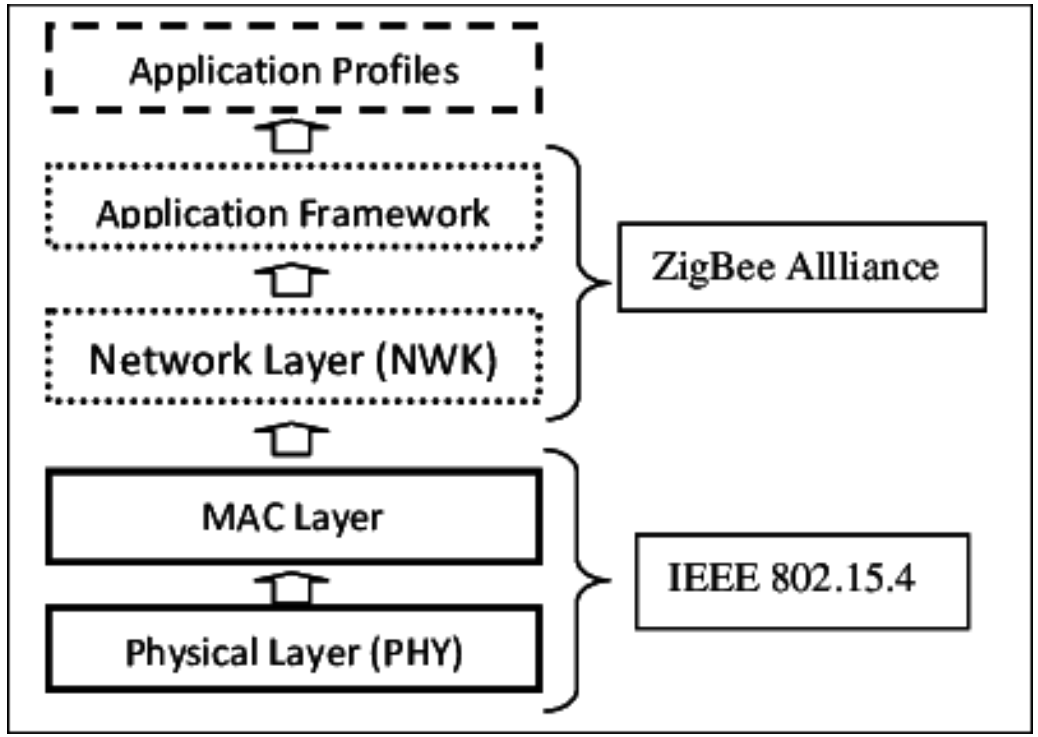
IOT Data link layer and Network layer protocols

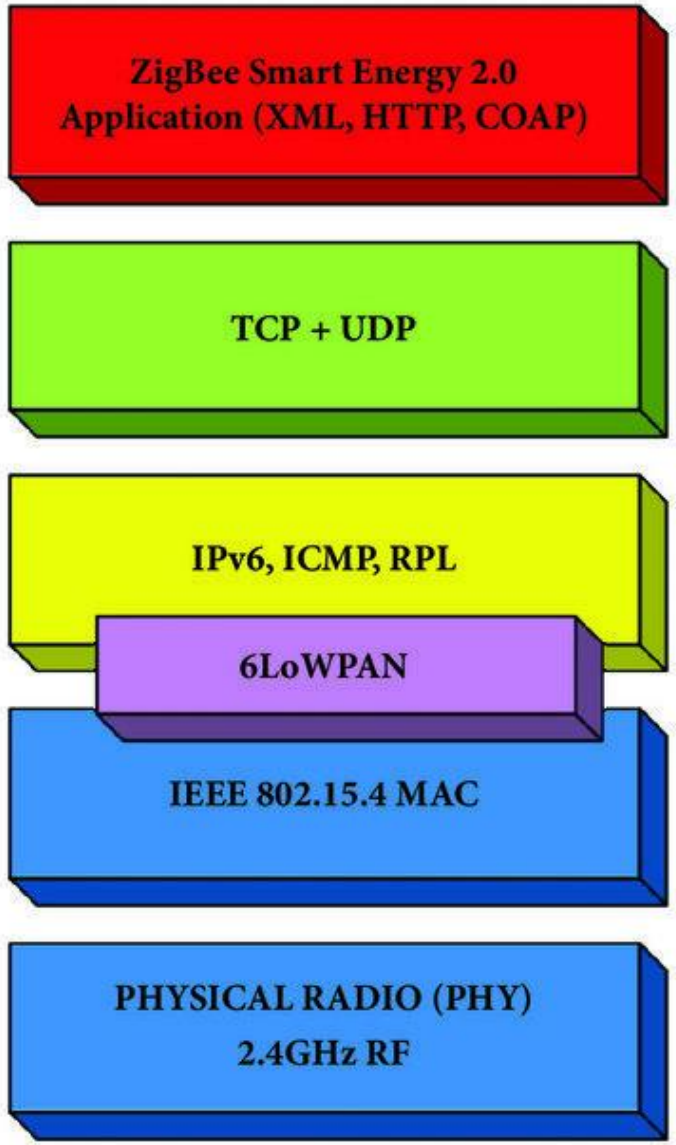
Architecture of Zigbee



IOT Data link layer and Network layer protocols

Protocol Stack of Zigbee

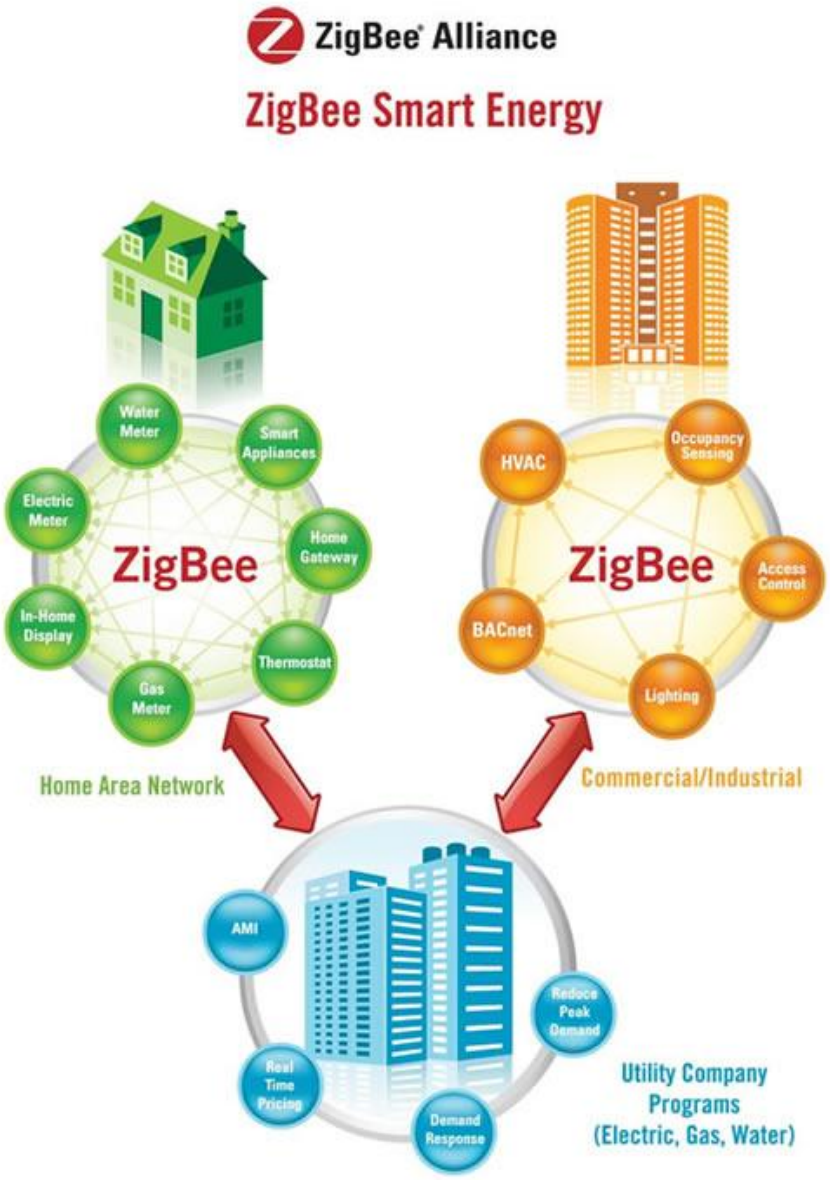




IOT Data link layer and Network layer protocols

Zigbee Smart Energy

- **Zigbee Smart Energy** is a profile designed for energy management and efficiency in smart grid and home automation systems.
- It leverages the Zigbee protocol to enable reliable, low-power communication between energy devices such as smart meters, thermostats, and appliances.



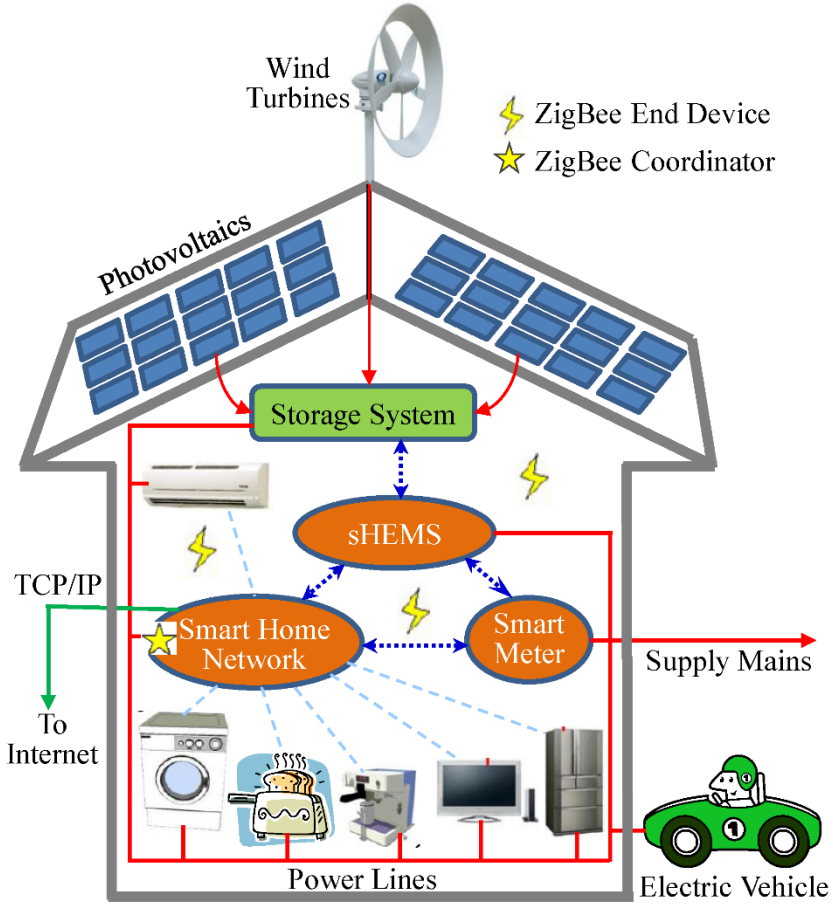
IOT Data link layer and Network layer protocols

Zigbee Smart Energy

Applications of Zigbee in Smart Energy:



•**Smart Meters**: Tracks electricity, gas, and water usage.



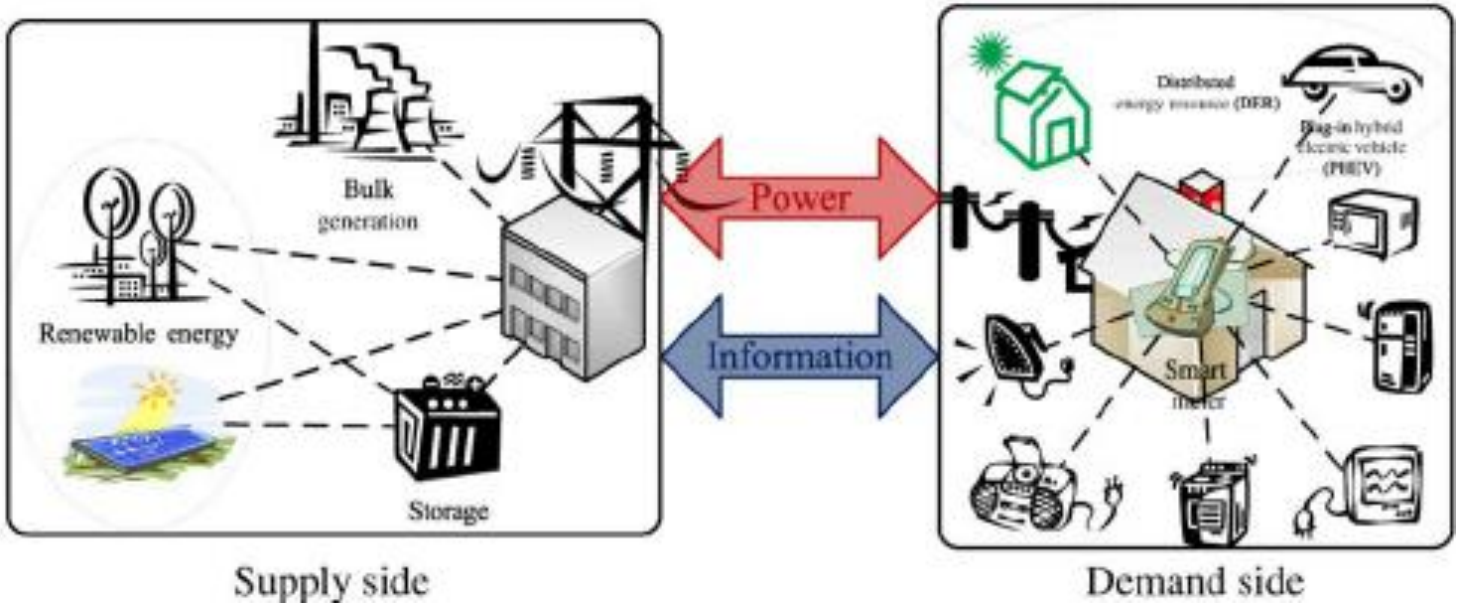
•**Home Energy Management Systems**: Controls appliances and lighting for energy efficiency.

IOT Data link layer and Network layer protocols

Zigbee Smart Energy

Applications of Zigbee in Smart Energy:

•**Demand Response Systems**: Enables utilities to manage energy demand by controlling device loads during peak times.

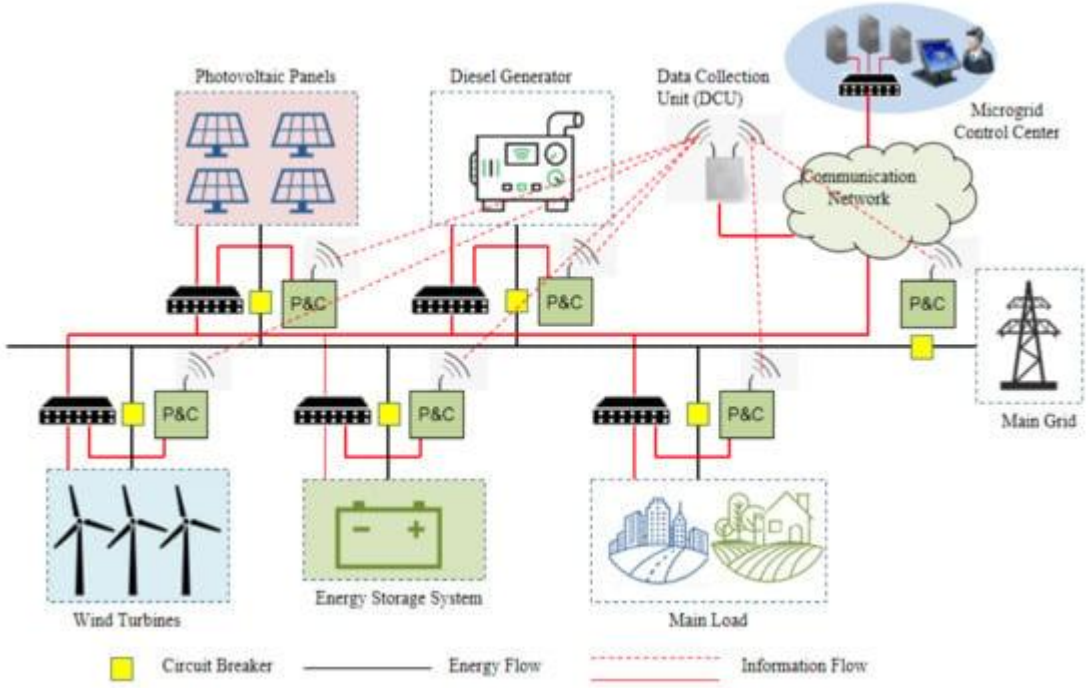


IOT Data link layer and Network layer protocols

Zigbee Smart Energy

Applications of Zigbee in Smart Energy:

•**Renewable Energy Integration**: Manages distributed energy sources like solar panels





•**Electric Vehicle Charging**: Coordinates charging schedules to minimize grid impact.

Key ZSE Function Clusters



Metering Cluster

Delivers real-time data on active energy (kWh), reactive power, gas, and water consumption to users and utilities.



Demand Response

Allows utility providers to issue load shedding events (DRLC) during critical grid peak hours.



Price Cluster

Transmits real-time tier pricing, Time-of-Use (TOU) rates, and currency conversion metrics directly to devices.

Smart Energy Endpoints



Smart Meter

Serves as the ESI coordinator & main grid utility gateway.



In-Home Display

Provides real-time feedback on wattage and billing rates.



Smart Thermostat

Adjusts HVAC setpoints automatically during peak hours.

IOT Data link layer and Network layer protocols

Zigbee Smart Energy Vs Bluetooth Low Energy

Parameter	BLE	Zigbee Smart Energy
Frequency	2.4 GHz	2.4 GHz
Data Rate	Up to 2 Mbps	Up to 250 kbps
Range	~10–100 m	~10–100 m per hop
Topology	Mainly Star	Mesh / Star
Power	Very Low	Very Low
Main Use	Smartphone & IoT devices	Smart meters & energy management
Smartphone Support	Direct	Usually through Gateway
Best Feature	Easy phone connectivity	Reliable mesh networking

IOT Data link layer and Network layer protocols

DASH7

Dynamic, Adaptive Streaming over HTTP

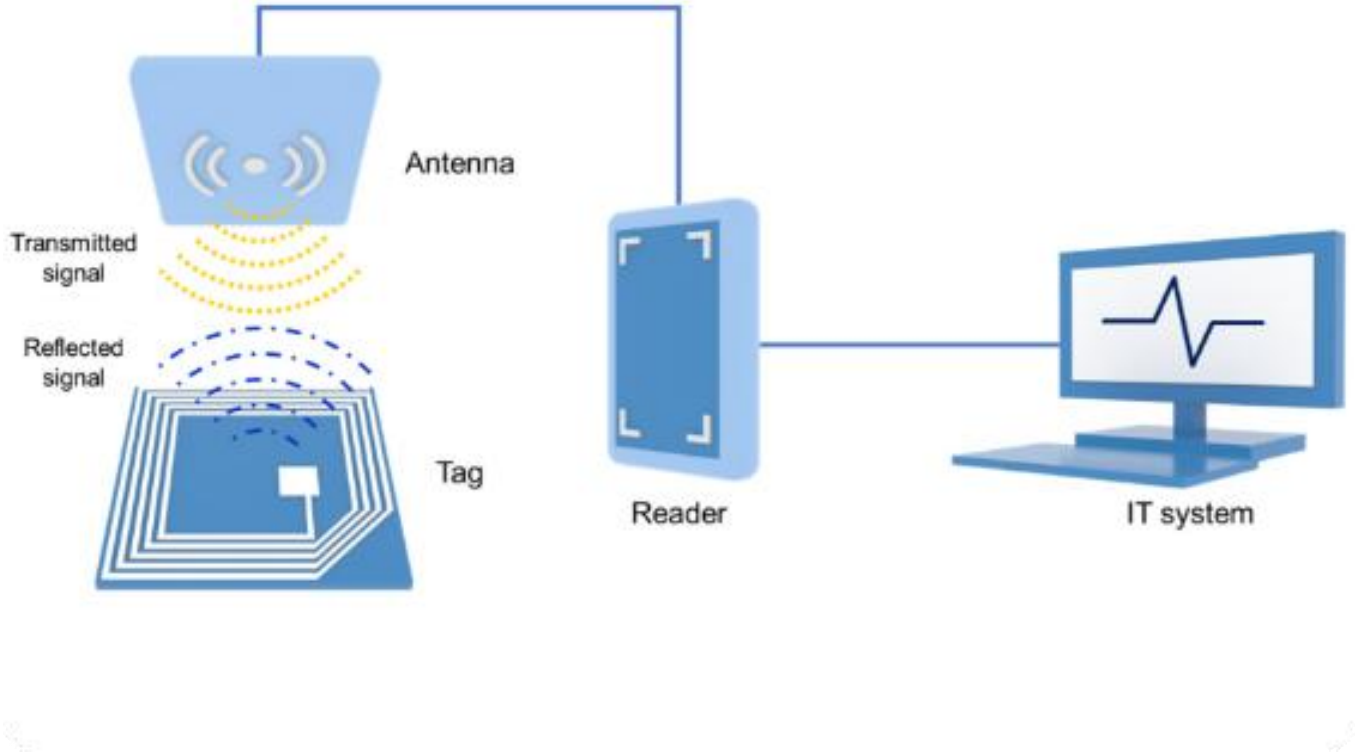
- An Open, Low-Power, Sub-GHz Wireless Protocol for Industrial IoT & Active RFID

DASH7 is a low-power, wireless communication protocol designed for long-range, low-data-rate, and low-latency applications.

It is based on the **ISO/IEC 18000-7** standard and is optimized for IoT use cases requiring efficient and reliable communication, such as asset tracking, industrial monitoring, and smart cities.

IOT Data link layer and Network layer protocols

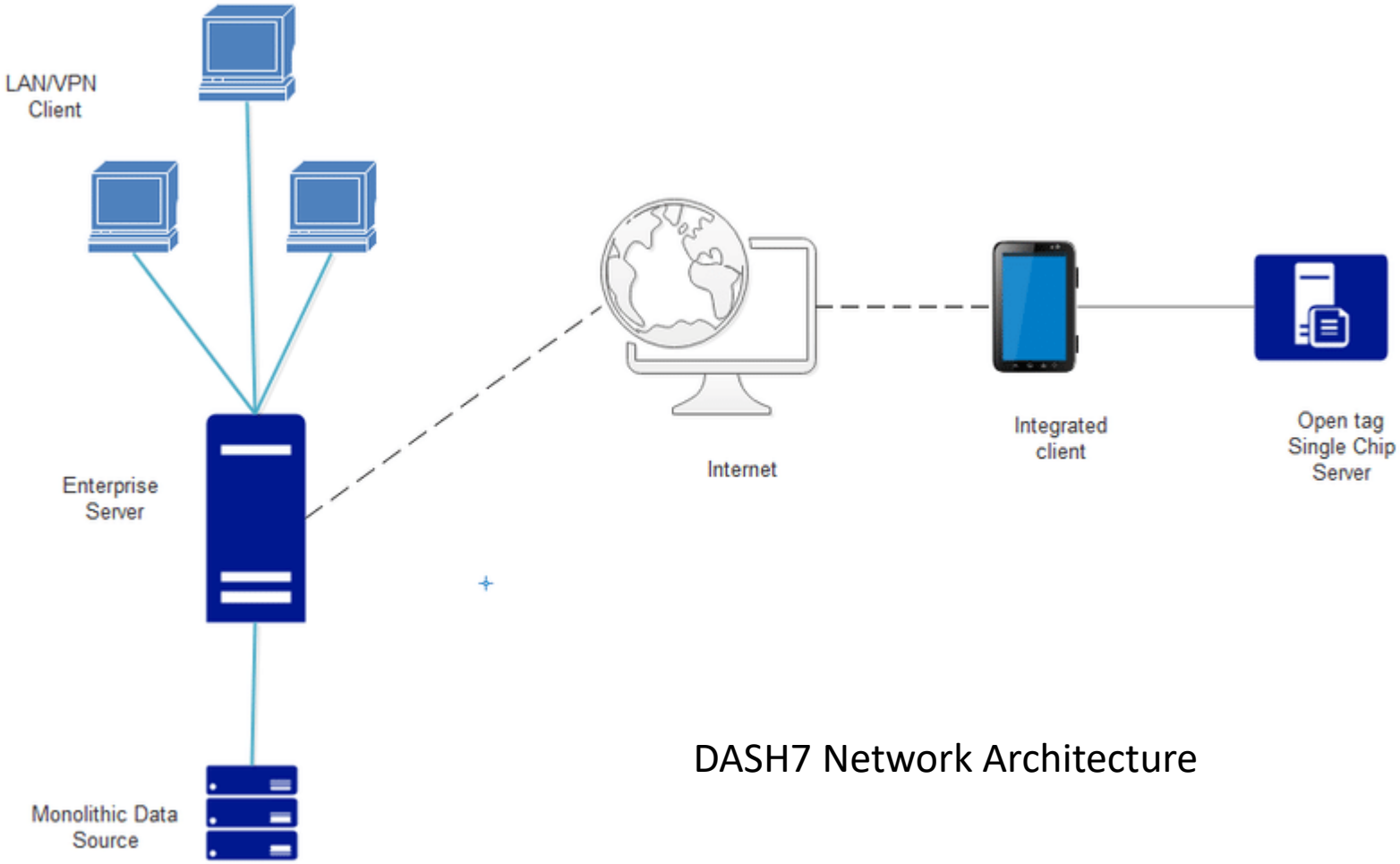
DASH7



DASH7 Working

IOT Data link layer and Network layer protocols

DASH7



DASH7 Network Architecture

DASH7 Protocol Stack Architecture

OSI Layer	DASH7 Protocol Layer	Key Responsibilities & Features
Application / Presentation	D7A File System (D7AFS)	Structured memory model; transparent local and remote read/write file access.
Session / Transport	Session & Transport Layer	Ad-hoc communication dialogs, reliable transactions, security, and retry policies.
Network / Data Link	NWK & MAC Layer	Single-hop & 2-hop routing, CSMA/CA channel access, wake-on-scan filtering.
Physical Layer (PHY)	Sub-GHz PHY Layer	GFSK modulation, 433 / 868 / 915 MHz ISM bands, data rates up to 166.67 kbps.

Physical Layer & Spectrum Allocation

Sub-GHz Spectrum & Modulation

Operates in license-free sub-GHz bands: **433 MHz**, **868 MHz** (EU), and **915 MHz** (US).

Uses **GFSK (Gaussian Frequency Shift Keying)** modulation for high spectral efficiency and low inter-channel interference.

DASH7 Channel Classes

Lo-Rate Class: 9.6 kbps (Maximum sensitivity, extended range up to several kilometers).

Normal-Rate Class: 55.55 kbps (Standard operational mode balancing speed and energy).

Hi-Rate Class: 166.67 kbps (High-speed data transfer for large firmware updates).

DASH7 Communication Models

Push Model (Tag-Initiated)

The sensor node autonomously initiates data transmission to the gateway upon a specific threshold or timer event.

Uses the **Beacon Transmit Series**. Ideal for periodic status updates, emergency alarms, and asset location reports.

Pull Model (Interrogator-Initiated)

The gateway or reader queries end devices within range using the **D7A Query Protocol**.

Subcontrollers or EndPoints respond dynamically without requiring pre-assigned time-slots or complex network join sessions.

Energy Efficiency & Longevity

10+

Years Operational Life

Why DASH7 is Exceptionally Low Power

Asynchronous Duty Cycling: Eliminates continuous clock synchronization and beacon overhead found in synchronous protocols.

Low Wake-Up Latency: Preamble sampling enables fast channel checks in under 1 millisecond.

Sub-GHz Efficiency: Requires less power than 2.4 GHz signals to achieve equivalent radio range and wall penetration.

Industrial IoT Application Scenarios



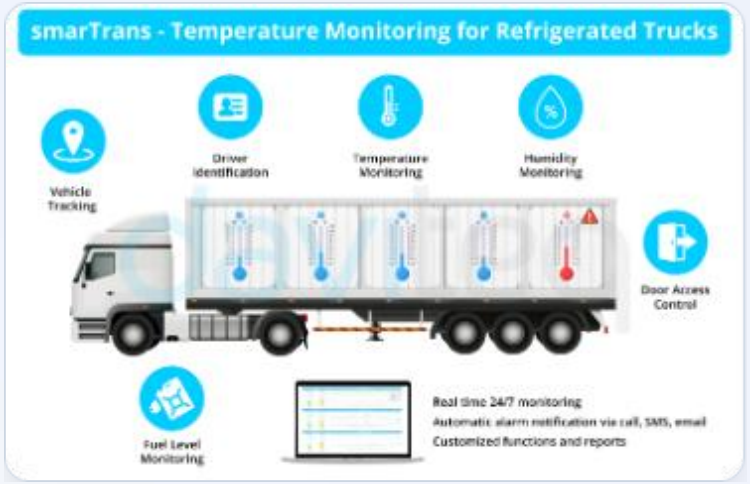
Logistics & Asset Tracking

Real-time location and status monitoring of warehouse pallets, containers, and mobile equipment.



Industrial Automation

Low-latency wireless monitoring of factory machinery, vibration sensors, and safety valves.



Cold Chain Monitoring

Continuous temperature tracking inside pharmaceutical and food transport vehicles without stopping tags.

IOT Data link layer and Network layer protocols

DASH7

